PARIKSHA365 — STUDY NOTES

Statistics — The JSO Mastery Book

SSC CGL Tier-2 Paper II (Junior Statistical Officer / Statistical Investigator) — 95%+ targeted

SSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

How to use this book

This is the **only** book a JSO aspirant should need.

It is built around four hard rules:

Rule	What it means in practice
Examiner mindset	For every topic we state how SSC frames it, the 3-5 angles they recycle, and the trap option they always plant.
Formula → derivation → drill	Every formula has a 1-line proof or intuition so you don't memorise blindly; then 8-15 worked PYQ-style questions per chapter.
Pedagogy first	Tables, mind maps, pitfall boxes, mnemonic boxes, recall prompts — no English paragraphs. Information is laid out like a publisher's textbook, not a blog.
Self-check	Every chapter ends with a mini-mock (10 Qs, fully solved) and active-recall prompts. Last chapter is a full 100-Q mock.

> ***The promise.*** *Read this book end-to-end with two passes + the embedded recall prompts and you will solve >95 of the 100 questions on JSO Paper II.*

Paper map and importance dashboard

Paper: SSC CGL Tier-2 Paper II (Statistics) — **Junior Statistical Officer / Statistical Investigator Grade-II**

Item	Spec
Total questions	100
Total marks	200
Marks per question	2
Negative marking	**−0.5** per wrong attempt
Time	120 minutes (2 hours)
Mode	Computer-Based (CBT)
Calculator	**Yes — built-in calculator** allowed in this paper
Medium	English / Hindi

Chapter-wise importance (averaged across last 6 JSO papers)

Ch	Topic	Avg Qs / paper	Marks (out of 200)	Importance	Difficulty
1	Collection, Classification, Presentation	8	16	⭐⭐⭐⭐ HIGH	Easy
2	Measures of Central Tendency	10	20	⭐⭐⭐⭐⭐ CRITICAL	Easy-Med
3	Measures of Dispersion	10	20	⭐⭐⭐⭐⭐ CRITICAL	Easy-Med
4	Moments, Skewness, Kurtosis	7	14	⭐⭐⭐⭐ HIGH	Med
5	Correlation & Regression	10	20	⭐⭐⭐⭐⭐ CRITICAL	Med
6	Probability Theory	8	16	⭐⭐⭐⭐⭐ CRITICAL	Med
7	Random Variable & Distributions	10	20	⭐⭐⭐⭐⭐ CRITICAL	Med-Hard
8	Sampling Theory	8	16	⭐⭐⭐⭐ HIGH	Easy-Med
9	Statistical Inference (Estimation + Testing)	12	24	⭐⭐⭐⭐⭐ CRITICAL	Med-Hard
10	Analysis of Variance	4	8	⭐⭐⭐ MEDIUM	Med
11	Time Series Analysis	7	14	⭐⭐⭐⭐ HIGH	Easy-Med
12	Index Numbers	6	12	⭐⭐⭐⭐ HIGH	Easy
	TOTAL	**100**	**200**		

What this dashboard tells you. *The five **CRITICAL** chapters (2, 3, 5, 6, 7, 9) carry **120 / 200 marks** = 60%. Mastering them alone gets you to ≈108/120. With the four **HIGH** chapters added (1, 4, 8, 11, 12) you cover **88% of the paper**. ANOVA is the only area where you can afford ≤80% mastery if time is tight.*

Cut-offs reality check

Historically, the General (UR) cut-off for JSO/Statistical Investigator Gr-II has hovered in the **70–75 % band**, i.e., roughly **140–150 out of 200 marks**. Securing **≥ 90 %** (≥ 180 / 200) practically guarantees a final selection in most recruitment cycles and is the aspirational target. The exact cut-off shifts cycle to cycle based on vacancy count and competition intensity — treat 75 % as your floor, not your ceiling.

A 95% target = **190/200**. That demands ≥96 correct out of 100 (with ≤2 wrongs). It is achievable **only** if you can recognise the type instantly and the formula is already in muscle memory.

PART 0 — FOUNDATIONS YOU CANNOT SKIP

Before any topic you must be fluent in these 5 micro-skills. None is hard. Skipping one will cost you 10–15 marks across the paper.

0.1 Sigma (Σ) algebra — the 8 rules

Let c be a constant, X, Y be variables, n the count of observations.

#	Rule	Example
1	$\sum c = nc$	$\sum_{i=1}^{4} 7 = 28$
2	$\sum cX_i = c \sum X_i$	constant pulls out
3	$\sum (X_i + Y_i) = \sum X_i + \sum Y_i$	sums distribute
4	$\sum (X_i - \bar{X}) = 0$	**always**, by definition of mean
5	$\sum (X_i - \bar{X})^2 = \sum X_i^2 - n\bar{X}^2$	computational form of variance
6	$\sum X_i^2 \neq (\sum X_i)^2$	**trap** — examiners love this
7	$\sum X_i Y_i \neq (\sum X_i)(\sum Y_i)$	another trap
8	$\overline{X+Y} = \bar{X} + \bar{Y}$ but $\overline{XY} \neq \bar{X}\bar{Y}$ (in general)	

× COMMON TRAP

Pitfall. Rule 4 is the single most-tested algebraic identity in JSO. Any "the sum of deviations from the mean is ..." question — answer is **0**. No computation needed.

0.2 The 3 means — when each one is asked

Mean	Formula (raw)	Formula (frequency)	When examiner uses it
Arithmetic (AM)	$\bar{X} = \frac{\sum X_i}{n}$	$\frac{\sum f_i X_i}{\sum f_i}$	Default. Anything with "average".
Geometric (GM)	$(X_1 \cdot X_2 \cdots X_n)^{1/n}$	$\exp(\frac{\sum f_i \log X_i}{\sum f_i})$	Ratios, growth rates, index numbers.
Harmonic (HM)	$\frac{n}{\sum (1/X_i)}$	$\frac{\sum f_i}{\sum f_i / X_i}$	Speeds-over-equal-distances, rates.

Inequality (always true for positive observations): $\mathrm{AM} \geq \mathrm{GM} \geq \mathrm{HM}$ with equality iff all values are equal.

Cross-link: $\mathrm{GM}^2 = \mathrm{AM} \times \mathrm{HM}$ **only for two positive numbers**.

0.3 Logarithm, antilog, exponent micro-refresher

Identity	Use
$\log(ab) = \log a + \log b$	GM, splicing
$\log(a/b) = \log a - \log b$	index deflation
$\log a^n = n \log a$	compounded growth
$e^x \cdot e^y = e^{x+y}$	Poisson, normal

Numerical anchors to memorise:

x	$\log_{10} x$
2	0.3010
3	0.4771
5	0.6990
7	0.8451
π	0.4971
e	0.4343

0.4 Combinatorics — only what you'll need

Symbol	Meaning	Formula
$n!$	factorial	$n \cdot (n-1) \cdots 1$, and $0! = 1$
${}^nP_r$	permutations	$\frac{n!}{(n-r)!}$
${}^nC_r$	combinations	$\frac{n!}{r!(n-r)!}$

Memorise the Pascal row up to $n = 10$ — it solves binomial probability and sampling questions in 5 seconds.

0.5 Greek-letter cheat-sheet (population vs sample)

Quantity	Population (parameter)	Sample (statistic)
Mean	μ	$\bar{X}$
Variance	σ^2	s^2
Std deviation	σ	s
Proportion	P (or π)	p
Correlation	ρ	r
Size	N	n

Examiner trap. *Whenever a question swaps N and n inside a variance formula, one of the four options will be the wrong-divisor answer. Always read which formula is asked:* ***population variance*** *uses N;* ***sample variance*** *uses $n-1$.*

CHAPTER 1 — Collection, Classification & Presentation of Data

Importance: ⭐⭐⭐⭐ HIGH (≈8 Qs / paper) **Difficulty:** Easy. Don't lose marks here — it's free fuel for your 95% target.

1.0 — Understanding Data Presentation from First Principles

✔ **QUICK RECALL**

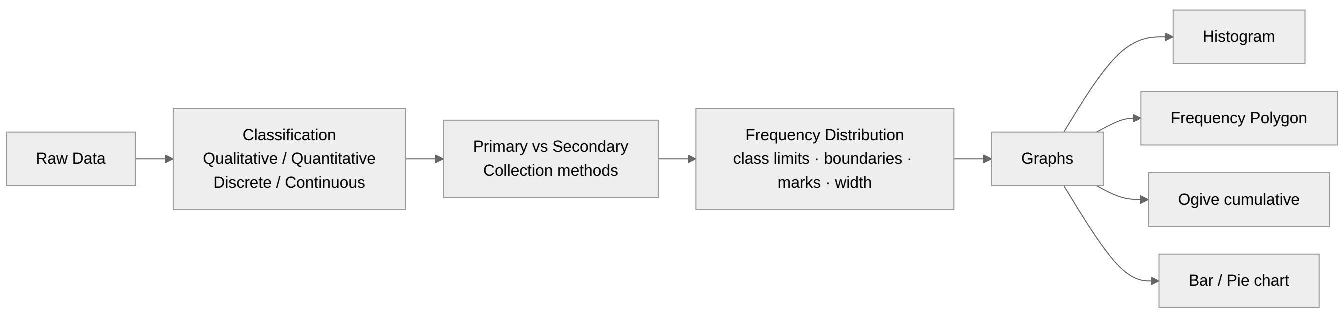
Why does data presentation exist — and why does statistics start here?

Suppose a government officer hands you a printout listing the ages of 10,000 school children, in the order they were recorded. Can you tell — just from staring at that list — what the typical age is? Whether most children fall in a narrow band or are spread across a wide range? Whether there are unusual outliers? No. Raw data in that form is nearly useless for decision-making.

Data presentation is the process of transforming raw data into an organised form that reveals its structure. The moment you group 10,000 ages into bins (5–7, 7–9, 9–11 years) and count how many fall in each bin, a pattern emerges. When you draw those counts as bars, the shape of the distribution becomes visible in seconds. An administrator can now see at a glance that most children are 8–10 years old, that there is a small cluster of over-age enrolments, and that the distribution is slightly skewed right.

The chain of chapters: Every statistical measure you will compute in Chapters 2–12 — mean, variance, correlation coefficient, regression line, test statistic — is computed from an organised frequency distribution, not from raw lists. **This chapter is the entry gate to all of statistics.**

Chapter roadmap — what you will learn:



Why multiple graph types? Each graph reveals a different property of the data:

Graph	What it reveals	When to use
Histogram	Shape of distribution (symmetric, skewed, bimodal)	Grouped continuous data
Frequency Polygon	Shape + easy to overlay two distributions	Same as histogram; comparison
Less-than Ogive	Cumulative %: "what % below value X?"	Reading median, quartiles graphically
More-than Ogive	Cumulative % from above	Crossing point of two ogives = median
Bar chart	Comparison between discrete categories	Discrete or categorical data
Pie chart	Part-of-whole (each slice = % of total)	Proportions add to 100%
Frequency Curve	Smooth theoretical shape	Large datasets; theoretical models

✔ KEY FACT

The histogram trap — unequal class widths. When class widths are different, you CANNOT draw bar heights proportional to frequency. You must use **frequency density** = $\text{frequency} \div \text{class width}$ on the Y-axis. Otherwise wider classes look artificially more frequent. This trap appears in at least one JSO question every two papers.

Concept: Inclusive vs Exclusive class intervals

Form	Example	Contains	How to convert
Exclusive	10–20, 20–30	$10 \le X < 20$	Ready to use directly
Inclusive	10–19, 20–29	$10 \le X \le 19$	Subtract 0.5 from lower, add 0.5 to upper boundary

So the class 10–19 (inclusive) has boundaries 9.5–19.5, class mark = $(9.5 + 19.5)/2 = 14.5$.

Solved Example 1.A: The heights (cm) of 30 students are recorded. Classes used: 150–155, 155–160, 160–165, 165–170, 170–175 with frequencies 3, 7, 12, 6, 2. Find: (i) class mark of 160–165, (ii) class boundaries if data were in inclusive form 160–164.

(i) Class mark = $(160 + 165)/2 = \mathbf{162.5}$ *cm.*

(ii) Boundaries: lower = $160 - 0.5 = 159.5$*; upper =* $164 + 0.5 = 164.5$*. Class width = 5.*

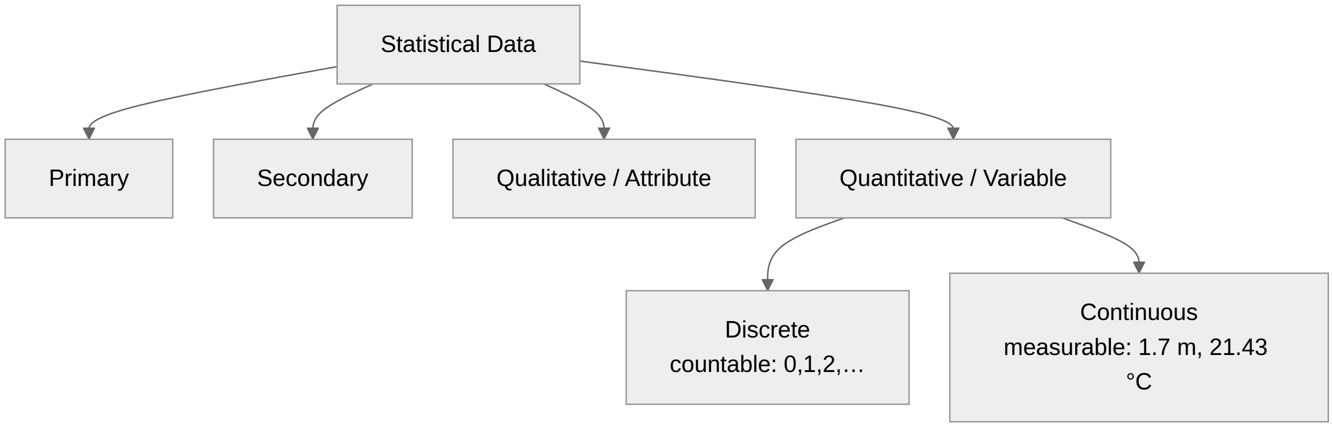
Solved Example 1.B: From the ogive below (less-than type), find the median graphically.

Locate $N/2$ *on the Y-axis (cumulative frequency). Draw a horizontal line to the ogive curve, then drop vertically to the X-axis. The value on the X-axis is the median. (In an exam, this is always a reading exercise — examiners give you a scale and expect you to read off the value at N/2.)*

1.1 Examiner mindset

Angle examiner uses	What you must know
Definition / type of data	Primary vs Secondary; Qualitative vs Quantitative; Discrete vs Continuous
Methods of primary data collection	Direct, indirect, mailed questionnaire, schedule, observation
Frequency-distribution mechanics	Class limits, class boundaries, class mark, width, inclusive vs exclusive
Diagrammatic & graphical presentation	Histogram, freq polygon, ogive, pie, bar (simple/multiple/sub-divided)
Which graph is best when	Histogram for grouped data; pie for parts-of-whole; bar for comparison; ogive to read median/quartiles

1.2 Data classification — the family tree



1.3 Primary vs Secondary — fact card

Aspect	Primary	Secondary
Origin	Collected first-hand by the investigator	Collected by someone else, used as-is
Cost	High	Low
Time	High	Low
Reliability	Higher	Depends on source
Examples	Census enumeration, your own field survey	RBI Bulletin, Economic Survey, NSS reports

1.4 Methods of primary data collection

Method	Used when	Pro	Con
Direct personal interview	Small area, sensitive issue	High accuracy	Costly
Indirect oral investigation	Witnesses, third parties	Good for crime/illness	Bias risk
Local correspondents	Routine area-wise reporting	Low cost	Non-uniform
Mailed questionnaire	Wide, literate population	Cheap, large scale	Low response
Schedule sent through enumerator	Mixed-literacy population	Higher response than mail	Costlier than mail
Observation method	Behaviour studies	Objective	Time-intensive

1.5 Frequency-distribution vocabulary

Suppose a class is written **20–30**.

Term	Meaning	Value here
Lower limit (L)	Smallest value the class includes	20
Upper limit (U)	Largest value the class includes	30
Class width (h)	$U - L$	10
Class mark / mid-value (m)	$(L + U)/2$	25
Class boundaries (continuous form)	$L - 0.5$ to $U + 0.5$ (for inclusive series)	19.5 – 30.5
Frequency (f)	Count of observations in the class	given

Term	Meaning	Value here
Cumulative frequency	Running total of f	computed
Relative frequency	$f/\sum f$	a proportion

Inclusive vs exclusive class:

Form	Looks like	Includes
Exclusive	20 – 30, 30 – 40	20 ≤ X < 30
Inclusive	20 – 29, 30 – 39	20 ≤ X ≤ 29

To convert inclusive → exclusive: subtract 0.5 from each lower limit and add 0.5 to each upper limit.

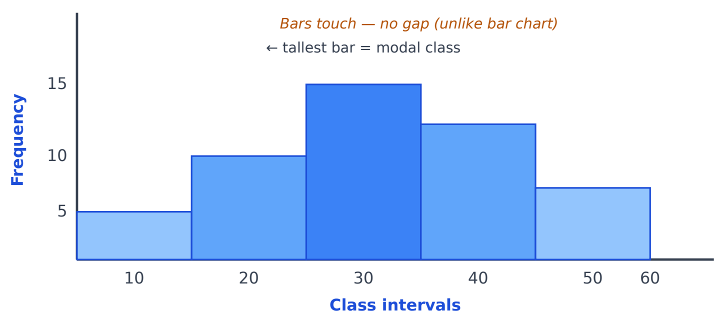
1.6 Graphs — when to use which

Graph	Best use	Key construction note
Histogram	Continuous frequency data	Bars touch; height × width = frequency density when class widths differ
Frequency polygon	Comparing distributions	Join class marks with straight lines; close to baseline at both ends
Frequency curve	Smoothed shape of distribution	Smooth the polygon
Ogive (cumulative)	Median, quartiles, percentiles	"Less-than" ogive rises; "more-than" ogive falls; intersection = median
Pie chart	Composition / parts of whole	Angle = (class freq / total) × 360°
Bar chart (simple/multiple/sub-divided)	Discrete categories, comparison	Bars do **not** touch

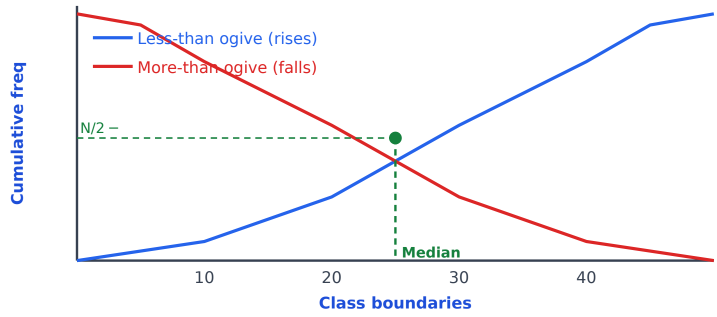
◎ **MNEMONIC**

Mnemonic — "HOPe to Be a STAR".
Histogram = grouped freq.
Ogive = cumulative freq → median.
Pie = parts of whole.
Bar = compare categories.

Histogram — what one looks like



Less-than Ogive and More-than Ogive — how they look



> ✔ **KEY FACT**
>
> **Key ogive fact:** The two ogives ALWAYS intersect at exactly the **median** point (cumulative frequency = N/2). Examiners show you the intersection and ask "what does it represent?" — answer: median.

1.7 Histogram with unequal class widths — the only trap

When class widths differ, you must plot **frequency density** (= f/h) on the y-axis, **not** raw frequency. Otherwise the visual impression lies.

Example. Classes 0–10 (f=20), 10–20 (f=30), 20–40 (f=40). Density = 20/10 = 2, 30/10 = 3, 40/20 = **2** (not 4).

1.8 Worked PYQ-style examples

> ✎ **WORKED EXAMPLE**
>
> **Q 1.1.** Class 30 – 40 has frequency 25. Its class mark is: **(a) 30 (b) 35 (c) 40 (d) 25**
>
> **Step 1 — Apply the class mark formula.**
>
> - Class mark = $(L + U)/2$
> - = $(30 + 40)/2$
> - = $70/2$
> - = **35**
>
> **Ans (b).**
>
> *Trap.* (a) lower limit, (c) upper limit, (d) the frequency itself.

> ✎ **WORKED EXAMPLE**
>
> **Q 1.2.** A pie chart shows education spending as 25 % of a state's outlay. The corresponding angle is: **(a) 25° (b) 90° (c) 25/360 ° (d) 100°**
>
> **Step 1 — Apply the pie-chart angle formula.**
>
> - Angle = share (%) × 360° / 100
> - = 25 × 360° / 100
> - = **90°**
>
> **Ans (b).**

✎ WORKED EXAMPLE

Q 1.3. Convert inclusive class **40–49** to exclusive form.

Soln. Lower boundary = 40 − 0.5 = 39.5. Upper boundary = 49 + 0.5 = **49.5**. Exclusive: **39.5 – 49.5**. Class width = 10.

✎ WORKED EXAMPLE

Q 1.4. Which graph is best for locating the median? **(a) Pie (b) Histogram (c) Ogive (d) Bar**

Soln. From a less-than ogive, drop a perpendicular at $N/2$ cumulative frequency to the x-axis → **median**. **Ans (c).**

✎ WORKED EXAMPLE

Q 1.5. Below is a frequency table.

Class	0–10	10–20	20–30	30–40
f	5	8	12	5

Find cumulative frequency of class 20–30 (less-than form).

Soln. "Less than 30" = 5 + 8 + 12 = **25**.

✎ WORKED EXAMPLE

Q 1.6. A study uses RBI Bulletin to compute inflation. The data is: **(a) Primary (b) Secondary (c) Internal (d) Manual**

Soln. Already published by RBI → secondary. **Ans (b).**

✎ WORKED EXAMPLE

Q 1.7. Which is **NOT** a method of primary data collection? **(a) Direct personal interview (b) Mailed questionnaire (c) NSS published reports (d) Schedule through enumerator**

Soln. NSS published reports = secondary. **Ans (c).**

✎ WORKED EXAMPLE

Q 1.8. A frequency distribution has class widths 5, 5, 10, 10. Frequencies 20, 25, 50, 30. To draw the histogram correctly, the bar heights should be (in same order): **(a) 20, 25, 50, 30 (b) 4, 5, 5, 3 (c) 4, 5, 10, 6 (d) 20, 25, 25, 15**

Step 1 — Compute frequency density = f / h for each class.

- Class 1 (width 5): 20 / 5 = 4
- Class 2 (width 5): 25 / 5 = 5
- Class 3 (width 10): 50 / 10 = 5
- Class 4 (width 10): 30 / 10 = 3

Heights = **4, 5, 5, 3**. **Ans (b).**

✎ WORKED EXAMPLE

Q 1.9. A frequency polygon is constructed by joining: **(a) Lower limits (b) Upper limits (c) Class marks (d) Class widths**

Soln. Class marks (mid-points). **Ans (c).**

✎ WORKED EXAMPLE

Q 1.10. In a "more-than" ogive, the curve generally: **(a) Rises (b) Falls (c) Is flat (d) Is U-shaped**

Soln. Cumulative frequencies decrease as the lower limit increases → **falls. Ans (b).**

1.9 Trap-recognition card

Trap option you'll see	Why wrong	Tell
"Histogram bars don't touch"	False — histogram bars **touch**; only bar charts have gaps.	Don't confuse the two.
"Pie angle = % directly"	Angle = % × 360° / 100, not the % itself.	Always multiply.
"Use raw f for unequal widths"	Must use density.	Question will explicitly mention different widths.
"Mailed questionnaire is secondary"	It's primary (you designed it).	Source-of-data ≠ medium.

1.10 Mini-mock (10 Qs, all solved)

#	Q	Ans
1	Class mark of 50–60 = ?	55
2	Total frequency = 200; sector for "Education" = 72°. % share?	72/360 × 100 = **20%**
3	Class 100–149 inclusive → exclusive lower boundary?	99.5
4	The graph used to show parts of a whole is?	Pie
5	Class width if classes are 0–4, 5–9, 10–14?	5
6	Height of histogram bar when classes have unequal widths?	Frequency density
7	"Less-than" ogive intersected with N/2 horizontal line gives?	Median
8	A schedule is filled by?	The enumerator
9	Census of India data when used in a study is?	Secondary
10	A bar chart's bars are?	Separated (do not touch)

1.11 Active-recall prompts (cover the answers, then test)

1. State the formula for class mark.
2. List four methods of collecting primary data.
3. Write the formula for the angle in a pie chart corresponding to a frequency f when total frequency is N.
4. Why do histogram bars touch but bar-chart bars don't?
5. What is plotted on the y-axis of a histogram with unequal class widths?

CHAPTER 2 — Measures of Central Tendency

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈10 Qs / paper) **Difficulty:** Easy–Medium. This is your foundation for Dispersion, Moments, Skewness — every later chapter assumes you can find a mean in 5 seconds.

2.0 — Understanding Central Tendency from First Principles

✔ **QUICK RECALL**

What is "central tendency" and why do we need more than one type of average?

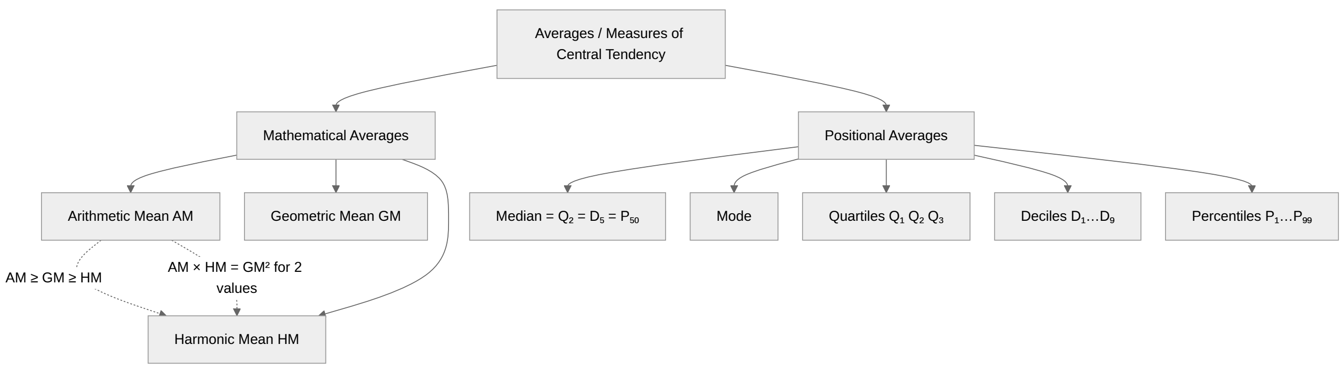
Any dataset can be summarised along two dimensions: *where it is centred* and *how spread out it is*. Central tendency addresses the first: what single number best represents the "middle" or "typical" value of the data?

The **arithmetic mean** is the most natural answer — add everything up and divide by count. But it has a fatal flaw: **it is sensitive to extreme values**. Consider 9 workers earning ₹20,000 each and 1 CEO earning ₹10,000,000. The mean salary is ≈ ₹1,018,000 — a number that describes nobody in the company. The **median** (the middle value when sorted: ₹20,000) is far more representative of what a typical employee earns.

The **mode** (most frequent value) reveals a different truth: in a bimodal distribution with two peaks, the mean and median both land somewhere in the valley between the peaks — describing the least common outcome! The mode correctly identifies both peaks.

Why three averages exist: Different types of phenomena need different summaries. - AM is best when the process is additive (total is meaningful — total marks, total production). - GM is best when the process is multiplicative (compound interest, growth rates, index ratios). - HM is best when the process involves rates and equal denominators (speeds over equal distances, costs per unit with equal budgets).

Concept map — the family of averages:



How AM, Median and Mode relate to distribution shape — a visual you must be able to recall instantly:

```
SYMMETRIC:          Mode = Median = Mean
                         ↑
                    _____|_____
                   /           \
                  /             \
_____________________________________

POSITIVE SKEW       Mo  Md   Mn →
(long right tail):   ↑   ↑    ↑
                  __|___|_____|___
                 /                \___
_____________________________________

NEGATIVE SKEW  ← Mn  Md   Mo
(long left tail):  ↑   ↑    ↑
               ___/____________|___
              /                |
_____________________________________
```

◆ FORMULA / KEY POINTS

Key relationships (memorise all four):

$$\bar{X} = A + \frac{\sum f_i u_i}{\sum f_i} \times h \quad \text{(step-deviation method — fastest for grouped data)}$$

$$\text{Median} = L + \frac{\frac{N}{2} - F}{f} \times h$$

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

$$\boxed{\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}} \quad \text{(empirical relation — used to find the third when two are known)}$$

Solved Example 2.A — Combined mean (very frequent exam question):

A factory has two shifts. Morning shift: 40 workers, average wage ₹600. Evening shift: 60 workers, average wage ₹500. What is the overall average wage?

Step 1 — Apply the combined-mean formula.

- $\bar{X}_{12} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$

Step 2 — Substitute the two shift sizes and averages.

- $= \dfrac{40 \times 600 + 60 \times 500}{40 + 60}$
- $= \dfrac{24000 + 30000}{100}$

Step 3 — Simplify to the final answer.

- $= \dfrac{54000}{100}$
- = **₹540**

Solved Example 2.B — When to use HM (the most common trap):

A car travels 120 km at 60 km/h and returns 120 km at 40 km/h. What is the average speed for the whole journey?

Step 1 — Identify the correct mean. Equal distances at different speeds ⇒ use **HM**, not AM.

Step 2 — Apply the HM formula for two values: HM = 2 / (1/a + 1/b).

- HM = $\frac{2}{\frac{1}{60}+\frac{1}{40}}$

Step 3 — Combine the reciprocals using a common denominator (LCM = 120).

- $\frac{1}{60}+\frac{1}{40}=\frac{2}{120}+\frac{3}{120}=\frac{5}{120}$

Step 4 — Compute the HM.

- HM = $\frac{2}{5/120}$
- = $\frac{2\times 120}{5}$
- = $\frac{240}{5}$
- = **48 km/h**

Why not AM? AM gives $(60+40)/2=50$ km/h. Verify: time = $120/60+120/40=2+3=5$ h; total distance = 240 km; average speed = $240/5=48$ km/h ✓ HM is correct.

Solved Example 2.C — Median for grouped data (step by step):

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	15	7	5

Find the median.

Step 1 — Compute total frequency N and N/2.

- $N=5+8+15+7+5=40$
- $N/2=20$

Step 2 — Build the cumulative-frequency column.

- CF: 5, 13, 28, 35, 40

Step 3 — Locate the median class — first class whose CF ≥ N/2.

- 28 ≥ 20 → **median class = 20–30**

Step 4 — Identify the formula symbols.

- $L=20$ (lower boundary), $F=13$ (CF before median class), $f=15$ (freq of median class), $h=10$

Step 5 — Apply the median formula.

- $\text{Median}=L+\frac{(N/2)-F}{f}\times h$
- = $20+\frac{20-13}{15}\times 10$
- = $20+\frac{70}{15}$
- = $20+4.67$
- = **24.67**

Solved Example 2.D — GM for growth rates:

A company's sales grew by 10% in Year 1 and 40% in Year 2. What is the average annual growth rate?

Step 1 — Convert growth rates to multipliers.

- Year 1: $1 + 10/100 = 1.10$
- Year 2: $1 + 40/100 = 1.40$

Step 2 — Apply the GM formula for two values: GM = √(a × b).

- GM = $\sqrt{1.10 \times 1.40}$
- = $\sqrt{1.54}$
- ≈ **1.2410**

Step 3 — Convert back to a percentage growth rate.

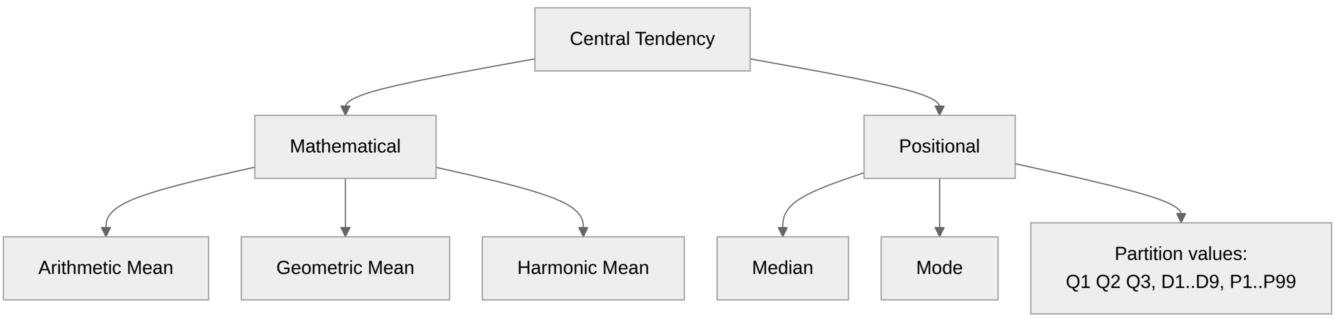
- Average annual growth $= 1.2410 - 1 = 0.2410$
- ≈ **24.1% per year**

Check: ₹100 → ₹110 (Year 1) → ₹154 (Year 2). Two-year growth = 54%. Annual rate $= \sqrt{1.54} - 1 \approx 24.1\%$ ✓

2.1 Examiner mindset

Angle	They ask it as	Marks-share
Pure formula	"Find mean of ... " (raw / discrete / grouped)	3-4 Qs
Properties of AM	"Sum of deviations from mean = ?" → 0; "If each value increased by k, mean = ?" → $\bar{X} + k$	1-2 Qs
Combined / weighted mean	Two groups with sizes n_1, n_2 and means $\bar{X}_1, \bar{X}_2$	1-2 Qs
Median / Mode of grouped data	Direct formula with class containing the median / mode	2-3 Qs
Empirical mode	Mode = 3 Median – 2 Mean	1 Q
Partition values	Quartiles, deciles, percentiles from ogive or formula	1-2 Qs
AM-GM-HM relation	AM ≥ GM ≥ HM; $GM^2 = AM \cdot HM$ (for two values)	1 Q

2.2 Mind map



2.3 Arithmetic Mean — three computational forms

Direct method

Data type	Formula
Raw / ungrouped	$\bar{X} = \dfrac{\sum X_i}{n}$

Data type	Formula
Discrete frequency	$\bar{X} = \dfrac{\sum f_i X_i}{\sum f_i}$
Continuous (grouped)	$\bar{X} = \dfrac{\sum f_i m_i}{\sum f_i}$, m_i = class mark

Assumed-mean (short-cut) method

Pick any A (preferably a middle class mark). Let $d_i = X_i - A$.

$$\bar{X} = A + \frac{\sum f_i d_i}{\sum f_i}$$

Step-deviation method

Pick A, let $u_i = (X_i - A)/h$ where h = common class width.

$$\bar{X} = A + \frac{\sum f_i u_i}{\sum f_i} \times h$$

This is the fastest for grouped data with equal class widths. **Memorise it.**

2.4 Five key properties of AM (very high-yield)

#	Property	One-line proof / use
P1	$\sum(X_i - \bar{X}) = 0$	by definition of mean
P2	$\sum(X_i - \bar{X})^2$ is **minimum** (less than for any other constant)	least-squares property
P3	If $Y_i = X_i + k$ then $\bar{Y} = \bar{X} + k$	shift property
P4	If $Y_i = cX_i$ then $\bar{Y} = c\bar{X}$	scale property
P5	Combined mean of two groups: $\bar{X}_{12} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$	weighted by size

◎ **MNEMONIC**

P1 is asked every alternate paper. If a question says "the algebraic sum of the deviations of values from their arithmetic mean is ...", the answer is **zero**. Move on.

2.5 Geometric and Harmonic means — when each is correct

Setup	Use	Why
Average growth rate over years (e.g., 10%, 20%, 30% in three years)	**GM**	Multiplicative process
Average ratio (price relative, index numbers)	**GM**	Symmetric in ratios
Avg speed when **equal distances** at different speeds	**HM**	Time = distance/speed; HM gives correct total-time average
Avg cost per unit when fixed expenditure spent at different prices	**HM**	Same as above with money
Average of measurements	**AM**	Default

Two-value identity: $\mathrm{AM} \times \mathrm{HM} = \mathrm{GM}^2$ (only for two positive numbers).

2.6 Median — formula for grouped data

$$\text{Median} = L + \frac{(N/2) - F}{f} \times h$$

Symbol	Meaning
L	lower boundary of the **median class**
N	total frequency, $\sum f_i$
F	cumulative frequency **before** the median class
f	frequency of the median class
h	class width

Median class = the class where cumulative frequency first reaches $N/2$.

2.7 Mode — formula for grouped data

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

Symbol	Meaning
L	lower boundary of the **modal class** (the class with maximum freq)
f_1	frequency of modal class
f_0	frequency of class **just before** modal class
f_2	frequency of class **just after** modal class
h	class width

2.8 Empirical relation (memorise both forms)

$$\boxed{\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}} \quad \Longleftrightarrow \quad \text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median})$$

Used when the question gives two of the three and asks the third.

2.9 Partition values

Quantity	What it splits	Formula (grouped data)
Median (Q_2)	bottom 50% / top 50%	$L + \frac{N/2 - F}{f} h$
Quartile Q_k ($k = 1, 2, 3$)	quarters	replace $N/2$ with $kN/4$
Decile D_k ($k = 1 \dots 9$)	tenths	replace with $kN/10$
Percentile P_k ($k = 1 \dots 99$)	hundredths	replace with $kN/100$

So $Q_2 = D_5 = P_{50}$ = Median.

2.10 Which average for which shape?

Distribution shape	Mean–Median–Mode order
Symmetric	Mean = Median = Mode
Positively skewed (long right tail)	**Mode < Median < Mean**

Distribution shape	Mean–Median–Mode order
Negatively skewed (long left tail)	**Mean < Median < Mode**

Visual:

```
positive skew (right-tailed):    Mo  Me  Mn  →
negative skew (left-tailed):     ←  Mn  Me  Mo
```

2.11 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 2.1. Mean of 5, 7, 9, 11, 13.

Step 1 — Apply the AM formula: $\bar{X} = \sum X_i/n$.

- $\bar{X} = (5 + 7 + 9 + 11 + 13)/5$

Step 2 — Add the values and divide.

- $= 45/5$
- **= 9**

✎ WORKED EXAMPLE

Q 2.2. A class has 30 students with mean weight 50 kg. Another has 20 with mean 60 kg. Combined mean?

Step 1 — Apply the combined-mean (weighted) formula.

- $\bar{X}_{12} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$

Step 2 — Substitute the two group sizes and means.

- $= \dfrac{30 \cdot 50 + 20 \cdot 60}{30 + 20}$
- $= \dfrac{1500 + 1200}{50}$

Step 3 — Simplify.

- $= \dfrac{2700}{50}$
- **= 54 kg**

✎ WORKED EXAMPLE

Q 2.3. The mean of 10 values is 25. If each value is multiplied by 4 and then 6 is subtracted, the new mean is:

Step 1 — Apply scale property P4: $\overline{cX} = c\bar{X}$.

- New mean after × 4 = $4 \cdot 25$
- **= 100**

Step 2 — Apply shift property P3: $\overline{X - k} = \bar{X} - k$.

- New mean after − 6 = $100 - 6$
- **= 94**

✎ WORKED EXAMPLE

Q 2.4. A man travels 60 km at 30 km/h and another 60 km at 60 km/h. Average speed?

Step 1 — Identify the correct mean. Equal distances at different speeds ⇒ use **HM**, not AM.

Step 2 — Apply HM formula for two values: HM = 2 / (1/a + 1/b).

- HM = $\dfrac{2}{(1/30) + (1/60)}$

Step 3 — Combine the reciprocals (LCM = 60).

- $\dfrac{1}{30} + \dfrac{1}{60} = \dfrac{2}{60} + \dfrac{1}{60} = \dfrac{3}{60}$

Step 4 — Compute the HM.

- HM = $\dfrac{2}{3/60} = \dfrac{2 \times 60}{3} = \dfrac{120}{3}$
- = **40 km/h**

Trap. AM gives 45 — wrong because the two trips take different times.

✎ WORKED EXAMPLE

Q 2.5. GM of 4 and 16.

Step 1 — Apply the two-value GM formula: GM = √(a · b).

- GM = $\sqrt{4 \cdot 16}$
- = $\sqrt{64}$
- = **8**

✎ WORKED EXAMPLE

Q 2.6. For the table below, find the median.

Class	0–10	10–20	20–30	30–40	40–50
f	5	8	15	7	5

Step 1 — Compute N and N/2.

- $N = 5 + 8 + 15 + 7 + 5 = 40$
- $N/2 = 20$

Step 2 — Build the cumulative-frequency column.

- CF: 5, 13, 28, 35, 40

Step 3 — Locate the median class — first CF that reaches or crosses N/2 = 20.

- 28 ≥ 20 → **median class = 20–30**

Step 4 — Identify the formula symbols.

- $L = 20, F = 13, f = 15, h = 10$

Step 5 — Apply the median formula.

- $\text{Median} = L + \dfrac{(N/2) - F}{f} \times h$
- $= 20 + \dfrac{20 - 13}{15} \times 10$
- $= 20 + \dfrac{70}{15}$
- $= 20 + 4.67$
- **= 24.67**

✎ WORKED EXAMPLE

Q 2.7. Same table, find mode.

Step 1 — Locate the modal class (class with the highest frequency).

- Highest $f = 15$ → **modal class = 20–30**

Step 2 — Identify the formula symbols.

- $L = 20, f_1 = 15, f_0 = 8, f_2 = 7, h = 10$

Step 3 — Apply the mode formula.

- $\text{Mode} = L + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$
- $= 20 + \dfrac{15 - 8}{2 \cdot 15 - 8 - 7} \times 10$
- $= 20 + \dfrac{7}{15} \times 10$
- $= 20 + 4.67$
- **= 24.67**

Coincidence here; usually Mode and Median differ.

✎ WORKED EXAMPLE

Q 2.8. Mean = 36, Median = 32. Mode by empirical relation?

Step 1 — Apply the empirical relation: Mode = 3 Median − 2 Mean.

- Mode = $3 \times 32 - 2 \times 36$

Step 2 — Simplify.

- = $96 - 72$
- = **24**

✎ WORKED EXAMPLE

Q 2.9. AM and HM of two positive numbers are 9 and 4. Their GM?

Step 1 — Apply the AM–GM–HM relation: GM² = AM × HM.

- GM² = $9 \times 4 = 36$
- GM = $\sqrt{36} = \mathbf{6}$

✎ WORKED EXAMPLE

Q 2.10. The mean of 50 observations was 30. Later it was discovered that an item 80 had been wrongly read as 40. Correct mean?

Step 1 — Find the original (incorrect) sum.

- Original sum = $n \times \bar{X} = 50 \times 30$
- = **1500**

Step 2 — Adjust the sum: add the correct value, remove the wrong one.

- Correct sum = $1500 + 80 - 40$
- = **1540**

Step 3 — Compute the correct mean.

- Correct mean = $1540/50$
- = **30.8**

✎ WORKED EXAMPLE

Q 2.11. For a positively skewed distribution, which is largest?

Soln. Order in positive skew: Mode < Median < Mean. **Mean** is largest. **Ans: Mean.**

✎ WORKED EXAMPLE

Q 2.12. D_5 of a distribution = ?

Soln. D_5 splits the bottom 5/10 from the top 5/10, which is exactly the median. **Ans: Median ($Q_2 = D_5 = P_{50}$).**

✎ WORKED EXAMPLE

Q 2.13. Sum of deviations of 10 observations from 17 is 30; from 20 is 0. The mean is:

Step 1 — Recall property P1: $\sum(X_i - c) = 0 \iff c = \bar{X}$.

- Sum of deviations from 20 = 0
- So the mean equals the constant that zeroes the deviation-sum
- **Mean = 20**

✎ WORKED EXAMPLE

Q 2.14. Which measure is **most affected by extreme values**?

Soln. The mean uses every value linearly, so a single outlier shifts it noticeably. Median and mode are positional and largely unaffected. **Ans: Mean.**

✎ WORKED EXAMPLE

Q 2.15. GM of 2, 4, 8.

Step 1 — Apply the n-value GM formula: GM = $(X_1 \cdot X_2 \cdots X_n)$^(1/n) with n = 3.

- GM = $(2 \cdot 4 \cdot 8)^{1/3}$
- = $64^{1/3}$

Step 2 — Compute the cube root.

- 4^3 = 64, so $64^{1/3}$ = **4**

2.12 Trap-recognition card

Trap	Wrong answer it produces	Defence
Equal distances, but candidate uses AM	Wrong faster-by-1 number	If "equal distance, different speeds" → HM.
Step-deviation forgetting × h	Off by factor 10 (or whatever h is)	Always end with "× h".
Median formula uses F of median class itself	Number is too low	F = CF **before** the median class.
Mode formula sign flip	Negative answer	Numerator: $f_1 - f_0$; denominator: $2f_1 - f_0 - f_2$.
Empirical relation memorised wrong	Mode = 2 Med − 3 Mean	Correct: Mode = 3 Med − 2 Mean.

2.13 Mini-mock (10 Qs, fully solved)

#	Question	Answer
1	Mean of 12, 18, 24, 30.	21
2	Combined mean: 40 boys (avg 60) + 60 girls (avg 50).	(40·60+60·50)/100 = 54
3	If mean of X is 50, mean of $Y = 2X - 5$?	95
4	Sum of deviations from mean = ?	0
5	AM = 12, GM = 6. HM?	GM² = AM·HM → HM = 36/12 = 3
6	A car covers 100 km @ 50 km/h, returns @ 25 km/h. Avg speed?	HM: 2/(1/50+1/25) = 2/(3/50) = 100/3 ≈ 33.33 km/h

#	Question	Answer
7	Mode = 30, Mean = 25. Median?	Mode = 3 Med − 2 Mean → 30 = 3 Med − 50 → Med = 80/3 ≈ 26.67
8	If each obs increased by 5, mean increases by?	5
9	If each obs multiplied by 3, mean multiplied by?	3
10	Median is the same as which decile?	D_5

2.14 Active-recall prompts

1. Write the step-deviation formula for AM (with all symbols).
2. State all 5 properties of AM.
3. When does a problem demand HM, not AM?
4. Write the median-from-grouped-data formula and label every symbol.
5. State the order of Mean, Median, Mode for a negatively skewed distribution.
6. State the empirical relation between Mean, Median, Mode.

CHAPTER 3 — Measures of Dispersion

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈10 Qs / paper) **Difficulty:** Easy–Medium. Pure formula. Once you nail the variance computational form, this entire chapter takes <5 min per question.

3.0 — Understanding Dispersion from First Principles

✔ QUICK RECALL

Why isn't the mean enough? The essential need for a measure of spread.

Two cricket batsmen both average 50 runs per innings. Batsman A scores: 48, 51, 49, 52, 50 (very consistent). Batsman B scores: 10, 90, 5, 95, 50 (wildly erratic). If you need a reliable performer for the final of a tournament, you'd choose Batsman A — even though both have the same mean. **Dispersion** captures this difference in consistency.

Dispersion measures how far the data points are scattered around the centre. Small dispersion = data bunched near the mean (consistent, predictable). Large dispersion = data spread far (variable, risky).

Why variance uses squared deviations: The obvious measure of spread would be the average deviation from the mean: $\frac{1}{n}\sum(X_i - \bar{X})$. But this always equals **zero** (Property P1 of the mean — positive and negative deviations cancel). The fix: square each deviation before averaging. This makes all terms non-negative, giving a meaningful measure. The standard deviation (SD) is then the square root of variance — bringing us back to the original units.

Why CV is needed: Is an SD of 5 large or small? Compared to a mean of 10, it's enormous (50% variation). Compared to a mean of 1000, it's negligible (0.5% variation). The **Coefficient of Variation** $= \frac{\text{SD}}{\text{Mean}} \times 100$ is unit-free and allows comparison across datasets with different scales or units.

Five dispersion measures — when each is appropriate:

Measure	Formula	Advantage	Limitation
Range	$X_{max} - X_{min}$	Simplest; easiest	Ignores all middle values; very sensitive to outliers
Quartile Deviation (QD)	$\frac{Q_3 - Q_1}{2}$	Not affected by extremes; works for open-ended classes	Ignores top and bottom 50%
Mean Deviation (MD)	$\frac{\sum f\lvert X_i - A\rvert}{N}$ (A = mean or median)	Uses all values; intuitive	Not algebraically tractable; ignores signs
Variance (σ^2)	$\frac{\sum f(X_i - \bar{X})^2}{N}$	Most powerful; algebraically tractable	Squared units; sensitive to outliers
Standard Deviation (σ)	$\sqrt{\text{Variance}}$	Same units as data	Sensitive to outliers
CV	$\frac{\sigma}{\bar{X}} \times 100$	Unit-free; enables cross-dataset comparison	Meaningless when mean ≈ 0

◆ FORMULA / KEY POINTS

The computational formula for variance (faster than the definitional formula):

$$\sigma^2 = \frac{\sum f_i X_i^2}{N} - \left(\frac{\sum f_i X_i}{N}\right)^2 = \overline{X^2} - (\bar{X})^2$$

This is always faster to compute than $\frac{\sum f_i (X_i - \bar{X})^2}{N}$ because you don't need $\bar{X}$ first.

Step-deviation form (for grouped data with equal class widths h):

$$\sigma^2 = h^2 \left[\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2\right], \quad u_i = \frac{X_i - A}{h}$$

Solved Example 3.A — Variance and SD from raw data:

Find the variance and SD of: 4, 7, 13, 2, 6, 8, 10.

Step 1 — Compute N, $\sum X$ and $\bar{X}$.

- $N = 7$
- $\sum X = 4 + 7 + 13 + 2 + 6 + 8 + 10 = 50$
- $\bar{X} = 50/7 \approx 7.14$

Step 2 — Compute $\sum X^2$.

- $\sum X^2 = 16 + 49 + 169 + 4 + 36 + 64 + 100$
- = **438**

Step 3 — Apply the computational formula: $\sigma^2 = \sum X^2/N - \bar{X}^2$.

- $\sigma^2 = \frac{438}{7} - \left(\frac{50}{7}\right)^2$
- $= 62.57 - 51.02$
- = **11.55**

Step 4 — Take the square root for SD.

- $\sigma = \sqrt{11.55}$
- ≈ **3.40**

Solved Example 3.B — Combined SD (the hard formula — appears 1-2 times per paper):

Group 1: $n_1 = 50$, $\bar{X}_1 = 113$, $\sigma_1 = 6$. Group 2: $n_2 = 100$, $\bar{X}_2 = 100$, $\sigma_2 = 7$. Find the combined SD.

Step 1 — Compute the combined mean.

- $\bar{X}_{12} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$
- $= \dfrac{50 \times 113 + 100 \times 100}{150}$
- $= \dfrac{5650 + 10000}{150}$
- $= \dfrac{15650}{150}$
- = **104.33**

Step 2 — Compute the deviations of each group mean from the combined mean.

- $d_1 = \bar{X}_1 - \bar{X}_{12} = 113 - 104.33 = 8.67$
- $d_2 = \bar{X}_2 - \bar{X}_{12} = 100 - 104.33 = -4.33$

Step 3 — Apply the combined-variance formula.

- $\sigma_{12}^2 = \dfrac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$
- $= \dfrac{50(36 + 75.17) + 100(49 + 18.75)}{150}$
- $= \dfrac{50 \times 111.17 + 100 \times 67.75}{150}$
- $= \dfrac{5558.5 + 6775}{150}$
- $= \dfrac{12333.5}{150}$
- = **82.22**

Step 4 — Take the square root for combined SD.

- $\sigma_{12} = \sqrt{82.22}$
- ≈ **9.07**

Solved Example 3.C — Coefficient of Variation for comparison:

Company A: mean profit ₹60,000, SD ₹15,000. Company B: mean profit ₹80,000, SD ₹18,000. Which company is more consistent?

Step 1 — Compute CV for Company A.

- $\mathrm{CV}_A = \dfrac{\sigma_A}{\bar{X}_A} \times 100$
- $= \dfrac{15000}{60000} \times 100$
- = **25 %**

Step 2 — Compute CV for Company B.

- $\mathrm{CV}_B = \dfrac{\sigma_B}{\bar{X}_B} \times 100$
- $= \dfrac{18000}{80000} \times 100$
- = **22.5 %**

Step 3 — Compare: smaller CV ⇒ more consistent.

- 22.5 % < 25 % → **Company B is more consistent**, despite its larger absolute SD. CV strips out the scale.

3.1 Examiner mindset

Angle	Their pet question
Range, QD, MD, SD basics	Direct formula
Variance — computational form	Given $\sum X, \sum X^2$, find σ^2
Effect of change of origin / scale on SD	Origin doesn't change SD; scale scales it
Combined SD (two groups)	One formula they expect verbatim
Coefficient of variation (CV)	"Which series is more consistent / less variable?" → smaller CV
Lorenz curve	Concept-only Q on inequality measurement

3.2 Five measures, one table

Measure	Formula (raw data, n obs)	Frequency / grouped
Range	$\max - \min$	upper limit of last class – lower limit of first class
Quartile deviation (Semi-IQR)	$(Q_3 - Q_1)/2$	same
Mean deviation about $\bar{X}$	$\frac{\sum \lvert X_i - \bar{X}\rvert}{n}$	$\frac{\sum f_i \lvert X_i - \bar{X}\rvert}{N}$
Variance σ^2	$\frac{\sum (X_i - \bar{X})^2}{n}$	$\frac{\sum f_i (X_i - \bar{X})^2}{N}$
Std deviation σ	$\sqrt{\sigma^2}$	same

3.3 Computational form of variance — memorise this

$$\boxed{\sigma^2 = \frac{\sum X^2}{n} - \bar{X}^2 = \frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2}$$

For frequency:

$$\sigma^2 = \frac{\sum fX^2}{N} - \left(\frac{\sum fX}{N}\right)^2$$

Step-deviation (when class widths equal h, $u = (X - A)/h$):

$$\sigma^2 = h^2 \left[\frac{\sum fu^2}{N} - \left(\frac{\sum fu}{N}\right)^2\right]$$

This is the single most-used formula in the entire JSO paper — it appears directly or indirectly in **at least 6 questions per paper**.

3.4 Properties of variance / SD (very high-yield)

Property	Result	Memory hook
Origin-invariant: if $Y = X + k$	$\sigma_Y = \sigma_X$, $\sigma_Y^2 = \sigma_X^2$	shifting all values doesn't change spread
Scale-multiplied: if $Y = cX$	$\sigma_Y = \lvert c\rvert\sigma_X$, $\sigma_Y^2 = c^2\sigma_X^2$	scale × |c|
Linear: if $Y = aX + b$	$\sigma_Y = \lvert a\rvert\sigma_X$	only *a* matters
Variance of constant	0	no spread
Sum of independent X, Y	$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$	only when independent

Property	Result	Memory hook
Difference of independent X, Y	$\sigma^2_{X-Y} = \sigma^2_X + \sigma^2_Y$	yes, **plus** — variances always add

× COMMON TRAP

Trap. Var(X − Y) = Var(X) + Var(Y) when independent. Many candidates write minus. Variance is non-negative, so subtraction would risk negatives — that's the giveaway.

3.5 Combined Standard Deviation

Two groups, sizes n_1, n_2, means $\bar{X}_1, \bar{X}_2$, SDs σ_1, σ_2. Let $\bar{X}_{12}$ be the combined mean. Then:

$$\boxed{\sigma^2_{12} = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

where $d_1 = \bar{X}_1 - \bar{X}_{12}$, $d_2 = \bar{X}_2 - \bar{X}_{12}$.

Memory hook: each group contributes its own variance **plus** its squared deviation from the overall mean, weighted by size.

3.6 Coefficient of Variation (CV)

$$\mathrm{CV} = \frac{\sigma}{\bar{X}} \times 100\%$$

Use	Why
Compare variability of two series with **different units** or different means	SD alone can mislead — a SD of 5 on a mean of 10 is huge; SD of 5 on mean of 1000 is tiny.
Smaller CV → more consistent / more uniform / less risky	Lower spread relative to size.

Examiners ALWAYS phrase this as "which is more consistent / homogeneous / stable / uniform" — that's your trigger word for CV.

3.7 Coefficients of dispersion (relative measures)

Measure	Coefficient
Range	$(X_{\max} - X_{\min})/(X_{\max} + X_{\min})$
Quartile deviation	$(Q_3 - Q_1)/(Q_3 + Q_1)$
Mean deviation	MD/Mean (or median, depending on which MD is taken)
SD (CV)	$\sigma/\bar{X} \times 100\%$

3.8 Empirical relation between MD and SD (for normal-ish data)

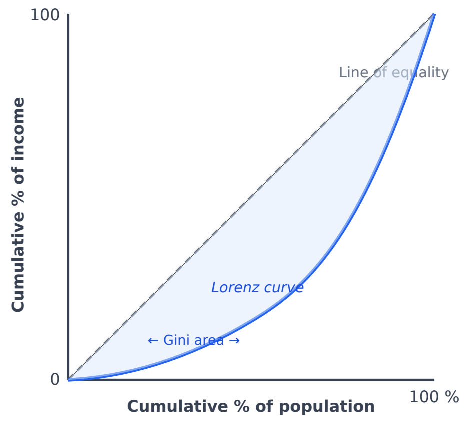
$$\mathrm{QD} : \mathrm{MD} : \mathrm{SD} \;\approx\; \frac{2}{3}\sigma : \frac{4}{5}\sigma : \sigma \;\approx\; 10 : 12 : 15$$

i.e., $\mathrm{QD} \approx 0.6745\sigma$, $\mathrm{MD} \approx 0.7979\sigma$. Asked as a one-mark fact question.

3.9 Lorenz curve

What	Used for	Reading
Plot of cumulative % of population (x) vs cumulative % of variable (y)	Income / wealth inequality	Closer to 45° "line of equality" → less inequality. Bigger bow → more inequality.

The **Gini coefficient** = area between the Lorenz curve and the line of equality / total area below the line. 0 = perfect equality, 1 = perfect inequality.



✔ KEY FACT

3 things examiners ask about the Lorenz curve: - Perfect equality → Lorenz curve **coincides with** the line of equality - Perfect inequality → Lorenz curve **follows the axes** (right angle at bottom-right) - Gini = 0 means **equal** distribution; Gini = 1 means **completely unequal**

3.10 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 3.1. Find variance of 4, 6, 8, 10, 12.

Step 1 — Compute the mean.

- $\bar{X} = (4 + 6 + 8 + 10 + 12)/5$
- $= 40/5$
- = **8**

Step 2 — Compute squared deviations $(X_i - \bar{X})^2$**.**

- Deviations: $-4, -2, 0, +2, +4$
- Squared: $16, 4, 0, 4, 16$
- Sum of squared deviations = **40**

Step 3 — Compute the variance.

- $\sigma^2 = \sum(X_i - \bar{X})^2/n = 40/5$
- = **8**

Step 4 — Compute the standard deviation.

- $\sigma = \sqrt{8} = 2\sqrt{2} \approx \mathbf{2.83}$

✎ WORKED EXAMPLE

Q 3.2. $\sum X = 50, \sum X^2 = 510$ for $n = 10$. Find SD.

Step 1 — Compute the mean.

- $\bar{X} = \sum X/n = 50/10$
- **= 5**

Step 2 — Apply the computational form of variance.

- $\sigma^2 = \dfrac{\sum X^2}{n} - \bar{X}^2$
- $= 510/10 - 5^2$
- $= 51 - 25$
- **= 26**

Step 3 — Compute the standard deviation.

- $\sigma = \sqrt{26} \approx \mathbf{5.10}$

✎ WORKED EXAMPLE

Q 3.3. SD of X is 6. SD of $Y = 3X + 7$?

Step 1 — Apply scaling property: SD$(aX + b) = |a|\cdot$ SD(X). The constant +7 (shift) contributes nothing.

- SD$(Y) = |3| \cdot 6$
- **= 18**

✎ WORKED EXAMPLE

Q 3.4. Two series: A has mean 50, SD 10; B has mean 80, SD 12. More consistent?

Step 1 — Compute the Coefficient of Variation (CV) for each series.

- $\mathrm{CV}_A = \sigma_A/\bar{X}_A \times 100 = 10/50 \times 100$
- **= 20 %**
- $\mathrm{CV}_B = \sigma_B/\bar{X}_B \times 100 = 12/80 \times 100$
- **= 15 %**

Step 2 — Compare: smaller CV means more consistent.

- CV of B (15 %) < CV of A (20 %) → **Series B is more consistent.**

✎ WORKED EXAMPLE

Q 3.5. Variance of first n natural numbers is:

Step 1 — Recall the standard result.

- $\sigma^2 = \dfrac{n^2 - 1}{12}$ — **memorise.**

Proof sketch. $\sum k = n(n+1)/2$, $\sum k^2 = n(n+1)(2n+1)/6$; substitute into $\sigma^2 = \sum k^2/n - (\sum k/n)^2$ and simplify.

✎ WORKED EXAMPLE

Q 3.6. $Q_1 = 25, Q_3 = 45$. Quartile deviation? Coefficient?

Step 1 — Compute the Quartile Deviation (QD).

- $\text{QD} = (Q_3 - Q_1)/2$
- = $(45 - 25)/2$
- = $20/2$
- = **10**

Step 2 — Compute the Coefficient of Quartile Deviation.

- Coefficient = $(Q_3 - Q_1)/(Q_3 + Q_1)$
- = $(45 - 25)/(45 + 25)$
- = $20/70$
- = **2/7 ≈ 0.286**

✎ WORKED EXAMPLE

Q 3.7. Var(X) = 9, Var(Y) = 16, X and Y independent. Var(X − Y)?

Step 1 — Apply the independent-sum rule: Var(aX + bY) = a²Var(X) + b²Var(Y). With a = 1, b = −1, both squares are 1.

- Var(X − Y) = $1 \cdot 9 + 1 \cdot 16$
- = **25** (variances add even on subtraction)

✎ WORKED EXAMPLE

Q 3.8. A group of 50 has mean 60, SD 5. A group of 30 has mean 60, SD 6. Combined SD?

Step 1 — Note that the two group means are equal.

- $\bar{X}_1 = \bar{X}_2 = 60 \to d_1 = d_2 = 0$

Step 2 — Apply the combined-variance formula (with d-terms zero).

- $\sigma_{12}^2 = \dfrac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$
- = $\dfrac{50 \times 25 + 30 \times 36}{80}$
- = $\dfrac{1250 + 1080}{80}$
- = $\dfrac{2330}{80}$
- = **29.125**

Step 3 — Take the square root for combined SD.

- SD = $\sqrt{29.125}$
- ≈ **5.40**

✎ WORKED EXAMPLE

Q 3.9. A group has mean 50 SD 4 (n = 60); another mean 55 SD 5 (n = 40). Combined SD?

Step 1 — Compute the combined mean.

- $\bar{X}_{12} = (60 \cdot 50 + 40 \cdot 55)/100$
- $= (3000 + 2200)/100$
- = **52**

Step 2 — Compute the deviations of each group mean from the combined mean.

- $d_1 = 50 - 52 = -2$
- $d_2 = 55 - 52 = 3$

Step 3 — Apply the combined-variance formula.

- $\sigma_{12}^2 = \dfrac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$
- $= \dfrac{60(16 + 4) + 40(25 + 9)}{100}$
- $= \dfrac{60 \cdot 20 + 40 \cdot 34}{100}$
- $= \dfrac{1200 + 1360}{100}$
- = **25.6**

Step 4 — Take the square root for combined SD.

- SD = $\sqrt{25.6}$
- ≈ **5.06**

✎ WORKED EXAMPLE

Q 3.10. For a normal-ish distribution, MD/SD ≈ ?

Soln. Empirical ratio for normal data: MD ≈ 0.7979 σ ≈ **4/5 σ**. So MD/SD ≈ **4/5** (memorise).

✎ WORKED EXAMPLE

Q 3.11. Range of the series 12, 18, 25, 7, 30, 14.

Step 1 — Identify max and min.

- max = 30, min = 7

Step 2 — Apply Range = max − min.

- Range = 30 − 7
- = **23**

✎ WORKED EXAMPLE

Q 3.12. SD of 10 observations is 4. If each is increased by 5, SD becomes:

Soln. SD is **origin-invariant** — adding a constant to every observation shifts the mean but leaves spread unchanged. SD remains **4**.

✎ WORKED EXAMPLE

Q 3.13. Variance of $2X + 3Y$ where Var(X) = 4, Var(Y) = 9, X and Y independent?

Step 1 — Apply Var(aX + bY) = a²Var(X) + b²Var(Y) (independent X, Y).

- Var$(2X + 3Y) = 2^2 \cdot 4 + 3^2 \cdot 9$
- = $16 + 81$
- = **97**

✎ WORKED EXAMPLE

Q 3.14. The CV is used to compare:

Soln. CV is unit-free, so it lets you compare **variability across series with different means or different units**.

✎ WORKED EXAMPLE

Q 3.15. A perfectly equal distribution gives a Lorenz curve that:

Soln. Coincides with the **45° line of equality** (Gini coefficient = 0).

3.11 Trap-recognition card

Trap	Why students fall	Fix
"SD changes when origin changes"	They confuse with mean	SD is **origin-invariant**.
"Var(X − Y) = Var(X) − Var(Y)"	Algebraic instinct	Variances **add** even on subtraction (independent case).
Forgot the h^2 in step-deviation	They put just h	Variance scales with **square** of factor.
Compared raw SDs to claim consistency	When means differ	Use **CV**, not SD.

3.12 Mini-mock

#	Q	Ans
1	Var(2X+5)? Var(X) = 9.	36
2	SD of constant 7 = ?	0
3	Combined mean of n_1=20 ($\bar{X}$=15) and n_2=30 ($\bar{X}$=25)?	21
4	Series with smaller CV is …?	More consistent
5	Variance of first 10 natural numbers?	(100−1)/12 = 99/12 = 8.25
6	If each obs is doubled, SD becomes?	doubled
7	Sum of squared deviations from mean is …?	minimum
8	QD ≈ ? × σ	2/3 σ (≈ 0.6745 σ)

#	Q	Ans
9	Var(X) = 4, Var(Y) = 5, X & Y independent. Var(X+Y)?	9
10	Coefficient of QD formula?	$(Q_3-Q_1)/(Q_3+Q_1)$

3.13 Active-recall prompts

1. Write the computational form of variance. Why is it preferred?
2. Effect on SD when (a) origin changes, (b) scale changes by factor c.
3. Write the combined SD formula in full.
4. State the empirical ratio QD : MD : SD.
5. Why use CV instead of SD when comparing two series?

CHAPTER 4 — Moments, Skewness & Kurtosis

Importance: ⭐⭐⭐⭐ HIGH (≈7 Qs / paper) **Difficulty:** Medium. Definitions are mechanical; the trap is in **central vs raw** moments and the **β_1, β_2 vs γ_1, γ_2** distinction.

4.0 — Understanding Moments, Skewness & Kurtosis

✔ QUICK RECALL

Beyond mean and SD: describing the shape of a distribution.

The mean tells you *where* a distribution is centred. The standard deviation tells you *how wide* it is. But two distributions can have identical means and SDs and still look completely different. Moments, skewness, and kurtosis capture this shape information mathematically.

Moments are a systematic way to compute higher-order summaries. The r-th central moment is $\mu_r = \frac{\sum f_i (X_i - \bar{X})^r}{N}$: - $\mu_1 = 0$ always (by definition of mean). - $\mu_2 = \sigma^2$ — the variance. Captures width/spread. - μ_3 — if positive, the right tail is heavier (positive skew); if negative, left tail is heavier. Captures **lopsidedness**. - μ_4 — captures **peakedness**: how concentrated values are near the mean vs how fat the tails are.

Skewness — why it matters: Salary distributions at companies are positively skewed — most employees earn moderate amounts, a handful of executives earn millions, pulling the mean far above what most people earn. The key diagnostic: *in a positively skewed distribution, Mode < Median < Mean* — the mean is dragged toward the long tail.

Kurtosis — why it matters: Two investment funds have the same mean return and the same SD. Fund A is leptokurtic ($\beta_2 > 3$): mostly gives small steady returns, but occasionally delivers extreme outcomes. Fund B is platykurtic ($\beta_2 < 3$): returns are spread more uniformly. Same mean and SD — but Fund A has "fat tails" (higher probability of extreme events). Kurtosis captures this tail-risk difference.

Visual guide to skewness directions:

```
POSITIVE SKEW (right-tailed):
  Frequency
     |
     |  ████
     | ██████
     |████████
     |██████████▓▓▓░░
  ───┼────────────────────── Value
     Mode Median Mean →

NEGATIVE SKEW (left-tailed):
  Frequency
     |
     |          ████
     |        ██████
     |      ████████
     |  ░░░▓▓██████████
  ───┼────────────────────── Value
     ← Mean Median Mode
```

◆ FORMULA / KEY POINTS

The four moment-based shape measures:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \quad \text{(measure of skewness squared; } \beta_1 = 0 \text{ for symmetric)}$$

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \pm\sqrt{\beta_1} \quad \text{(signed skewness; positive = right-skewed)}$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} \quad \text{(kurtosis; normal distribution: } \beta_2 = 3\text{)}$$

$$\gamma_2 = \beta_2 - 3 \quad \text{(excess kurtosis; leptokurtic: } \gamma_2 > 0\text{; platykurtic: } \gamma_2 < 0\text{)}$$

The three Karl Pearson skewness coefficients (all used in exams):

Formula	When to use
$\text{Sk} = \dfrac{\text{Mean} - \text{Mode}}{\sigma}$	When mode is well-defined
$\text{Sk} = \dfrac{3(\text{Mean} - \text{Median})}{\sigma}$	When mode is ill-defined (uses empirical relation)
$\text{Sk} = \dfrac{\mu_3}{\sigma^3} = \gamma_1$	Moment-based (most rigorous)

Solved Example 4.A: For a distribution: Mean = 25, Median = 24, SD = 5. Find Karl Pearson's coefficient of skewness.

Step 1 — Apply the median-based Karl Pearson skewness formula.

- $\text{Sk} = 3(\bar{X} - \text{Median})/\sigma$
- = $3(25 - 24)/5$
- = $3/5$
- = **0.6** (positive skew — right-tailed)

Solved Example 4.B: $\mu_2 = 16$, $\mu_3 = 0$, $\mu_4 = 512$. Find β_1 and β_2.

Step 1 — Compute β₁ (skewness).

- $\beta_1 = \mu_3^2/\mu_2^3$
- = $0^2/16^3$
- = $0/4096$
- = **0** (symmetric distribution)

Step 2 — Compute β₂ (kurtosis).

- $\beta_2 = \mu_4/\mu_2^2$
- = $512/16^2$
- = $512/256$
- = **2** (platykurtic since $\beta_2 < 3$)

$\gamma_2 = 2 - 3 = -1$ (flatter than normal)

4.1 Examiner mindset

Angle	Pet question
Definition of r-th raw and central moment	"Which of these expressions is ...?"
Relationship between raw and central moments	Convert μ_r' into μ_r and back
Karl Pearson's coefficient of skewness	Mean – Mode formula or Mean – Median ×3 / σ
Bowley's coefficient of skewness	$(Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$
Range of β_1, β_2 and what they signify	Symmetric, leptokurtic, etc.

4.2 Moments — definitions

Raw (about origin) moments μ_r'

$$\mu_r' = \frac{\sum f_i X_i^r}{N}$$

So $\mu_1' = \bar{X}$ (the mean), $\mu_0' = 1$, etc.

Central (about mean) moments μ_r

$$\mu_r = \frac{\sum f_i (X_i - \bar{X})^r}{N}$$

r	Value of μ_r
0	1
1	0 (always)
2	σ^2 (the variance)
3	measures skewness numerator
4	measures kurtosis numerator

Conversion (raw → central)

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$$

$$\mu_4 = \mu_4' - 4\mu_1'\mu_3' + 6(\mu_1')^2\mu_2' - 3(\mu_1')^4$$

Moments about an arbitrary point A

If $d = X - A$, and $M_r = \sum f d^r / N$, then:

$$\mu_2 = M_2 - M_1^2, \quad \mu_3 = M_3 - 3M_1M_2 + 2M_1^3, \quad \text{etc.}$$

4.3 Skewness — five formulas (with when to use)

Formula	When examiner gives you
$\mathrm{Sk}_p = \dfrac{\text{Mean} - \text{Mode}}{\sigma}$	Mean, Mode, SD (Karl Pearson direct)
$\mathrm{Sk}_p = \dfrac{3(\text{Mean} - \text{Median})}{\sigma}$	Mean, Median, SD (when Mode is hard to find)
Bowley's $\mathrm{Sk}_B = \dfrac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$	Quartiles only

Formula	When examiner gives you
Kelly's $Sk_K = \dfrac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$	Percentiles
Coefficient $\beta_1 = \dfrac{\mu_3^2}{\mu_2^3}$; $\gamma_1 = \dfrac{\mu_3}{\sigma^3}$	Moments based

Range tells:

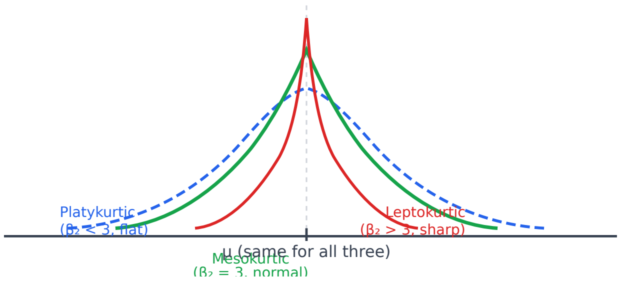
Sign	Shape
Sk = 0	Symmetric
Sk > 0	Positively skewed (right tail)
Sk < 0	Negatively skewed (left tail)

Bowley's Sk lies in **[−1, 1]**. Karl Pearson's lies typically in **[−3, 3]** (theoretically).

4.4 Kurtosis — peakedness

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$

β_2	γ_2	Curve	Name
= 3	= 0	normal-shaped	**Mesokurtic**
> 3	> 0	thinner peak, fatter tails	**Leptokurtic**
< 3	< 0	flatter peak	**Platykurtic**



◎ **MNEMONIC**

Memory hook. "**Lepto** = **L**eaping high (sharper peak)." "**Platy** = **P**lateau (flat)." "**Meso** = middle = normal."

4.5 Sheppard's correction (rare but asked)

When data is grouped, raw moments overstate variance slightly. Correction:

$$\mu_2(\text{corrected}) = \mu_2 - \frac{h^2}{12}$$

$$\mu_4(\text{corrected}) = \mu_4 - \frac{1}{2}h^2\mu_2 + \frac{7}{240}h^4$$

h = class width. The 2nd-moment correction is most commonly asked; the 4th rarely.

4.6 Worked PYQ-style examples

✎ **WORKED EXAMPLE**

Q 4.1. $\mu_1' = 4, \mu_2' = 25$. Find variance.

Step 1 — Apply the raw→central conversion: $\mu_2 = \mu_2' - (\mu_1')^2$.

- $\mu_2 = 25 - 4^2$
- $= 25 - 16$
- **= 9**

✎ **WORKED EXAMPLE**

Q 4.2. $\mu_2 = 4, \mu_3 = 8$. Coefficient of skewness β_1?

Step 1 — Apply the β_1 formula.

- $\beta_1 = \mu_3^2/\mu_2^3$
- $= 8^2/4^3$
- $= 64/64$
- **= 1**

✎ **WORKED EXAMPLE**

Q 4.3. Mean = 50, Mode = 47, SD = 10. Karl Pearson Sk?

Step 1 — Apply the mode-based Karl Pearson formula.

- $\text{Sk} = (\bar{X} - \text{Mode})/\sigma$
- $= (50 - 47)/10$
- $= 3/10$
- = **0.3** → mildly positively skewed

✎ **WORKED EXAMPLE**

Q 4.4. $Q_1 = 14, Q_2 = 21, Q_3 = 26$. Bowley's Sk?

Step 1 — Apply Bowley's skewness formula.

- $\text{Sk} = (Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$
- $= (26 + 14 - 2 \times 21)/(26 - 14)$
- $= (40 - 42)/12$
- $= -2/12$
- $= \mathbf{-1/6} \approx -0.167$ → mildly negative skew

✎ **WORKED EXAMPLE**

Q 4.5. $\beta_2 = 5$. Curve is:

Soln. $\beta_2 > 3$ → **leptokurtic** (sharper peak, fatter tails than normal).

✎ **WORKED EXAMPLE**

Q 4.6. First moment about origin is 5. Raw moments μ_2', μ_3' are 35 and 295 respectively. Find μ_3.

Step 1 — Recall the raw→central conversion: $\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$.

Step 2 — Substitute the given raw moments.

- $\mu_3 = 295 - 3 \times 5 \times 35 + 2 \times 5^3$
- $= 295 - 525 + 250$
- **= 20**

✎ **WORKED EXAMPLE**

Q 4.7. Sum of central moments of order 1 in any distribution is:

Soln. $\mu_1 = \sum f(X - \bar{X})/N = 0$ — by the very definition of the mean. Always **0**.

✎ **WORKED EXAMPLE**

Q 4.8. A leptokurtic curve has γ_2:

Soln. Leptokurtic ⇒ $\beta_2 > 3 \Rightarrow \gamma_2 = \beta_2 - 3 > 0$. Hence **positive**.

✎ **WORKED EXAMPLE**

Q 4.9. For a symmetric distribution, every odd-order central moment is:

Soln. Positive and negative deviations cancel pairwise in any odd power. So $\mu_1 = \mu_3 = \mu_5 = \cdots =$ **0**.

✎ **WORKED EXAMPLE**

Q 4.10. Class width 5. $\mu_2 = 35.5$. Sheppard-corrected variance?

Step 1 — Apply Sheppard's correction: $\mu_2^{\text{corr}} = \mu_2 - h^2/12$.

- $\mu_2^{\text{corr}} = 35.5 - 5^2/12$
- $= 35.5 - 25/12$
- $= 35.5 - 2.083$
- **= 33.42**

4.7 Computation drills — the 8 problem types the exam sets

These 8 types cover every numerical question on Chapter 4 in the last 6 JSO papers. Solve each before reading the solution.

✎ WORKED EXAMPLE

CD-4.1. Karl Pearson Sk — mode based. Mean = 45, Mode = 38, SD = 7. Find Pearson's coefficient of skewness.

Step 1 — Apply formula: $\mathrm{Sk} = (\bar{X} - \mathrm{Mo})/\sigma$.

- = $(45 - 38)/7$
- = $7/7$
- = **+1.0** (moderate positive skew — mode is below mean)

Trap. Divide by SD, never by variance. Sk = 1 means "one standard deviation" of skew — purely a ratio, not a probability.

✎ WORKED EXAMPLE

CD-4.2. Karl Pearson Sk — median based (mode ill-defined). Mean = 62, Median = 60, SD = 6. Find skewness.

Step 1 — Apply: $\mathrm{Sk} = 3(\bar{X} - \mathrm{Md})/\sigma$.

- = $3 \times (62 - 60)/6$
- = $3 \times 2/6$
- = $6/6$
- = **+1.0**

Note. Both forms give the same number here by design. Median-based uses factor 3 because the empirical relation Mode ≈ 3·Median − 2·Mean was substituted.

✎ WORKED EXAMPLE

CD-4.3. Bowley's coefficient of skewness. Q_1 = 28, Q_2 (median) = 40, Q_3 = 58. Find Bowley's Sk.

Step 1 — Apply formula: $S_B = (Q_1 + Q_3 - 2Q_2)/(Q_3 - Q_1)$.

- Numerator: $28 + 58 - 2 \times 40 = 86 - 80 = 6$
- Denominator: $58 - 28 = 30$

Step 2 — Divide.

- $S_B = 6/30 = +\mathbf{0.2}$

Interpretation. Slight positive skew. Range: $-1 \le S_B \le +1$.

✎ **WORKED EXAMPLE**

CD-4.4. Compute β₁ (skewness measure) and γ₁ (signed skewness). $\mu_2 = 9$, $\mu_3 = -54$.

Step 1 — Compute $\beta_1 = \mu_3^2/\mu_2^3$.

- $= (-54)^2/9^3$
- $= 2916/729$
- $= \mathbf{4.0}$

Step 2 — Compute $\gamma_1 = \mu_3/\mu_2^{3/2}$ (signed version).

- $\mu_2^{3/2} = 9^{3/2} = 27$
- $\gamma_1 = -54/27 = -\mathbf{2.0}$ (negative skew — long left tail)

Trap. β_1 is always ≥ 0 (it's squared). The sign of skewness lives in γ_1.

✎ **WORKED EXAMPLE**

CD-4.5. Compute β₂ (kurtosis) and γ₂ (excess kurtosis). $\mu_2 = 16$, $\mu_4 = 1024$.

Step 1 — Compute $\beta_2 = \mu_4/\mu_2^2$.

- $= 1024/16^2$
- $= 1024/256$
- $= \mathbf{4.0}$

Step 2 — Compute $\gamma_2 = \beta_2 - 3$.

- $= 4.0 - 3 = +\mathbf{1.0}$ → **leptokurtic** (sharper peak than normal)

Check. Normal distribution has $\beta_2 = 3$, $\gamma_2 = 0$. Any positive γ_2 → leptokurtic.

✎ **WORKED EXAMPLE**

CD-4.6. Convert raw moments to central moments. Raw moments about origin: $\mu_1' = 4$, $\mu_2' = 22$, $\mu_3' = 140$.

Step 1 — Compute μ_2 (central moment).

- $\mu_2 = \mu_2' - (\mu_1')^2 = 22 - 16 = \mathbf{6}$

Step 2 — Compute μ_3.

- $\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$
- $= 140 - 3 \times 22 \times 4 + 2 \times 64$
- $= 140 - 264 + 128$
- $= \mathbf{4}$

Step 3 — Determine skewness direction.

- $\gamma_1 = \mu_3/\mu_2^{3/2} = 4/6^{1.5} = 4/14.70 \approx +0.27$ → slight positive skew.

✎ WORKED EXAMPLE

CD-4.7. Back-calculate SD from skewness. Karl Pearson's Sk = +0.6, Mean = 50, Mode = 44. Find SD.

Step 1 — Rearrange formula: $\sigma = (\bar{X} - \text{Mo})/\text{Sk}$.

- $= (50 - 44)/0.6$
- $= 6/0.6$
- $= \mathbf{10}$

Exam note. Examiners give Sk and two of {Mean, Mode, SD} and ask for the third. Always identify which is unknown, rearrange, plug in.

✎ WORKED EXAMPLE

CD-4.8. Identify curve type from β_1 and β_2.

μ_2	μ_3	μ_4	What is it?
4	0	48	$\beta_1 = 0^2/4^3 = 0$; $\beta_2 = 48/16 = 3$ → **Symmetric mesokurtic**
9	27	324	$\beta_1 = 729/729 = 1$; $\beta_2 = 324/81 = 4$ → **Positive skew, leptokurtic**
25	0	1250	$\beta_1 = 0$; $\beta_2 = 1250/625 = 2$ → **Symmetric, platykurtic**

Pattern. $\beta_1 = 0$ → symmetric. $\beta_2 = 3$ → mesokurtic. $\beta_2 > 3$ → lepto. $\beta_2 < 3$ → platy.

4.8 Trap-recognition card

Trap	Defence
$\beta_2 = 3$ marked as "leptokurtic"	It's mesokurtic; lepto needs $\beta_2 > 3$.
Confusing μ_r with μ_r'	Central = about mean. Raw = about origin. $\mu_1 = 0$ but $\mu_1' = \bar{X}$.
Pearson Sk denominator without σ	The denominator is **σ** (SD), not variance.
Bowley sign flipped	Numerator is $Q_3 + Q_1 - 2Q_2$. If positive → positive skew.

4.8 Mini-mock

#	Q	Ans
1	μ_1 for any distribution = ?	0
2	μ_2 is also called?	Variance
3	If $\beta_2 = 2.5$, distribution is?	Platykurtic
4	Pearson Sk if Mean = Mode?	0
5	Bowley Sk range?	[−1, 1]
6	A symmetric distribution: Mean − Mode = ?	0
7	Mean = 60, Mode = 54, σ = 12. Pearson Sk?	0.5
8	Sheppard's correction for variance?	subtract $h^2/12$
9	$\mu_3 = 0, \mu_2 = 4$. Sk?	0 (symmetric)
10	"Lepto" peak shape?	Sharp / tall

4.9 Active-recall prompts

1. Write μ_2, μ_3, μ_4 in terms of raw moments.
2. State all three Karl Pearson Sk formulas.
3. State Bowley's range.
4. Define mesokurtic, leptokurtic, platykurtic with their β_2 cutoffs.
5. State Sheppard's correction for the second central moment.

CHAPTER 5 — Correlation & Regression

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈10 Qs / paper) **Difficulty:** Medium. Half the questions are formula plug-ins; the rest test the famous **two regression lines** properties.

5.0 — Understanding Correlation & Regression from First Principles

✔ QUICK RECALL

What is correlation? What does it actually measure — and what doesn't it measure?

Correlation answers: "Do two variables tend to move together?" Height and weight move together (taller people tend to be heavier) — **positive correlation**. Temperature and hot-drink sales move oppositely — **negative correlation**. Shoe size and exam score have no connection — **zero correlation**.

Karl Pearson's r quantifies this as a number between −1 and +1, measuring the *strength and direction of the linear relationship*: - $r = +1$: perfect positive straight-line relationship (all points lie exactly on an upward line) - $r = -1$: perfect negative straight-line relationship - $r = 0$: no *linear* relationship

Critical warning that every JSO student must remember: $r = 0$ does NOT mean "no relationship." It means no *linear* relationship. The dataset $Y = X^2$ has $r = 0$ between X and Y — but Y is perfectly determined by X (it's a perfect parabola). Correlation only measures *linear* association.

From correlation to regression: Correlation tells you *whether* two variables are related and how strongly. Regression tells you *the equation* to predict one from the other. If height and weight have $r = 0.85$, the regression line gives: predicted weight = $a + b \times \text{height}$.

Why two regression lines? When you minimise errors in predicting Y from X, you get the "Y on X" regression line (minimises vertical squared distances from points to line). When you minimise errors in predicting X from Y, you get the "X on Y" line (minimises horizontal squared distances). These are *geometrically different lines* — the direction of error minimisation is different. They coincide only when $|r| = 1$ (perfect linear relationship). Both always pass through $(\bar{X}, \bar{Y})$.

Geometric picture of the two regression lines:

```
Y
|             . .  .
|          .  Byx (Y on X line — steep when r is high)
|       . Bxy (X on Y line — less steep)
|    .×.   ← both pass through (X̄, Ȳ)
|  .
|.
└────────────────────── X
   (angle between lines ↓ as |r| → 1; lines coincide at |r|=1)
```

◆ FORMULA / KEY POINTS

The two regression coefficients and their link to r:

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x} \quad \text{(slope of Y on X line)}$$

$$b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y} \quad \text{(slope of X on Y line)}$$

$$\boxed{r = \pm\sqrt{b_{yx} \times b_{xy}}} \quad \text{(sign = common sign of the two regression coefficients)}$$

$$\text{Karl Pearson's } r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{N\,\sigma_x\,\sigma_y} = \frac{N\sum X_iY_i - (\sum X_i)(\sum Y_i)}{\sqrt{[N\sum X_i^2 - (\sum X_i)^2][N\sum Y_i^2 - (\sum Y_i)^2]}}$$

Solved Example 5.A — Find r from regression coefficients:

If $b_{yx} = 0.8$ and $b_{xy} = 0.2$, find r.

Step 1 — Apply the regression-slope identity: $r = \pm\sqrt{b_{yx} \cdot b_{xy}}$.

- $r = \sqrt{0.8 \times 0.2}$
- = $\sqrt{0.16}$
- = **0.4**

Step 2 — Fix the sign. Both regression coefficients are positive ⇒ r is positive ⇒ $r = +0.4$.

Solved Example 5.B — Find missing regression coefficient:

$r = 0.6$, $\sigma_x = 4$, $\sigma_y = 3$. Find b_{yx} and b_{xy}.

Step 1 — Compute the regression coefficient of Y on X.

- $b_{yx} = r \times (\sigma_y/\sigma_x)$
- = $0.6 \times (3/4)$
- = **0.45**

Step 2 — Compute the regression coefficient of X on Y.

- $b_{xy} = r \times (\sigma_x/\sigma_y)$
- = $0.6 \times (4/3)$
- = **0.8**

Step 3 — Verify using $r = \sqrt{b_{yx} \times b_{xy}}$.

- $r = \sqrt{0.45 \times 0.8} = \sqrt{0.36} = 0.6$ ✓

Solved Example 5.C — Spearman's rank correlation:

Two judges rank 5 students: Judge A gives ranks 1,2,3,4,5; Judge B gives ranks 2,1,4,3,5. Find Spearman's ρ.

Step 1 — Compute the rank differences $d_i = R_A - R_B$ for each student.

- d: $1-2=-1$, $2-1=1$, $3-4=-1$, $4-3=1$, $5-5=0$

Step 2 — Square the differences and sum.

- d^2: $1, 1, 1, 1, 0$
- $\sum d^2 = 4$

Step 3 — Apply Spearman's rank-correlation formula.

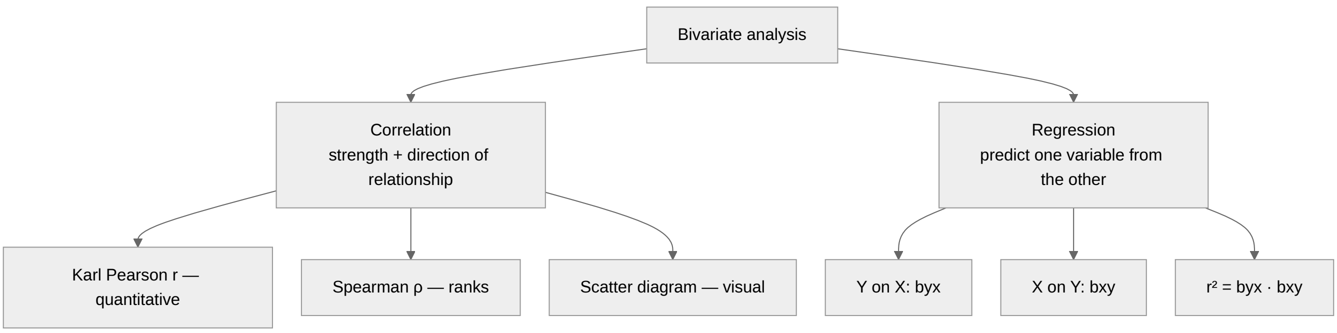
- $\rho = 1 - \dfrac{6\sum d^2}{n(n^2-1)}$
- $= 1 - \dfrac{6 \times 4}{5 \times (25-1)}$
- $= 1 - \dfrac{24}{5 \times 24}$
- $= 1 - \dfrac{24}{120}$
- $= 1 - 0.2$
- $=$ **0.8**

Strong positive agreement between the two judges.

5.1 Examiner mindset

Angle	Pet question
Range and meaning of r	"Range of correlation coefficient is ..." → [−1, 1]
Karl Pearson formula	direct plug-in given covariance and SDs
Spearman rank correlation	with and without ties
Properties of regression coefficients	product = r^2, same sign, geometric mean = ±r
Two regression lines, point of intersection	$(\bar{X}, \bar{Y})$ — always
Angle between regression lines	when $r = 0$ → 90°; $r = \pm 1$ → 0°
Multiple / partial correlation	basic 2-variable question

5.2 Mind map



5.3 Correlation — Karl Pearson

$$r = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \cdot \sum (Y_i - \bar{Y})^2}}$$

Computational form (preferred for plug-in questions):

$$\boxed{r = \frac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}}$$

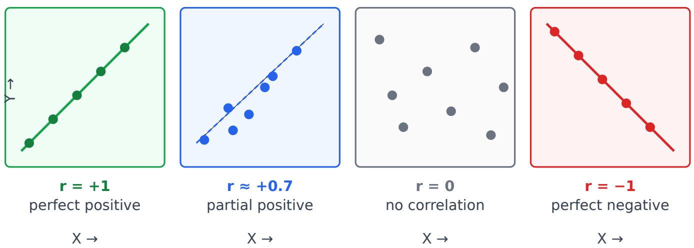
Range: $-1 \le r \le +1$.

r	Meaning
+1	perfect direct (positive) correlation
0 < r < 1	partial direct
0	no linear correlation
−1 < r < 0	partial inverse
−1	perfect inverse

Properties of r:

Property	Note
Symmetric	$r_{XY} = r_{YX}$
Independent of origin	r of $(X+a, Y+b)$ = r of (X, Y)
Independent of scale (when both positive)	r of (cX, dY) with $c, d > 0$ = r
Sign flips when one scale flips	r of $(-X, Y) = -r_{XY}$
Pure number	unitless

Scatter-plot patterns — recognise r at a glance:



⚠ EXAM ALERT

Exam shortcut. If all scatter points fall exactly on a line → |r| = 1. If they form a wide cloud with no discernible trend → r ≈ 0. The question always asks which of the four panels a given correlation value matches — memorise these four shapes.

5.4 Spearman's rank correlation

When data is ordinal or you only have ranks:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

d_i = difference in ranks of the i-th pair, n = number of pairs.

With tied ranks, add correction $\dfrac{m(m^2-1)}{12}$ for each set of m tied ranks to $\sum d^2$:

$$\rho = 1 - \frac{6[\sum d^2 + \sum \frac{m(m^2-1)}{12}]}{n(n^2-1)}$$

5.5 Regression — the two lines

Y on X (used to predict Y given X):

$$Y - \bar{Y} = b_{YX}(X - \bar{X}), \qquad b_{YX} = r\frac{\sigma_Y}{\sigma_X} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X^2}$$

X on Y (used to predict X given Y):

$$X - \bar{X} = b_{XY}(Y - \bar{Y}), \qquad b_{XY} = r\frac{\sigma_X}{\sigma_Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_Y^2}$$

5.6 The 7 sacred properties of regression coefficients

#	Property	Why useful
1	$b_{YX} \cdot b_{XY} = r^2$	Lets you find r from the two slopes (sign from common sign of slopes).
2	$r = \pm\sqrt{b_{YX} \cdot b_{XY}}$	Geometric mean of slopes.
3	b_{YX} and b_{XY} have the **same sign** as r.	If signs differ, you've made an error.
4	If $\lvert b_{YX}\rvert > 1$ then $\lvert b_{XY}\rvert < 1$ (and vice-versa).	because $r^2 \le 1$
5	The two regression lines **intersect at** $(\bar{X}, \bar{Y})$.	Always; never doubt.
6	When $r = \pm 1$ the two lines **coincide**.	perfect correlation
7	When $r = 0$ the two lines are **perpendicular** (one horizontal, one vertical).	no linear relationship

◎ **MNEMONIC**

Memorise this picture: imagine two pencils crossing at the means (\bar X, \bar Y). When correlation is strong, they nearly stack on each other; when weak, they spread apart; at zero, they form a + sign.

5.7 Recovering $r, \bar{X}, \bar{Y}$ from the two equations

Given $aX + bY + c = 0$ and $dX + eY + f = 0$:

1. **Solve simultaneously** for the intersection — that gives $(\bar{X}, \bar{Y})$.
2. Decide which line is **Y on X**: the one with smaller $\lvert b_{YX}\rvert$ is the Y-on-X line **if** $\lvert r\rvert \le 1$; test by computing slopes.
3. From the two coefficients, compute $r = \pm\sqrt{b_{YX} \cdot b_{XY}}$, with sign = common sign of the two slopes.

5.8 Multiple and partial correlation (3 variables)

Symbol	Meaning
$r_{12.3}$	partial correlation between X_1 and X_2 keeping X_3 constant
$R_{1.23}$	multiple correlation of X_1 on X_2, X_3

$$r_{12.3} = \frac{r_{12} - r_{13}r_{23}}{\sqrt{(1 - r_{13}^2)(1 - r_{23}^2)}}$$

$$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2}$$

Range	Note
Partial $r_{12.3}$	$-1 \le r_{12.3} \le 1$

Range	Note
Multiple $R_{1.23}$	$0 \le R_{1.23} \le 1$ (never negative)

5.9 Coefficient of determination

$$r^2 = \text{fraction of variance of } Y \text{ explained by } X.$$

If $r = 0.8$, $r^2 = 0.64$ → 64% of Y's variation explained by X.

5.10 Standard error of estimate

$$S_{Y \cdot X} = \sigma_Y \sqrt{1 - r^2}, \qquad S_{X \cdot Y} = \sigma_X \sqrt{1 - r^2}$$

When $r = \pm 1$, SE = 0 (perfect prediction). When $r = 0$, SE = σ (no improvement over mean).

5.11 Worked PYQ-style examples

✎ **WORKED EXAMPLE**

Q 5.1. $b_{YX} = 0.8, b_{XY} = 0.5$. Find r.

Step 1 — Apply $r = \pm\sqrt{b_{YX} \cdot b_{XY}}$.

- $r = \sqrt{0.8 \times 0.5}$
- $= \sqrt{0.4}$
- $\approx$ **0.632**

Step 2 — Fix the sign. Both slopes positive ⇒ r positive ⇒ $r = +0.632$.

✎ **WORKED EXAMPLE**

Q 5.2. Two regression lines: $2X + 3Y - 8 = 0$ and $X + 4Y - 7 = 0$. Find $\bar{X}, \bar{Y}$.

Step 1 — Use the fact that both regression lines pass through $(\bar{X}, \bar{Y})$. So solve the two equations simultaneously.

Step 2 — Express X from line 2.

- $X = 7 - 4Y$

Step 3 — Substitute into line 1 and solve for Y.

- $2(7 - 4Y) + 3Y = 8$
- $14 - 8Y + 3Y = 8$
- $-5Y = -6$
- $Y = 1.2$

Step 4 — Back-substitute to find X.

- $X = 7 - 4(1.2)$
- $= 7 - 4.8$
- = **2.2**

Final: $\bar{X} = 2.2$, $\bar{Y} = 1.2$.

✎ **WORKED EXAMPLE**

Q 5.3. $r = 0.6, \sigma_X = 5, \sigma_Y = 10$. Find b_{YX} and b_{XY}.

Step 1 — Compute the regression coefficient of Y on X.

- $b_{YX} = r \times \sigma_Y / \sigma_X$
- $= 0.6 \times 10/5$
- **= 1.2**

Step 2 — Compute the regression coefficient of X on Y.

- $b_{XY} = r \times \sigma_X / \sigma_Y$
- $= 0.6 \times 5/10$
- **= 0.3**

Check: $b_{YX} \times b_{XY} = 1.2 \times 0.3 = 0.36 = 0.6^2$ ✓

✎ **WORKED EXAMPLE**

Q 5.4. Spearman: 5 students, ranks in two subjects: d^2-values 1, 0, 4, 1, 4. Sum = 10. Find ρ.

Step 1 — Apply Spearman's formula: $\rho = 1 - 6\sum d^2/(n(n^2-1))$. Here $n = 5$, $\sum d^2 = 10$.

- $\rho = 1 - \frac{6 \times 10}{5 \times (25-1)}$
- $= 1 - \frac{60}{5 \times 24}$
- $= 1 - \frac{60}{120}$
- $= 1 - 0.5$
- **= 0.5**

✎ **WORKED EXAMPLE**

Q 5.5. If $r = 0$, the two regression lines are:

Soln. Property: $r = 0 \Rightarrow$ the two regression lines are **perpendicular** (one horizontal, one vertical).

✎ **WORKED EXAMPLE**

Q 5.6. $r_{XY} = 0.8$. Then correlation of $X' = X + 5$ and $Y' = 2Y - 3$ is:

Soln. Karl Pearson r is invariant under shift (origin change) and unaffected by a positive scale factor. So $r' =$ **0.8**.

✎ **WORKED EXAMPLE**

Q 5.7. $r = 0.5$. Coefficient of determination?

Step 1 — Coefficient of determination = r^2.

- $r^2 = 0.5^2 = 0.25$
- = **25 %** of Y's variation explained by X

✎ WORKED EXAMPLE

Q 5.8. $b_{YX} = 1.5, b_{XY} = 0.9$. Is this possible?

Step 1 — Use the identity $b_{YX} \cdot b_{XY} = r^2$ with the constraint $r^2 \leq 1$.

- Product = $1.5 \times 0.9 = 1.35 > 1$
- $r^2 > 1$ is impossible → **No, this configuration cannot occur.**

✎ WORKED EXAMPLE

Q 5.9. A regression line passes through (5, 7) and the means are $\bar{X} = 5, \bar{Y} = ?$.

Soln. Property: **every** regression line passes through $(\bar{X}, \bar{Y})$. Given $\bar{X} = 5$ and the line passes through $(5, 7)$, the means are $(5, 7)$, so $\bar{Y} = 7$.

✎ WORKED EXAMPLE

Q 5.10. If $\sigma_X = \sigma_Y$, then $b_{YX} = ?$

Step 1 — Substitute into $b_{YX} = r \cdot \sigma_Y / \sigma_X$.

- With $\sigma_X = \sigma_Y$, the ratio = 1
- $b_{YX} = r$ (and similarly $b_{XY} = r$)

✎ WORKED EXAMPLE

Q 5.11. Karl Pearson r is independent of:

Soln. Karl Pearson r is independent of the **origin** (shift) and of the **positive scale** of measurement. It is a pure (unitless) number.

✎ WORKED EXAMPLE

Q 5.12. A scatter plot showing all points on a straight line with positive slope indicates:

Soln. All points exactly on an upward line ⇒ perfect direct linear association ⇒ $r = +1$.

✎ WORKED EXAMPLE

Q 5.13. $n = 10, \sum X = 50, \sum Y = 100, \sum X^2 = 300, \sum Y^2 = 1100, \sum XY = 525$. Find r.

Step 1 — Compute the numerator of Karl Pearson's r.

- Numerator = $n \sum XY - (\sum X)(\sum Y)$
- = $10 \times 525 - 50 \times 100$
- = $5250 - 5000$
- = **250**

Step 2 — Compute the denominator.

- Factor 1 = $n \sum X^2 - (\sum X)^2 = 10 \times 300 - 2500 = 3000 - 2500 = 500$
- Factor 2 = $n \sum Y^2 - (\sum Y)^2 = 10 \times 1100 - 10000 = 11000 - 10000 = 1000$
- Denominator = $\sqrt{500 \times 1000} = \sqrt{500000} \approx 707.1$

Step 3 — Compute r.

- $r = 250/707.1 \approx \mathbf{0.354}$

✎ WORKED EXAMPLE

Q 5.14. Two regression equations: $3Y = 2X + 5$ and $15X = 10Y + 8$. Find b_{YX}, b_{XY}, r.

Step 1 — Extract the slope b_{YX} from the Y-on-X line.

- Rearrange: $Y = (2/3)X + 5/3$
- $b_{YX} = \mathbf{2/3}$

Step 2 — Extract the slope b_{XY} from the X-on-Y line.

- Rearrange: $X = (10/15)Y + 8/15 = (2/3)Y + 8/15$
- $b_{XY} = \mathbf{2/3}$

Step 3 — Compute r from the two regression slopes.

- $r = +\sqrt{b_{YX} \times b_{XY}} = +\sqrt{(2/3)(2/3)} = 2/3 \approx \mathbf{0.667}$

✎ WORKED EXAMPLE

Q 5.15. Two regression coefficients are 0.5 and −0.4. Then r is:

Soln. Property: b_{YX} and b_{XY} must share the **same sign** (the sign of r). One positive and one negative ⇒ impossible. Such a pair cannot both be valid regression coefficients.

✎ WORKED EXAMPLE

Q 5.16. Standard error of estimate of Y on X = ?

Soln. Standard formula: $S_{Y \cdot X} = \sigma_Y \sqrt{1 - r^2}$. (When $r = \pm 1$ ⇒ SE = 0; when $r = 0$ ⇒ SE = σ_Y.)

✎ **WORKED EXAMPLE**

Q 5.17. When $r = 1$, $b_{YX} \cdot b_{XY} = ?$

Step 1 — Apply $b_{YX} \cdot b_{XY} = r^2$.

- With $r = 1$: product = 1^2 = **1**
- The two slopes are reciprocals of each other.

✎ **WORKED EXAMPLE**

Q 5.18. Multiple correlation $R_{1.23}$ is always:

Soln. $R_{1.23} \in [0, 1]$ — **non-negative** by definition (it is a square root of a squared-correlation quantity).

5.12 Trap-recognition card

Trap	Why wrong	Defence
Both slopes > 1	impossible ($r^2 > 1$)	Test product.
Slopes have different signs	impossible	They must agree with r's sign.
"Regression lines never meet"	They always meet at $(\bar{X}, \bar{Y})$.	Always.
"Spearman uses raw values"	No, ranks.	Re-read formula.
Scale flips and student keeps same r sign	Negative scale → sign flips.	Track sign of scale factor.
Forgot r is unitless	They keep units.	It's a pure number.

5.13 Mini-mock

#	Q	Ans
1	Range of r?	[−1, 1]
2	$b_{YX} = 0.4, b_{XY} = 0.9.$ r?	√0.36 = 0.6
3	Two regression lines intersect at?	$(\bar{X}, \bar{Y})$
4	r = 0 → angle between lines?	90°
5	r^2 is called?	Coefficient of determination
6	Spearman with d^2 sum 30, n=10?	1 − 6·30/(10·99) = 1 − 180/990 ≈ 0.818
7	If σx = 4, σy = 8, r = 0.5 → bYX?	1
8	r between (X+5) and (Y−3) when r(X,Y)=0.7?	0.7 (origin invariant)
9	Multiple correlation R is in?	[0, 1]
10	Standard error of estimate when r=1?	0

5.14 Active-recall prompts

1. Write the computational formula for r.
2. State the 7 properties of regression coefficients.
3. Where do the two regression lines meet?
4. Write Spearman's formula and the tie correction.

5. Write the formula for $r_{12.3}$ and $R^2_{1.23}$.

CHAPTER 6 — Probability Theory

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈8 Qs / paper) **Difficulty:** Medium. Almost every wrong answer here is from a single error: confusing **mutually exclusive** with **independent**. Pin that down and you score full.

6.0 — Understanding Probability from First Principles

✔ QUICK RECALL

What is probability — and why did mathematicians need to formalise it?

Before the 17th century, gamblers relied on intuition and superstition to assess chances. Pascal and Fermat formalised probability in 1654 while solving a gambling problem. Today probability is the mathematical language of uncertainty — used in insurance, medicine, finance, quality control, and statistics (Chapters 8–9 depend entirely on it).

Three definitions — all useful, each for different situations:

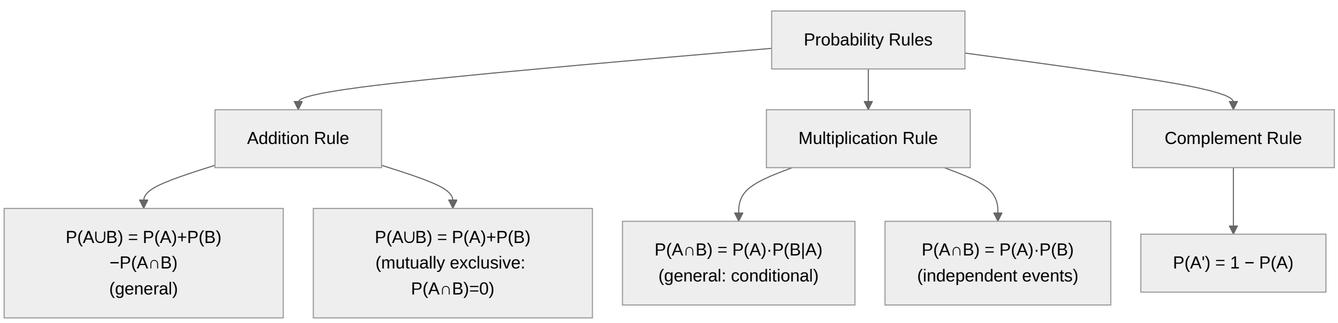
1. **Classical (Laplace):** $P(A) = \dfrac{\text{number of favourable outcomes}}{\text{total number of equally likely outcomes}}$. Works for coins, dice, cards where outcomes are symmetric.
2. **Frequentist:** $P(A) = \lim_{n\to\infty} \dfrac{f_A}{n}$ where f_A is the number of times A occurs in n trials. Used when you can repeat experiments — "if we toss this biased coin 10,000 times, heads will appear about 60% of the time."
3. **Axiomatic (Kolmogorov):** P is a function satisfying: (i) $P(A) \geq 0$; (ii) $P(\Omega) = 1$; (iii) for mutually exclusive events, $P(A \cup B) = P(A) + P(B)$. This is the mathematically rigorous definition that makes all proofs watertight.

The most critical distinction in this chapter: "Mutually exclusive" ≠ "Independent".

- **Mutually exclusive:** A and B cannot both happen. $P(A \cap B) = 0$. Example: a single die roll showing 3 and showing 5 simultaneously.
- **Independent:** Knowing A happened gives no information about B. $P(A \cap B) = P(A) \times P(B)$. Example: coin flip 1 (Heads) and coin flip 2 (Heads) — these are independent.

Can two events be both mutually exclusive AND independent? Only if at least one has probability 0. Otherwise, mutually exclusive events are actually *negatively dependent* — knowing one happened tells you the other definitely didn't.

Concept map — probability rules:



◆ FORMULA / KEY POINTS

Bayes' Theorem (appears every paper):

$$\boxed{P(H_i|E) = \frac{P(H_i) \cdot P(E|H_i)}{\sum_j P(H_j) \cdot P(E|H_j)}}$$

H_i are mutually exclusive, exhaustive hypotheses (causes). E is the observed evidence (effect). Bayes flips the conditioning: from $P(E|H_i)$ (known) to $P(H_i|E)$ (what you want to know).

Solved Example 6.A — Bayes' Theorem (the classic medical test type):

A factory has 3 machines producing bolts: Machine A (30% of output), Machine B (50%), Machine C (20%). Defect rates: A gives 1% defective, B gives 2%, C gives 3%. A bolt is found defective. What is the probability it came from Machine B?

Step 1 — List the prior probabilities for the three machines.

- $P(A) = 0.3,\ \ P(B) = 0.5,\ \ P(C) = 0.2$

Step 2 — List the likelihoods (defect rate given each machine).

- $P(D|A) = 0.01,\ \ P(D|B) = 0.02,\ \ P(D|C) = 0.03$

Step 3 — Compute the total probability of a defective bolt (denominator).

- $P(D) = 0.3 \times 0.01 + 0.5 \times 0.02 + 0.2 \times 0.03$
- $= 0.003 + 0.010 + 0.006$
- $=$ **0.019**

Step 4 — Apply Bayes' theorem to find $P(B|D)$.

- $P(B|D) = \frac{P(B) \cdot P(D|B)}{P(D)}$
- $= \frac{0.5 \times 0.02}{0.019}$
- $= \frac{0.010}{0.019}$
- $\approx$ **0.526**

Solved Example 6.B — Independence vs mutual exclusivity:

$P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.12$. Are A and B independent?

Step 1 — Test the independence criterion $P(A) \cdot P(B) \stackrel{?}{=} P(A \cap B)$.

- $P(A) \cdot P(B) = 0.4 \times 0.3 = 0.12$
- $P(A \cap B) = 0.12$ ✓
- **Yes — A and B are independent.**

Step 2 — Check mutual exclusivity.

- $P(A \cap B) = 0.12 \neq 0$ ⇒ **not mutually exclusive.**

6.1 Examiner mindset

Angle	Pet question
Definitions (classical / empirical / axiomatic)	One-line MCQ

Angle	Pet question
Addition theorem (with / without ME)	"P(A ∪ B) = ?"
Multiplication theorem (with / without independence)	"P(A ∩ B) = ?"
Conditional probability	"P(A given B) = ?"
Independence	Test $P(A \cap B) = P(A)P(B)$
Bayes' theorem	"If event already happened, find probability of cause" — exactly 1 Q every paper

6.2 Definitions

Type	Definition
Classical	$P(A) = \dfrac{\text{favourable outcomes}}{\text{total equally likely outcomes}}$
Empirical (frequentist)	$P(A) = \lim_{n\to\infty} \dfrac{\text{number of times A occurs}}{n}$
Axiomatic (Kolmogorov)	$P \geq 0,\ P(\text{sample space}) = 1,\ P(\bigcup A_i) = \sum P(A_i)$ when A_i disjoint

6.3 Set-theory shortcuts

For events A, B in a sample space:

Quantity	Formula
$P(A^c)$ (complement)	$1 - P(A)$
$P(A \cup B)$	$P(A) + P(B) - P(A \cap B)$
$P(A \cup B \cup C)$	$\sum P - \sum P(\text{pair}) + P(\text{triple})$ (inclusion - exclusion)
$P(A \cap B^c)$	$P(A) - P(A \cap B)$
$P(\text{exactly one})$	$P(A) + P(B) - 2P(A \cap B)$

6.4 Mutually exclusive vs Independent — the only confusion that matters

Property	Mutually exclusive	Independent
Definition	Cannot occur together: $A \cap B = \varnothing$	One does not affect the other: $P(A \cap B) = P(A)P(B)$
$P(A \cap B)$	0	$P(A)P(B)$
$P(A \cup B)$	$P(A) + P(B)$	$P(A) + P(B) - P(A)P(B)$
Can both hold simultaneously?	Only if $P(A) = 0$ or $P(B) = 0$. Otherwise **never both**.	

× COMMON TRAP

Trap. "If A and B are mutually exclusive, are they independent?" — typically **no** (unless one has zero probability). Examiners ask this exactly to catch lazy candidates.

6.5 Conditional probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0$$

Multiplication theorem:

$$P(A \cap B) = P(B) \cdot P(A|B) = P(A) \cdot P(B|A)$$

For three events: $P(A \cap B \cap C) = P(A)P(B|A)P(C|A \cap B)$.

6.6 Bayes' theorem (high-yield — 1 Q every paper)

If $B_1, B_2, \ldots, B_k$ partition the sample space (mutually exclusive + exhaustive) and A is any event:

$$\boxed{P(B_i|A) = \frac{P(B_i) \cdot P(A|B_i)}{\sum_{j=1}^{k} P(B_j) \cdot P(A|B_j)}}$$

Reading the formula in plain English: - Numerator = (chance B_i was the cause) × (chance it produced effect A). - Denominator = total chance of A summed over all causes (this is the **theorem of total probability**).

6.7 Quick reference: counting

Setup	Count
1 die thrown	6 outcomes
2 dice thrown	36 outcomes
1 coin tossed n times	2^n outcomes
Pack of cards	52 cards (4 suits × 13 ranks; 26 red, 26 black; 12 face cards Q J K only — note: face cards exclude Ace)
Selecting r from n	${}^nC_r$
Arranging r from n	${}^nP_r$

6.8 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 6.1. Two dice rolled. P(sum = 7)?

Step 1 — Count favourable ordered pairs that sum to 7.

- (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) → **6 pairs**

Step 2 — Total sample space = 6 × 6 = 36.

Step 3 — Compute P.

- $P = 6/36 =$ **1/6**

✎ WORKED EXAMPLE

Q 6.2. A card is drawn. P(king or red)?

Step 1 — Identify the individual probabilities.

- P(king) = 4/52; P(red) = 26/52; P(king ∩ red) = 2/52

Step 2 — Apply the addition rule.

- P(king ∪ red) = 4/52 + 26/52 − 2/52
- = 28/52
- = **7/13**

✎ WORKED EXAMPLE

Q 6.3. P(A) = 0.3, P(B) = 0.4, P(A ∩ B) = 0.12. Are A, B independent?

Step 1 — Test the independence criterion $P(A) \cdot P(B) \stackrel{?}{=} P(A \cap B)$.

- $0.3 \times 0.4 = 0.12 = P(A \cap B)$ ✓
- **Yes — independent.**

✎ WORKED EXAMPLE

Q 6.4. A bag has 5 white, 3 black balls. Two drawn without replacement. P(both white)?

Step 1 — P(1st white) = 5/8 (5 white out of 8 total).

Step 2 — P(2nd white | 1st white) = 4/7 (after removing one white).

Step 3 — Multiply (chain rule).

- $P = \frac{5}{8} \cdot \frac{4}{7} = \frac{20}{56}$ = **5/14**

✎ WORKED EXAMPLE

Q 6.5. P(A) = 0.4, P(B) = 0.3, P(A ∪ B) = 0.6. P(A ∩ B)?

Step 1 — Rearrange the addition rule: $P(A \cap B) = P(A) + P(B) - P(A \cup B)$.

- $= 0.4 + 0.3 - 0.6$
- = **0.1**

✎ WORKED EXAMPLE

Q 6.6. Toss 3 coins. P(at least 2 heads)?

Step 1 — Total outcomes = $2^3 = 8$.

Step 2 — Count outcomes with ≥ 2 heads.

- HHH, HHT, HTH, THH → **4 favourable**

Step 3 — Compute P.

- $P = 4/8 =$ **1/2**

✎ WORKED EXAMPLE

Q 6.7. Two events with $P(A) = 0.5, P(B) = 0.6, A, B$ mutually exclusive. P(A ∩ B)?

Soln. Definition of mutually exclusive: A and B cannot both happen ⇒ $P(A \cap B) = \mathbf{0}$. (Note: the values of $P(A), P(B)$ are irrelevant once we know they are ME.)

✎ **WORKED EXAMPLE**

Q 6.8. A box has 60% A-grade and 40% B-grade items. 5% of A and 8% of B are defective. An item picked at random is defective. Probability it was A-grade? **(Bayes')**

Step 1 — Compute the numerator: $P(A) \cdot P(D|A)$.

- $= 0.6 \times 0.05 = 0.03$

Step 2 — Compute the total-probability denominator $P(D)$.

- $P(D) = 0.6 \times 0.05 + 0.4 \times 0.08$
- $= 0.03 + 0.032$
- = **0.062**

Step 3 — Apply Bayes': $P(A|D) = \text{numerator}/P(D)$.

- $= 0.03/0.062$
- ≈ **0.484**

✎ **WORKED EXAMPLE**

Q 6.9. P(A) = 0.6, P(B|A) = 0.4. P(A ∩ B)?

Step 1 — Apply the multiplication theorem: $P(A \cap B) = P(A) \cdot P(B|A)$.

- $= 0.6 \times 0.4$
- = **0.24**

✎ **WORKED EXAMPLE**

Q 6.10. Three independent events with probabilities 0.5, 0.4, 0.3. P(all happen)?

Step 1 — For independent events, P(all) = product of individual probabilities.

- $P = 0.5 \times 0.4 \times 0.3$
- = **0.06**

✎ **WORKED EXAMPLE**

Q 6.11. Same as above, P(none happen)?

Step 1 — P(none) = product of (1 – p_i) for independent events.

- $P = (1 - 0.5)(1 - 0.4)(1 - 0.3)$
- $= 0.5 \times 0.6 \times 0.7$
- = **0.21**

✎ WORKED EXAMPLE

Q 6.12. Two cards drawn without replacement. P(both kings)?

Step 1 — P(1st king) = 4/52 (4 kings in 52 cards).

Step 2 — P(2nd king | 1st king) = 3/51 (after removing one king).

Step 3 — Multiply.

- $P = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652}$ = **1/221**

✎ WORKED EXAMPLE

Q 6.13. A fair die thrown. P(prime number)?

Step 1 — Identify primes in {1,…,6}: {2, 3, 5} → 3 favourable outcomes.

Step 2 — Compute P.

- $P = 3/6 =$ **1/2**

✎ WORKED EXAMPLE

Q 6.14. Two events are exhaustive iff:

Soln. Exhaustive ⇔ their union covers the entire sample space: $A \cup B = S$, i.e., $P(A \cup B) = 1$.

✎ WORKED EXAMPLE

Q 6.15. A box has 4 defective and 6 good bulbs. 3 drawn together. P(exactly 1 defective)?

Step 1 — Count favourable outcomes (1 defective, 2 good).

- Favourable = ${}^{4}C_1 \times {}^{6}C_2 = 4 \times 15 = 60$

Step 2 — Count total outcomes.

- Total = ${}^{10}C_3 = 120$

Step 3 — Compute the probability.

- $P = 60/120 = \mathbf{1/2}$

6.9 Trap-recognition card

Trap	Defence
"ME implies independent"	Almost never. They are usually opposite ideas.
"P(A ∪ B) = P(A) + P(B) always"	Only if ME. Else subtract overlap.
Replacement vs without replacement	Read carefully. Sample-without-replacement changes denominator.
"Face cards include aces"	They don't. Face cards = J, Q, K only.
Bayes' denominator missed	Don't divide by P(A
Confusing P(A	B) with P(B

6.10 Mini-mock

#	Q	Ans
1	Sum of 2 dice = 8. P?	5/36
2	Card = ace. P?	4/52 = 1/13
3	P(A)=0.5, P(B)=0.5, indep. P(A∩B)?	0.25
4	P(A)=0.5, P(B)=0.5, ME. P(A∪B)?	1.0
5	One coin tossed twice. P(both H)?	1/4
6	Box: 7 R, 3 G. 1 drawn. P(R)?	7/10
7	Two indep events: P(A)=0.4, P(B)=0.6. P(A or B)?	0.4+0.6−0.24 = 0.76
8	Bag has 3R, 4B. 2 drawn no rep. P(2R)?	3/7·2/6 = 1/7
9	P(A∩B) = 0, P(A) > 0, P(B) > 0. Indep?	No (mutually exclusive)
10	Three coins. P(at least 1 H)?	1 − 1/8 = 7/8

6.11 Active-recall prompts

1. State the addition theorem for two events (with and without ME).
2. State the multiplication theorem (with and without independence).
3. Write Bayes' theorem with all symbols.
4. Why are ME and independent typically incompatible?
5. Write $P(\text{exactly one of A or B})$.

CHAPTER 7 — Random Variables & Probability Distributions

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈10 Qs / paper) **Difficulty:** Medium-Hard. The 4 named distributions (Binomial, Poisson, Normal, Hypergeometric) cover every distribution Q. Memorise mean and variance for each.

7.0 — Understanding Random Variables & Distributions

✔ QUICK RECALL

What is a "random variable" and why do we need distributions?

In probability, the sample space Ω can contain abstract outcomes — {Head, Tail}, {Red, Blue, Green}, {Defective, Good}. To do mathematics, we need numbers. A **random variable** X is a function that maps each outcome to a real number. For a die roll: $X =$ "number shown" maps the outcome 3 to the number 3. Simple.

But why stop at assigning numbers? Because once we have numbers, we can ask: "How probable is each value?" This is the **probability distribution** — the probability law governing which values X takes and with what probability. For a fair die: $P(X = k) = 1/6$ for $k = 1, \dots, 6$ — a uniform discrete distribution.

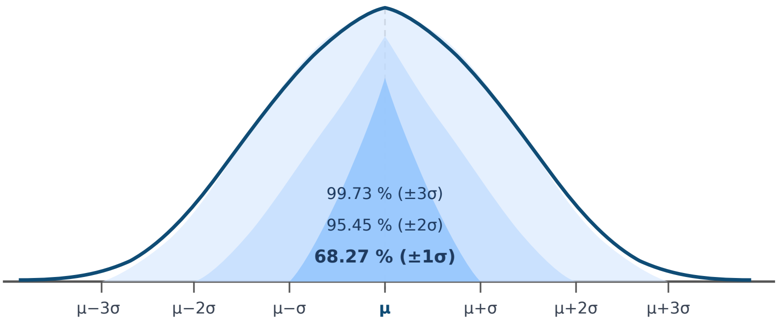
Why do specific named distributions arise? Because many different real-world phenomena follow the same mathematical pattern:

- **Binomial:** Whenever you repeat the same experiment (trial) n times independently, each with probability p of "success," the number of successes follows Binomial(n, p). Tossing a coin 10 times and counting heads. Inspecting 100 items and counting defectives. Testing 20 patients and counting who respond to treatment.
- **Poisson:** Whenever you count rare events in a fixed time or space interval — phone calls per minute to a helpdesk (mean λ calls/minute), typographical errors per page, cars arriving at a toll booth per hour. The Poisson distribution is the limiting case of Binomial when $n \to \infty$, $p \to 0$, $np = \lambda$ stays fixed.
- **Normal:** The bell curve. Its central role comes from the Central Limit Theorem: *no matter what distribution the original data follows, the distribution of the sample mean becomes approximately Normal for large enough sample sizes.* This is why the Normal distribution underpins all of Chapters 8–9.

The four named distributions — at a glance:

Distribution	Parameters	PMF / PDF	Mean	Variance	Key identifying feature
Binomial	n, p	$\binom{n}{r}p^r(1-p)^{n-r}$	np	$np(1-p) = npq$	Fixed n trials, constant p, count successes
Poisson	λ	$\frac{e^{-\lambda}\lambda^r}{r!}$	λ	λ	Rare events; mean = variance
Normal	μ, σ^2	$\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	Symmetric bell curve; 68-95-99.7 rule
Hypergeometric	N, K, n	$\frac{\binom{K}{r}\binom{N-K}{n-r}}{\binom{N}{n}}$	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{N^2(N-1)}$	Sampling without replacement from finite population

The Normal distribution — 68-95-99.7 rule (must know):



$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$
$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$
$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$

Solved Example 7.A — Binomial probability:

A biased coin has $P(\text{Heads}) = 0.4$. It is tossed 5 times. Find $P(X = 3)$.

Step 1 — Identify the parameters: $n = 5$, $p = 0.4$, $q = 0.6$, $r = 3$.

Step 2 — Apply the Binomial pmf: $P(X = r) = {}^nC_r \cdot p^r \cdot q^{n-r}$.

- $P(X = 3) = \binom{5}{3}(0.4)^3(0.6)^2$

Step 3 — Compute each factor.

- $\binom{5}{3} = 10$
- $(0.4)^3 = 0.064$
- $(0.6)^2 = 0.36$

Step 4 — Multiply.

- $= 10 \times 0.064 \times 0.36$
- $= 10 \times 0.02304$
- = **0.2304**

Solved Example 7.B — Poisson approximation to Binomial:

In a production run of 1000 items, each has $P(\text{defective}) = 0.003$. Find the probability that exactly 2 are defective.

Step 1 — Check the Poisson-approximation conditions: n **large,** p **small.**

- $n = 1000$ (large), $p = 0.003$ (small) ⇒ approximation valid.

Step 2 — Compute $\lambda = np$.

- $\lambda = 1000 \times 0.003 = 3$

Step 3 — Apply the Poisson pmf: $P(X = r) = e^{-\lambda}\lambda^r / r!$ **with** $r = 2$.

- $P(X = 2) = \dfrac{e^{-3} \cdot 3^2}{2!}$

Step 4 — Substitute $e^{-3} \approx 0.0498$, $3^2 = 9$, $2! = 2$.

- $= \dfrac{0.0498 \times 9}{2}$
- $= \dfrac{0.4482}{2}$
- = **0.2241**

Solved Example 7.C — Normal distribution (standardisation):

$X \sim N(50, 100)$ (mean = 50, variance = 100, so $\sigma = 10$). Find $P(X > 65)$.

Step 1 — Identify the SD.

- $\sigma = \sqrt{100} = 10$

Step 2 — Standardise: $Z = (X - \mu)/\sigma$.

- $Z = (65 - 50)/10$
- = **1.5**

Step 3 — Look up $\Phi(1.5)$ from the standard normal table.

- $\Phi(1.5) = 0.9332$

Step 4 — Compute the right-tail probability.

- $P(X > 65) = P(Z > 1.5) = 1 - \Phi(1.5)$
- = $1 - 0.9332$
- = **0.0668**

7.1 Examiner mindset

Angle	Pet question
Discrete vs continuous RV — pmf vs pdf	"Which is a valid pmf?"
Expectation, variance	Plug-in
Binomial: P(r successes)	direct
Poisson: P(r events), Mean = Var = λ	direct, often as approximation to Binomial
Normal: standardise via z	use the z-table fact base (68/95/99.7)
Hypergeometric	sampling without replacement
Properties of expectation, variance	linearity, independence

7.2 Random variable, pmf, pdf

Type	Defining function	Conditions
Discrete	pmf $p(x) = P(X = x)$	$p(x) \geq 0,\ \sum p(x) = 1$
Continuous	pdf $f(x)$	$f(x) \geq 0,\ \int_{-\infty}^{\infty} f(x)\,dx = 1$
Either	CDF $F(x) = P(X \leq x)$	non-decreasing, right-continuous, $F(-\infty) = 0, F(\infty) = 1$

7.3 Expectation and variance — formulas

Quantity	Discrete	Continuous
$E[X]$	$\sum x\,p(x)$	$\int xf(x)dx$
$E[g(X)]$	$\sum g(x)p(x)$	$\int g(x)f(x)dx$
$\mathrm{Var}(X)$	$E[X^2] - (E[X])^2$	same

Linearity: $E[aX + b] = aE[X] + b$. Always true (independence not required).

Variance scaling: $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$.

For independent X, Y: $E[XY] = E[X]E[Y]$ and $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

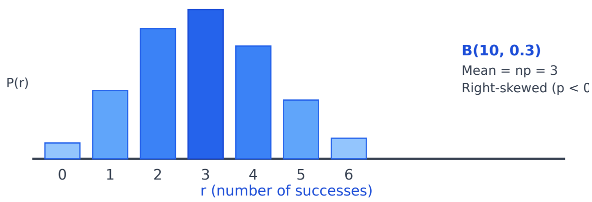
7.4 The four named distributions — master card

Binomial $B(n, p)$

Quantity	Formula
pmf	$P(X = r) = {}^nC_r\, p^r (1-p)^{n-r},\ r = 0, 1, \ldots, n$
Mean	np
Variance	$npq,\ q = 1 - p$
SD	$\sqrt{npq}$
Skewness	$(1-2p)/\sqrt{npq}$
Kurtosis (β_2)	$3 + (1-6pq)/(npq)$

When to use: n independent Bernoulli trials, fixed p, count of successes.

Binomial B(10, 0.3) — shape (right-skewed when p < 0.5):

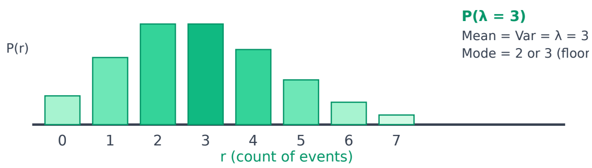


Poisson $P(\lambda)$

Quantity	Formula
pmf	$P(X = r) = \dfrac{e^{-\lambda}\lambda^r}{r!},\ r = 0, 1, 2, \ldots$
Mean	λ
Variance	λ
SD	$\sqrt{\lambda}$

When to use: Rare events in a fixed interval (calls per minute, defects per metre). Also approximates $B(n, p)$ when $n \to \infty, p \to 0, np = \lambda$.

Poisson P(λ = 3) — shape (right-skewed for small λ):



Normal $N(\mu, \sigma^2)$

Quantity	Formula
pdf	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$
Mean / Median / Mode	all = μ

Quantity	Formula
Variance	σ^2
Skewness	0
Kurtosis (β_2)	3 (mesokurtic)
Standardisation	$Z = (X - \mu)/\sigma \sim N(0, 1)$

The 68–95–99.7 rule:

Range	Probability
$\mu \pm \sigma$	0.6826 (≈ 68 %)
$\mu \pm 2\sigma$	0.9544 (≈ 95 %)
$\mu \pm 3\sigma$	0.9974 (≈ 99.7 %)

Common z-values to memorise: $z_{0.05} = 1.645$ (one-tail), $z_{0.025} = 1.96$ (two-tail 95%), $z_{0.005} = 2.58$ (two-tail 99%).

Hypergeometric $H(N, K, n)$

Sampling n items **without replacement** from a population of N containing K successes.

Quantity	Formula
pmf	$P(X = k) = \dfrac{{}^{K}C_k \cdot {}^{N-K}C_{n-k}}{{}^{N}C_n}$
Mean	nK/N
Variance	$n\dfrac{K}{N} \cdot \dfrac{N-K}{N} \cdot \dfrac{N-n}{N-1}$ (the last factor is the **finite-population correction**)

Uniform (continuous) $U(a, b)$

Quantity	Formula
pdf	$1/(b - a)$ on $[a, b]$, else 0
Mean	$(a + b)/2$
Variance	$(b - a)^2/12$

7.5 Recognising which distribution to use

Wording cue	Distribution
"n trials, fixed probability p per trial"	Binomial
"rate of λ per unit time/length"	Poisson
"approximately normally distributed", "symmetric, bell-shaped"	Normal
"without replacement, finite population"	Hypergeometric
"equally likely over an interval"	Uniform

7.6 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 7.1. Toss a fair coin 10 times. P(exactly 4 heads)?

Step 1 — Identify Binomial parameters: $n = 10$, $p = 0.5$, $r = 4$.

Step 2 — Apply pmf $P(X = r) = {}^nC_r\, p^r (1 - p)^{n-r}$.

- $P(X = 4) = {}^{10}C_4 (0.5)^4 (0.5)^6$
- $= 210 \times (0.5)^{10}$
- $= 210/1024$
- $\approx$ **0.205**

✎ WORKED EXAMPLE

Q 7.2. $X \sim B(8, 1/4)$. Mean and variance?

Step 1 — Compute the mean.

- Mean $= np = 8 \times 1/4$
- = **2**

Step 2 — Compute the variance.

- Var $= npq = 8 \times (1/4) \times (3/4)$
- $= 8 \times 3/16$
- = **1.5**

✎ WORKED EXAMPLE

Q 7.3. Number of defects per page is Poisson with mean 2. P(no defects on a page)?

Step 1 — Apply Poisson pmf with $\lambda = 2$, $r = 0$.

- $P(X = 0) = \dfrac{e^{-2} \cdot 2^0}{0!} = e^{-2}$

Step 2 — Numerical value.

- $e^{-2} \approx$ **0.135**

✎ WORKED EXAMPLE

Q 7.4. Calls arrive at rate 4/hour (Poisson). P(exactly 5 calls in 1 hour)?

Step 1 — Identify Poisson parameters: $\lambda = 4$, $r = 5$.

Step 2 — Apply pmf $P(X = r) = e^{-\lambda}\lambda^r / r!$.

- $P(X = 5) = \dfrac{e^{-4} \cdot 4^5}{5!}$

Step 3 — Substitute $e^{-4} \approx 0.0183$, $4^5 = 1024$, $5! = 120$.

- $= \dfrac{0.0183 \times 1024}{120}$
- $\approx$ **0.156**

✎ WORKED EXAMPLE

Q 7.5. $X \sim N(50, 100)$. P(40 < X < 60)?

Step 1 — Identify the standard deviation.

- Variance = 100, so $\sigma = \sqrt{100} = 10$

Step 2 — Standardise the bounds.

- $z_1 = (40 - 50)/10 = -1$
- $z_2 = (60 - 50)/10 = +1$

Step 3 — Apply the 68/95/99.7 rule.

- $P(-1 < Z < 1)$ = within ±1σ
- = **0.6826 (68.26 %)**

✎ WORKED EXAMPLE

Q 7.6. $X \sim N(20, 16)$. P(X > 28)?

Step 1 — Identify the standard deviation.

- Variance = 16, so $\sigma = \sqrt{16} = 4$

Step 2 — Standardise 28.

- $z = (28 - 20)/4 = 8/4$
- = **2**

Step 3 — Compute the tail probability.

- $P(X > 28) = P(Z > 2)$
- = $(1 - 0.9544)/2$
- = $0.0456/2$
- = **0.0228**

✎ WORKED EXAMPLE

Q 7.7. Binomial $n = 100, p = 0.02$. Approximate by Poisson with mean?

Step 1 — Poisson approximation uses $\lambda = np$.

- $\lambda = 100 \times 0.02 =$ **2**

✎ WORKED EXAMPLE

Q 7.8. Mean of $X \sim P(\lambda)$ is 9. SD?

Step 1 — For Poisson, Mean = Variance = λ.

- $\lambda = 9 \Rightarrow$ Variance = 9

Step 2 — Take the square root for SD.

- $\sigma = \sqrt{9} =$ **3**

✎ WORKED EXAMPLE

Q 7.9. Lot of 20 has 5 defectives. 4 picked without replacement. P(exactly 2 defective)?

Step 1 — Identify Hypergeometric parameters.

- $N = 20$ (lot size), $K = 5$ (defectives), $n = 4$ (drawn), $k = 2$

Step 2 — Apply Hypergeometric pmf: $P(X = k) = \dfrac{{}^{K}C_k \cdot {}^{N-K}C_{n-k}}{{}^{N}C_n}$.

- $P(X = 2) = \dfrac{{}^{5}C_2 \cdot {}^{15}C_2}{{}^{20}C_4}$

Step 3 — Compute each combination.

- ${}^{5}C_2 = 10$
- ${}^{15}C_2 = 105$
- ${}^{20}C_4 = 4845$

Step 4 — Substitute and simplify.

- $P = \dfrac{10 \times 105}{4845} = \dfrac{1050}{4845}$
- ≈ **0.217**

✎ WORKED EXAMPLE

Q 7.10. X uniform on [0, 10]. P(X > 7)?

Step 1 — For Uniform on [a, b], $P(X > x_0) = (b - x_0)/(b - a)$.

- $P(X > 7) = (10 - 7)/(10 - 0)$
- $= 3/10$
- = **0.3**

✎ WORKED EXAMPLE

Q 7.11. For a Binomial with mean = 8 and variance = 4, find n and p.

Step 1 — Set up the system of equations.

- Mean: $np = 8$
- Variance: $npq = 4$

Step 2 — Divide variance by mean to find q.

- $q = npq/np = 4/8$
- **= 0.5**

Step 3 — Find p and n.

- $p = 1 - q = 1 - 0.5 = \mathbf{0.5}$
- $n = 8/p = 8/0.5 = \mathbf{16}$

✎ WORKED EXAMPLE

Q 7.12. Mode of normal distribution = ?

Soln. The Normal pdf peaks at $x = \mu$, and by symmetry Mean = Median = Mode = μ.

✎ WORKED EXAMPLE

Q 7.13. $E[X] = 5, \mathrm{Var}(X) = 4.$ $E[2X + 3]$ and $\mathrm{Var}(2X + 3)$?

Step 1 — Compute E[2X+3] using linearity of expectation.

- $E[2X + 3] = 2 \cdot E[X] + 3$
- $= 2 \times 5 + 3$
- **= 13**

Step 2 — Compute Var(2X+3) using the scaling rule.

- $\mathrm{Var}(2X + 3) = 2^2 \cdot \mathrm{Var}(X)$
- $= 4 \times 4$
- **= 16**

✎ WORKED EXAMPLE

Q 7.14. Normal: 95% of data lies within how many σ?

Soln. The standard normal critical value for a two-tail 95 % region is **±1.96σ** (often rounded to ±2σ via the 68/95/99.7 rule).

✎ WORKED EXAMPLE

Q 7.15. A discrete RV X has pmf p(0) = 0.3, p(1) = 0.4, p(2) = 0.2, p(3) = c. Find c, mean, variance.

Step 1 — Find c (probabilities must sum to 1).

- $c = 1 - (0.3 + 0.4 + 0.2) = 1 - 0.9 = \mathbf{0.1}$

Step 2 — Compute E[X].

- $E[X] = 0(0.3) + 1(0.4) + 2(0.2) + 3(0.1)$
- $= 0 + 0.4 + 0.4 + 0.3 = \mathbf{1.1}$

Step 3 — Compute Var(X) via $E[X^2] - (E[X])^2$.

- $E[X^2] = 0^2(0.3) + 1^2(0.4) + 2^2(0.2) + 3^2(0.1)$
- $= 0 + 0.4 + 0.8 + 0.9 = 2.1$
- Var $= E[X^2] - (E[X])^2 = 2.1 - 1.1^2 = 2.1 - 1.21 = \mathbf{0.89}$

7.7 Trap-recognition card

Trap	Defence
Binomial mean = npq (wrong)	Mean = np. Variance = npq.
For Poisson, "mean ≠ variance"	They are equal: both = λ. That's the **identifying property**.
Normal density value vs probability	The pdf height is not a probability; areas under it are.
Forgot finite-population correction in hypergeometric	Multiply variance by (N − n)/(N − 1).
Mixing up z for one-tail and two-tail at 5%	One-tail: 1.645. Two-tail: 1.96.

7.8 Mini-mock

#	Q	Ans
1	Mean of Binomial(20, 0.3)?	6
2	Var of Binomial(20, 0.3)?	4.2
3	Mean of Poisson(λ=5)?	5
4	SD of Poisson(λ=9)?	3
5	Normal: P(	Z
6	If E[X]=10, E[2X−3]?	17
7	Var(3X) when Var(X)=4?	36
8	Mode of N(50,100)?	50
9	A pmf must have Σ p(x) = ?	1
10	Continuous uniform on [2,8]: variance?	$(8-2)^2/12 = 3$

7.9 Active-recall prompts

1. State the pmf, mean, variance of Binomial.
2. State the pmf, mean, variance of Poisson. Why mean = variance?

3. Write the standardising transformation for normal.

4. Write the 68/95/99.7 rule.

5. State the hypergeometric pmf and where the finite-population correction sits.

CHAPTER 8 — Sampling Theory

Importance: ⭐⭐⭐⭐ HIGH (≈8 Qs / paper) **Difficulty:** Easy–Medium. Most questions are conceptual / identification; only a few demand a formula.

8.0 — Understanding Sampling Theory from First Principles

✔ QUICK RECALL

Why sample? Why not just measure the whole population?

India has 1.4 billion people. To estimate average household income, a complete census would cost thousands of crores and take years to complete. Instead, the NSSO surveys about 100,000 households — and the result is almost as accurate as a census. How is this possible?

Two fundamental theorems guarantee it:

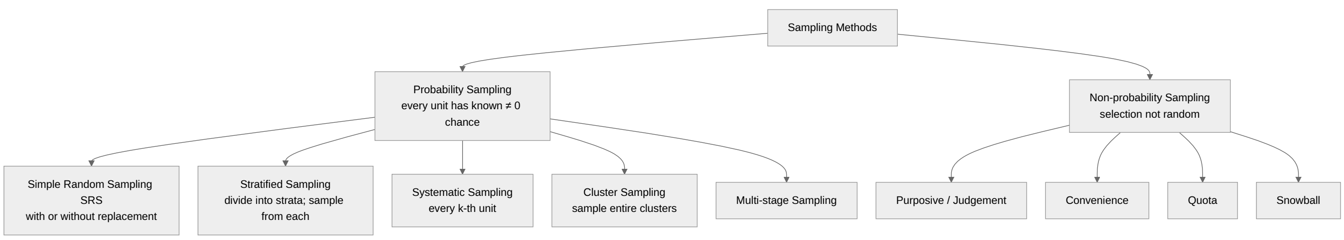
Law of Large Numbers: As sample size n increases, the sample mean $\bar{X}$ converges (in probability) to the population mean μ. The more people you survey, the closer your estimate is to the truth.

Central Limit Theorem (CLT): For any population (normal or not), the distribution of $\bar{X}$ across all possible samples of size n is approximately $N(\mu, \sigma^2/n)$ for large n. This is remarkable — no matter how weird the original population distribution is, sample means are bell-shaped. This is what makes confidence intervals and hypothesis tests work.

Standard Error (SE): The SD of the sampling distribution of $\bar{X}$ is $\text{SE} = \sigma/\sqrt{n}$. Note: SE decreases as $\sqrt{n}$. To halve the SE, you need 4× the sample size. This is why going from $n = 100$ to $n = 400$ is worth it, but going from $n = 10000$ to $n = 40000$ gives diminishing returns.

Population parameter (Greek letter: μ, σ, p) = true value in the entire population — usually unknown. **Sample statistic** (Roman letter: $\bar{X}$, s, $\hat{p}$) = computed from the sample — our estimate of the parameter.

Sampling methods — classification and when to use each:



Method	Best when	Advantage	Disadvantage
SRSWOR	Population is homogeneous, complete frame available	Simple; unbiased; theory clean	Expensive if spread geographically
Stratified	Population has distinct subgroups (strata)	More precise than SRS; ensures all strata represented	Need to know stratum sizes
Systematic	Ordered population, no periodic pattern	Simpler than SRS; good spread	Periodicity in population → biased
Cluster	Population naturally grouped; no complete frame	Cheaper when clusters are geographic	Less precise than SRS

Solved Example 8.A — SE calculation:

Population: $\mu = 80$, $\sigma = 20$. Random sample of $n = 100$. Find the SE of $\bar{X}$ and $P(78 < \bar{X} < 82)$.

Step 1 — Compute the Standard Error of the mean.

- $SE = \dfrac{\sigma}{\sqrt{n}}$
- $= \dfrac{20}{\sqrt{100}}$
- $= \dfrac{20}{10}$
- = **2**

Step 2 — Standardise the probability bounds.

- Lower $z = (78 - 80)/2 = -1$
- Upper $z = (82 - 80)/2 = +1$

Step 3 — Use the 68/95/99.7 rule.

- $P(78 < \bar{X} < 82) = P(-1 < Z < 1)$
- = **0.6827**

8.1 Examiner mindset

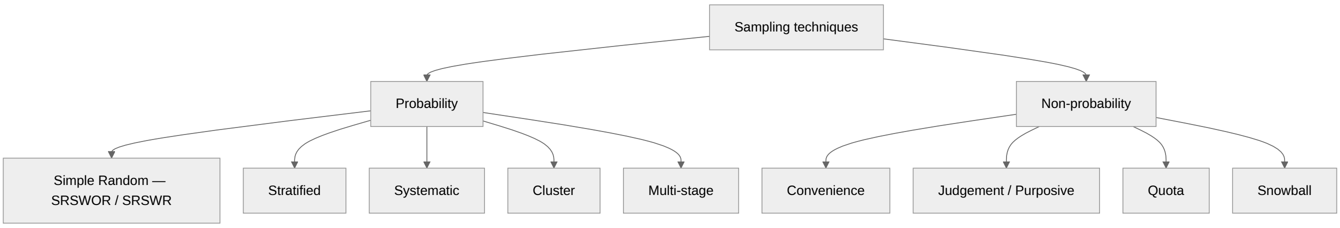
Angle	Pet question
Census vs sample	conceptual MCQ
Probability vs non-probability sampling	name the technique
Standard error, sampling distribution	formula plug-in
When to use which sampling design	"If population is heterogeneous in groups, what to use?" → stratified
Sample size determination	margin of error formula
Sampling vs non-sampling errors	classify the error

8.2 Population, sample, parameter, statistic

Term	Symbol	Description
Population	size N	Entire group of interest
Sample	size n	Subset chosen for study
Parameter	μ, σ, P	Numerical value of population
Statistic	$\bar{X}, s, p$	Numerical value of sample
Sampling		Process of drawing a sample
Sampling frame		List of all units in the population

Why sample? *Census is exhaustive but expensive, slow, often impossible (destructive testing, infinite/conceptual populations). A well-designed sample is fast, cheap, and provides quantifiable margin of error.*

8.3 Family tree of sampling techniques



8.4 Probability sampling — when to use what

Technique	Pick when ...	Key feature
Simple random (SRS)	Population is homogeneous, complete frame available	Each unit has equal chance
Stratified	Population has clear sub-groups (strata) with within-strata homogeneity	Sample drawn from each stratum; gains precision
Systematic	Population is ordered, cheap implementation needed	Pick every k-th unit; $k = N/n$
Cluster	Geographically dispersed population, no full frame	Population split into clusters; some clusters fully sampled
Multi-stage	Very large populations	Cluster + further sampling within (e.g., state → district → block → village)

Stratified vs cluster — examiners' favourite contrast:

	Stratified	Cluster
Purpose	Reduce sampling error	Reduce cost / handle no-frame
Within-group	Homogeneous	Heterogeneous (mini-population)
Between-group	Heterogeneous	Homogeneous
Sample drawn from	Every stratum	Selected clusters only

8.5 Non-probability sampling

Technique	Description	Use case
Convenience	Whoever's easiest to reach	Pilot studies
Judgement / purposive	Researcher picks "typical" units	Expert opinion polls
Quota	Fix counts per category, then pick conveniently	Market research
Snowball	Existing subjects refer next subjects	Hidden populations (drug users, rare diseases)

> ***Big disadvantage of non-probability:*** *No mathematical estimate of sampling error possible. Bias risk is high.*

8.6 Sampling distribution and standard error

The **sampling distribution** of a statistic is the distribution of its values across all possible samples of the same size.

Standard error of the mean

For a population with SD σ, sample of size n:

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{(infinite population)}$$

$$\mathrm{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}} \quad \text{(finite population, FPC)}$$

Standard error of a proportion

$$\mathrm{SE}(p) = \sqrt{\frac{P(1-P)}{n}}$$

Standard error of difference of means (independent samples)

$$\mathrm{SE}(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

8.7 Central Limit Theorem (CLT)

> *If $X_1, X_2, \ldots, X_n$ are i.i.d. with mean μ and finite variance σ^2, then for large n, $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$,* ***regardless of the original distribution****.*

Rule of thumb: $n \geq 30$ is usually "large enough".

This is the single most important theorem in sampling — it's why we can do Z-tests on means even when the underlying data is not normal.

8.8 Sampling vs non-sampling errors

Type	Cause	How to control
Sampling error	Random; you didn't see whole population	Larger n, better design
Non-sampling error	Frame errors, response errors, processing errors, non-response bias	Better questionnaire, training, follow-up

Key fact: Non-sampling error tends to **increase** with sample size (more data, more chances to mess up); sampling error **decreases**.

8.9 Sample-size determination (mean)

Required sample size for margin of error E at confidence level $1-\alpha$:

$$n = \left(\frac{z_{\alpha/2} \cdot \sigma}{E}\right)^2$$

For a proportion (worst-case $P = 0.5$):

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \cdot 0.25$$

8.10 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 8.1. Σ = 16, sample size 64. SE of mean (infinite pop)?

Step 1 — Apply the SE formula.

- $\mathrm{SE} = \sigma/\sqrt{n}$
- $= 16/\sqrt{64}$
- $= 16/8$
- = **2**

✎ WORKED EXAMPLE

Q 8.2. A frame of 1000 units, sample 100, σ = 20. SE with FPC?

Step 1 — Compute SE without FPC.

- $\mathrm{SE} = \sigma/\sqrt{n} = 20/\sqrt{100}$
- $= 20/10$
- = **2**

Step 2 — Compute the Finite Population Correction (FPC) factor.

- $\mathrm{FPC} = \sqrt{(N-n)/(N-1)}$
- $= \sqrt{(1000-100)/(1000-1)}$
- $= \sqrt{900/999}$
- ≈ 0.949

Step 3 — Apply the FPC to get the corrected SE.

- $\mathrm{SE\ with\ FPC} = 2 \times 0.949 \approx \mathbf{1.90}$

✎ WORKED EXAMPLE

Q 8.3. Population is geographically scattered with no list of households. Best sampling?

Soln. With no full frame and geographic dispersion, build clusters (e.g., villages/blocks) and sample within selected clusters → **Cluster sampling**.

✎ WORKED EXAMPLE

Q 8.4. A factory has machine A producing 70% of items and B producing 30%. To estimate overall defect rate, you sample 70% of items from A's output and 30% from B. This is:

Soln. Dividing into homogeneous sub-groups (machine A, machine B) and sampling each in proportion to size = **Stratified sampling, proportional allocation**.

✎ WORKED EXAMPLE

Q 8.5. From a list of 1000 employees, every 10th is selected. Sampling type?

Soln. Picking every k-th element from an ordered list is **systematic sampling** with $k = N/n = 1000/100 = 10$.

✎ WORKED EXAMPLE

Q 8.6. As n increases, SE of mean:

Soln. $\mathrm{SE} = \sigma/\sqrt{n}$ — proportional to $1/\sqrt{n}$. So SE **decreases** as n rises. (To halve SE you need 4× the sample.)

✎ WORKED EXAMPLE

Q 8.7. SE(p) when $n = 100, P = 0.4$?

Step 1 — Apply the SE formula for a proportion: $\mathrm{SE}(p) = \sqrt{P(1-P)/n}$.

- $\mathrm{SE}(p) = \sqrt{0.4 \times 0.6/100}$

Step 2 — Simplify.

- $= \sqrt{0.24/100}$
- $= \sqrt{0.0024}$
- ≈ **0.049**

✎ WORKED EXAMPLE

Q 8.8. σ known to be 30. Want margin 5 at 95% confidence. Sample size?

Step 1 — Use the sample-size formula: $n = (z_{\alpha/2} \cdot \sigma/E)^2$.

Step 2 — Substitute $z_{0.025} = 1.96, \sigma = 30, E = 5$.

- $n = (1.96 \times 30/5)^2$
- $= (58.8/5)^2$
- $= (11.76)^2$
- ≈ **138.3**

Step 3 — Round up to a whole number (always round up for sample size).

- $n =$ **139**

✎ WORKED EXAMPLE

Q 8.9. A non-probability technique relying on existing respondents to refer new ones is:

Soln. Existing subjects refer new subjects → **Snowball sampling**. Used for hidden / hard-to-reach populations.

✎ WORKED EXAMPLE

Q 8.10. As sample size increases, non-sampling error generally:

Soln. More data ⇒ more measurement/processing/non-response opportunities ⇒ **non-sampling error tends to increase**. (In contrast, sampling error falls as $1/\sqrt{n}$.)

8.11 Computation drills — the 7 problem types the exam sets

✎ WORKED EXAMPLE

CD-8.1. Standard error of the mean. Population SD σ = 12. Sample size n = 36. Find SE($\bar{X}$).

Step 1 — Apply $\mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}$**.**

- $= 12/\sqrt{36}$
- $= 12/6$
- **= 2**

Interpretation. The sample mean will fluctuate ± 2 units around the true mean on average.

✎ WORKED EXAMPLE

CD-8.2. Standard error of proportion. n = 400, sample proportion $\hat{p} = 0.35$. Find SE($\hat{p}$).

Step 1 — Apply $\mathrm{SE}(\hat{p}) = \sqrt{\hat{p}(1 - \hat{p})/n}$**.**

- $= \sqrt{0.35 \times 0.65/400}$
- $= \sqrt{0.2275/400}$
- $= \sqrt{0.00056875}$
- $\approx \mathbf{0.0239}$

✎ WORKED EXAMPLE

CD-8.3. 95 % confidence interval for the mean. $\bar{X} = 82$, σ = 10, n = 100. Construct 95% CI.

Step 1 — Compute SE.

- $\mathrm{SE} = 10/\sqrt{100} = 10/10 = 1$

Step 2 — Apply CI formula: $\bar{X} \pm Z_{0.025} \cdot \mathrm{SE}$**.**

- $Z_{0.025} = 1.96$ (memorise)
- Lower: $82 - 1.96 \times 1 = 82 - 1.96 = \mathbf{80.04}$
- Upper: $82 + 1.96 \times 1 = 82 + 1.96 = \mathbf{83.96}$

Ans: 95% CI = **(80.04, 83.96)**.

✎ WORKED EXAMPLE

CD-8.4. Sample size for estimating the mean. Margin of error e = 3, σ = 15, 95% confidence (Z = 1.96). Find n.

Step 1 — Apply $n = \left(\frac{Z\sigma}{e}\right)^2$**.**

- $= \left(\frac{1.96 \times 15}{3}\right)^2$
- $= (9.8)^2$
- $= \mathbf{96.04}$

Round up → n = 97 (always round UP for sample size).

✎ WORKED EXAMPLE

CD-8.5. Sample size for estimating a proportion. Desired margin ±0.04 at 95% confidence. P unknown (worst case). Find n.

Step 1 — Use worst-case P = 0.5 (maximises PQ = 0.25).

Step 2 — Apply $n = Z^2PQ/e^2$.

- $= (1.96)^2 \times 0.5 \times 0.5/(0.04)^2$
- $= 3.8416 \times 0.25/0.0016$
- $= 0.9604/0.0016$
- **= 600.25 → n = 601**

✎ WORKED EXAMPLE

CD-8.6. Finite population correction (FPC). Population N = 500, sample n = 100, σ = 20. Find corrected SE($\bar{X}$).

Step 1 — Compute the infinite-population SE.

- $SE_\infty = \sigma/\sqrt{n} = 20/\sqrt{100} = 2$

Step 2 — Apply FPC factor: $SE = SE_\infty \times \sqrt{(N-n)/(N-1)}$.

- $= 2 \times \sqrt{(500-100)/(500-1)}$
- $= 2 \times \sqrt{400/499}$
- $= 2 \times \sqrt{0.8016}$
- $= 2 \times 0.8953$
- $\approx$ **1.79** (smaller than uncorrected 2 — sampling 20% of pop gains precision)

Rule of thumb. Apply FPC when sample fraction $n/N > 0.05$ (sampling > 5% of population).

✎ WORKED EXAMPLE

CD-8.7. Stratified sampling — proportional allocation. Population: Stratum 1 (N_1 = 600), Stratum 2 (N_2 = 300), Stratum 3 (N_3 = 100). Total n = 200. Find n per stratum.

Step 1 — Total population N = 600 + 300 + 100 = 1000.

Step 2 — Proportional allocation: $n_i = n \times (N_i/N)$.

- $n_1 = 200 \times (600/1000) = \mathbf{120}$
- $n_2 = 200 \times (300/1000) = \mathbf{60}$
- $n_3 = 200 \times (100/1000) = \mathbf{20}$

Check: 120 + 60 + 20 = 200 ✓

8.12 Trap-recognition card

Trap	Defence
Quota called "probability"	It's non-probability — selection is by convenience within a quota.
SRSWR vs SRSWOR formula confusion	Without replacement uses FPC; with replacement does not.
"More sample size → no error"	Sampling error decreases; non-sampling error rises. Total can go either way.

Trap	Defence
Stratified ≠ cluster	Read mini-population vs sub-group cue.

8.12 Mini-mock

#	Q	Ans
1	A complete enumeration is called?	Census
2	List of all units in population?	Sampling frame
3	$n=400$, $\sigma=20$. SE of mean?	1
4	Strata are: a) homogeneous b) heterogeneous?	a (within); heterogeneous between
5	If $P=0.5$, $n=400$, SE(p)?	$\sqrt{(0.25/400)}=0.025$
6	Quota sampling is which type?	Non-probability
7	CLT applies for $n \geq$?	usually 30
8	Population SD 12, $n=144$. SE?	1
9	Method that picks every kth element?	Systematic
10	Formula for sample size at margin E?	$(z \cdot \sigma/E)^2$

8.13 Active-recall prompts

1. Difference between parameter and statistic.
2. State the Central Limit Theorem.
3. Write the SE formulas (mean, proportion, difference of means).
4. When do we use stratified vs cluster sampling?
5. Why does non-sampling error grow with n?

CHAPTER 9 — Statistical Inference (Estimation + Testing of Hypotheses)

Importance: ⭐⭐⭐⭐⭐ CRITICAL (≈12 Qs / paper) **Difficulty:** Medium-Hard. The biggest chapter in the paper. Master the **decision rule** template and you'll never lose marks here.

9.0 — Understanding Statistical Inference from First Principles

✔ QUICK RECALL

What is "statistical inference" and what is a hypothesis test actually doing?

Inference uses sample data to draw conclusions about the unknown population. Two tasks:

1. **Estimation:** "The sample mean is 75 kg. What is the population mean?" The answer is not just "75 kg" — it is a **confidence interval**: "We are 95% confident the population mean lies between 72.1 and 77.9 kg." This interval accounts for sampling variability.

2. **Hypothesis Testing:** "Someone claims the population mean is 80 kg. My sample gives 75 kg. Could this 5-unit gap just be chance sampling variation, or is it strong evidence against the claim?"

The null hypothesis framework — how to think about it:

Hypothesis testing works like a court trial: the defendant (null hypothesis H_0) is presumed innocent (true) until proven guilty beyond reasonable doubt. The "evidence" is your sample statistic. The "beyond reasonable doubt" threshold is your significance level α (usually 5%).

We compute: "If H_0 were true, how likely is it to observe a sample statistic this extreme or more extreme?" This probability is the **p-value**. If $p < \alpha$, the evidence against H_0 is strong enough to reject it.

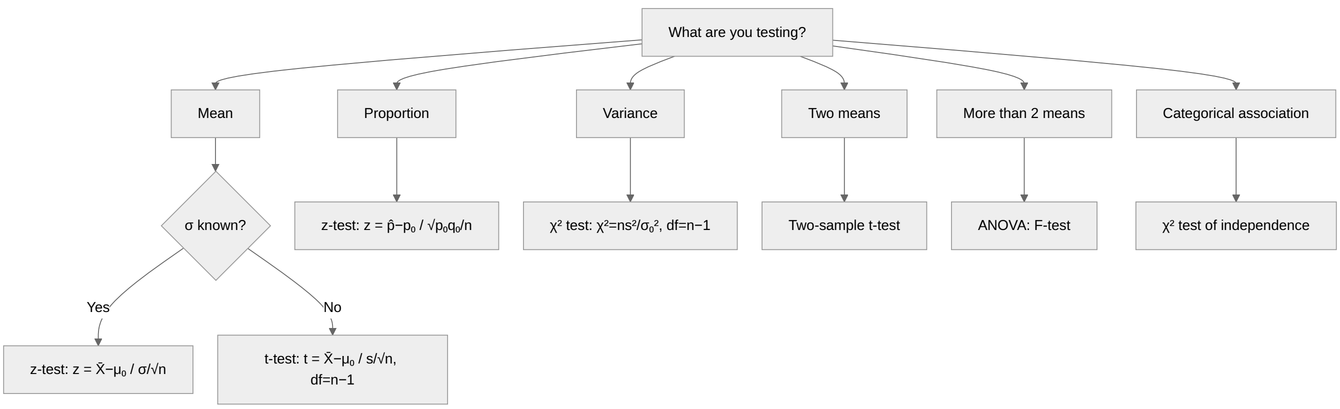
What p-value IS and IS NOT: - IS: $P(\text{data this extreme} \mid H_0 \text{ is true})$ — probability of the observed result assuming the null is true. - IS NOT: $P(H_0 \text{ is true})$ — the probability that the null hypothesis is true. This is a common misconception.

Type I vs Type II errors:

Decision	H_0 is actually TRUE	H_0 is actually FALSE
Reject H_0	Type I error (α) — false alarm	Correct — detected real effect
Fail to reject H_0	Correct — no false alarm	Type II error (β) — missed detection

Power of test = $1-\beta$ = probability of correctly detecting a real effect. Increasing sample size increases power.

Test-selection flowchart:



◆ FORMULA / KEY POINTS

Critical value decision rule (works for every test):

$$\text{Reject } H_0 \text{ if } |\text{test statistic}| > \text{critical value at } \alpha$$

For z-test at $\alpha = 5\%$: critical value = **1.96** (two-tail), **1.645** (one-tail). For t-test: read from t-table at $df = n - 1$. For χ^2 test: read from χ^2 table at appropriate df.

Solved Example 9.A — z-test for mean (σ known):

A company claims its bulbs last on average 1000 hours. A sample of 64 bulbs has $\bar{X} = 980$ hours. Population SD $\sigma = 80$ hours. Test at 5% significance (two-tail).

$H_0: \mu = 1000; H_1: \mu \neq 1000.$

Step 1 — Compute the standard error of the mean.

- $\text{SE} = \sigma/\sqrt{n}$
- $= 80/\sqrt{64}$
- $= 80/8$
- = **10**

Step 2 — Compute the z-statistic.

- $z = (\bar{X} - \mu_0)/\text{SE}$
- $= (980 - 1000)/10$
- $= -20/10$
- = **-2.0**

Step 3 — Compare to critical value and decide.

- Critical value at 5% two-tail: ± 1.96
- $|z| = 2.0 > 1.96$ → **Reject H_0**. Evidence suggests bulbs last less than 1000 hours.

Solved Example 9.B — χ² test of independence:

Survey data: 100 males (60 prefer product A, 40 prefer B), 80 females (30 prefer A, 50 prefer B). Test independence of gender and product preference at 5%.

Step 1 — State the hypotheses.

- H_0: gender and preference are **independent**
- H_1: they are **not independent**

Step 2 — Compute row, column and grand totals.

- Row totals: Males = 100, Females = 80
- Column totals: A = 90, B = 90
- Grand total N = 180

Step 3 — Compute expected frequencies: $E_{ij} = (\text{Row}_i \times \text{Col}_j)/N$.

- $E(\text{Male, A}) = 100 \times 90/180 = 50$
- $E(\text{Male, B}) = 100 \times 90/180 = 50$
- $E(\text{Female, A}) = 80 \times 90/180 = 40$
- $E(\text{Female, B}) = 80 \times 90/180 = 40$

Step 4 — Compute $\chi^2 = \sum(O - E)^2/E$ over the four cells.

- $(60 - 50)^2/50 = 100/50 = 2$
- $(40 - 50)^2/50 = 100/50 = 2$
- $(30 - 40)^2/40 = 100/40 = 2.5$
- $(50 - 40)^2/40 = 100/40 = 2.5$
- $\chi^2 = 2 + 2 + 2.5 + 2.5 =$ **9.0**

Step 5 — Find df and the critical value.

- $df = (r - 1)(c - 1) = 1 \times 1 = 1$
- $\chi^2_{0.05,1}$ = **3.84**

Step 6 — Decide.

- $9.0 > 3.84$ → **Reject H_0**. Gender and product preference are NOT independent.

9.1 Examiner mindset

Angle	Pet question
Properties of estimators	Unbiased, consistent, efficient, sufficient
Methods of estimation	MLE, MoM, Least squares — definition
Confidence intervals	Mean (σ known/unknown), proportion
Type I vs Type II error	Definition + symbols α and β
Z-test, t-test, χ^2-test, F-test	Pick the right test, state critical region
Power and OC curve	Concept

9.2 Estimation — point vs interval

Type	What you give	Example
Point estimate	A single number	$\hat{\mu} = \bar{X} = 50.2$
Interval estimate	A range with confidence level	"We're 95% confident μ ∈ [48.3, 52.1]"

9.3 Properties of a good estimator

Property	Meaning	Symbolically
Unbiasedness	Expected value of estimator = parameter	$E[\hat{\theta}] = \theta$
Consistency	Probability of being close to true value → 1 as n → ∞	$\hat{\theta} \xrightarrow{P} \theta$
Efficiency	Among unbiased estimators, smallest variance	min Var
Sufficiency	Estimator captures all sample information about θ	factorisation theorem

Unbiased estimators to memorise:

Parameter	Unbiased estimator
μ	$\bar{X}$
Population variance σ^2	sample variance with $n-1$ in denominator: $s^2 = \dfrac{\sum(X_i - \bar{X})^2}{n-1}$
Population proportion P	sample proportion $p = X/n$

Watch out: $\bar{X}^2$ is **not** an unbiased estimator of μ^2. And s (the SD) is a slightly biased estimator of σ even though s^2 is unbiased for σ^2.

9.4 Methods of estimation

Method	One-line description
Maximum Likelihood (MLE)	Pick θ that maximises the likelihood $L(\theta) = \prod f(x_i; \theta)$. Often most efficient asymptotically.
Method of Moments (MoM)	Equate sample moments to population moments and solve for θ. Easy but not always most efficient.
Least Squares (LS)	Minimise $\sum(X_i - \hat{\mu})^2$. Used in regression and ANOVA.

9.5 Confidence interval for the mean

σ known	σ unknown, n large (≥30)	σ unknown, n small
$\bar{X} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$	$\bar{X} \pm z_{\alpha/2}\dfrac{s}{\sqrt{n}}$	$\bar{X} \pm t_{\alpha/2,\, n-1}\dfrac{s}{\sqrt{n}}$

Common multipliers:

Confidence	$z_{\alpha/2}$
90 %	1.645
95 %	1.960
99 %	2.576

9.6 Hypothesis testing — the universal template

Step 1. State H_0 (null) and H_1 (alternative).

Step 2. Pick test statistic (Z, t, χ^2, F) based on what's being tested and what's known.

Step 3. Decide level of significance α (usually 5%).

Step 4. Find critical region (one-tail or two-tail).

Step 5. Compute test statistic from sample.

Step 6. Reject H_0 iff test stat falls in critical region. Otherwise fail to reject.

9.7 Type I and Type II errors

Decision \ Reality	H_0 true	H_0 false
Reject H_0	**Type I error**, prob = α	Correct (power = 1 − β)
Fail to reject	Correct	**Type II error**, prob = β

Memory hook: "α: rejecting a true H_0 (false alarm)." "β: keeping a false H_0 (missed alarm)."

9.8 The four big tests

(i) Z-test (large sample, σ known or n ≥ 30)

Hypothesis	Test stat
$H_0 : \mu = \mu_0$	$Z = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
Two means	$Z = \dfrac{\bar{X}_1 - \bar{X}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$
Proportion	$Z = \dfrac{p - P_0}{\sqrt{P_0(1 - P_0)/n}}$
Two proportions	$Z = \dfrac{p_1 - p_2}{\sqrt{\hat{P}(1 - \hat{P})(1/n_1 + 1/n_2)}}$, $\hat{P} = (X_1 + X_2)/(n_1 + n_2)$

(ii) t-test (small sample, σ unknown)

Hypothesis	Test stat	df
One mean, $H_0 : \mu = \mu_0$	$t = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}$	$n - 1$
Two means (independent, equal variances)	$t = \dfrac{\bar{X}_1 - \bar{X}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$, $s_p^2 = \dfrac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$	$n_1 + n_2 - 2$
Paired	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$	$n - 1$

(iii) Chi-square test

Application	Test statistic
Goodness of fit	$\chi^2 = \sum \dfrac{(O - E)^2}{E}$, df = (categories − 1 − parameters estimated)
Independence ($r \times c$ table)	same formula, df = $(r - 1)(c - 1)$

Application	Test statistic
Test of variance, $H_0 : \sigma^2 = \sigma_0^2$	$\chi^2 = (n-1)s^2/\sigma_0^2$, df = $n-1$

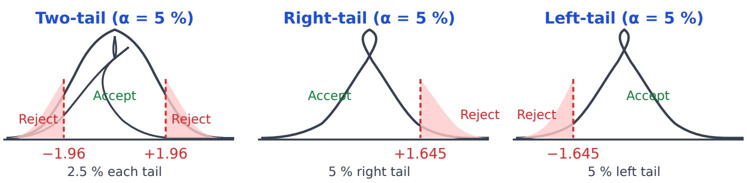
(iv) F-test

Application	Test statistic
Equality of two variances	$F = s_1^2/s_2^2$ (larger over smaller), df = $(n_1 - 1, n_2 - 1)$
ANOVA (Chapter 10)	$F = MSB/MSW$

9.9 Critical-region cheat sheet (5% significance)

Tail	Test	Reject H_0 if ...
Two-tail	Z	$\lvert Z\rvert > 1.96$
Right-tail	Z	$Z > 1.645$
Left-tail	Z	$Z < -1.645$
Two-tail	t	$\lvert t\rvert > t_{0.025,df}$ — table
Right-tail	χ²	$\chi^2 > \chi^2_{0.05,df}$ — table
F	upper tail	$F > F_{0.05,df_1,df_2}$ — table

Rejection zones — where to shade:



Critical values to memorise

α = 5 %: Two-tail Z = ±1.96 One-tail Z = ±1.645
α = 1 %: Two-tail Z = ±2.576 One-tail Z = ±2.326
α = 10%: Two-tail Z = ±1.645 One-tail Z = ±1.282
Rule: for α=5%, always check ±1.96 (two-tail) or ±1.645 (one-tail) first.

9.10 Power, level, OC curve

Quantity	Symbol	Meaning
Level of significance	α	Max tolerable Type I error
Power of test	1 − β	Probability of rejecting a false H_0
OC curve	β as a function of true parameter	shows test's discrimination
Best test	Maximises power for given α	Neyman-Pearson lemma

A "more powerful test" = lower β for the same α. Powerful tests are usually **one-tailed** (when alternative direction is known).

9.11 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 9.1. Sample of n=64 from a population with σ=8 gave $\bar{X} = 51$. Test $H_0 : \mu = 50$ at α = 5% (two-tail).

Step 1 — Compute the standard error.

- $\text{SE} = \sigma/\sqrt{n}$
- = $8/\sqrt{64}$
- = $8/8$
- **= 1**

Step 2 — Compute the Z-statistic.

- $Z = (\bar{X} - \mu_0)/\text{SE}$
- = $(51 - 50)/1$
- = $1/1$
- **= 1.0**

Step 3 — Compare to critical value.

- $|Z| = 1.0 < 1.96 \rightarrow$ **fail to reject** H_0.

✎ WORKED EXAMPLE

Q 9.2. $n = 25, \bar{X} = 102, s = 10$, test $H_0 : \mu = 100$ at α = 5% (two-tail).

Step 1 — Compute the standard error.

- $\text{SE} = s/\sqrt{n}$
- = $10/\sqrt{25}$
- = $10/5$
- **= 2**

Step 2 — Compute the t-statistic.

- $t = (\bar{X} - \mu_0)/\text{SE}$
- = $(102 - 100)/2$
- = $2/2$
- **= 1.0**

Step 3 — Compare to critical value (df = n – 1 = 24).

- $t_{0.025,\,24} \approx 2.064$
- $1.0 < 2.064 \rightarrow$ **fail to reject** H_0.

✎ WORKED EXAMPLE

Q 9.3. $n = 100, p = 0.45$. Test $H_0 : P = 0.5$ at α = 5% two-tail.

Step 1 — Compute the SE under H_0: $\sqrt{P_0(1 - P_0)/n}$.

- $\text{SE} = \sqrt{0.5 \times 0.5/100}$
- $= \sqrt{0.0025}$
- = **0.05**

Step 2 — Compute the Z-statistic.

- $Z = (p - P_0)/\text{SE}$
- $= (0.45 - 0.5)/0.05$
- $= -0.05/0.05$
- = **−1.0**

Step 3 — Compare to critical value.

- $|Z| = 1.0 < 1.96$ → **fail to reject** H_0 (no significant departure from $P = 0.5$).

✎ WORKED EXAMPLE

Q 9.4. $s_1^2 = 25, s_2^2 = 16, n_1 = n_2 = 11$. Test equality of variances at 5%.

Step 1 — Compute the F-statistic (larger variance over smaller).

- $F = s_1^2/s_2^2$
- $= 25/16$
- = **1.5625**

Step 2 — Compare to critical value (df = 10, 10).

- $F_{0.05,\,10,\,10} \approx 2.97$
- $1.5625 < 2.97$ → **fail to reject** H_0 (variances not significantly different).

✎ WORKED EXAMPLE

Q 9.5. A goodness-of-fit χ² = 12.6 with df = 5. At α = 5% ($\chi^2_{crit} = 11.07$), decision?

Step 1 — Compare test statistic to critical value.

- $\chi^2 = 12.6 > 11.07 = \chi^2_{0.05,5}$

Step 2 — Decision. → **Reject** H_0 (data does not fit the proposed distribution).

✎ WORKED EXAMPLE

Q 9.6. Type I error is the probability of:

Soln. Rejecting a true H_0 (false alarm). Symbol: α. Distinguish from Type II (β = failing to reject a false H_0).

✎ **WORKED EXAMPLE**

Q 9.7. Power of a test = ?

Soln. Power = $P(\text{reject } H_0 \mid H_0 \text{ false})$ = **1 – β**.

✎ **WORKED EXAMPLE**

Q 9.8. 95% CI for mean if $\bar{X} = 50, s = 4, n = 64$:

Step 1 — Apply the CI formula: $\bar{X} \pm z_{\alpha/2} \cdot s/\sqrt{n}$. With $n = 64$ large, use $z_{0.025} = 1.96$.

- $\text{CI} = 50 \pm 1.96 \times \dfrac{4}{\sqrt{64}}$

Step 2 — Compute the SE.

- $s/\sqrt{n} = 4/8 = 0.5$

Step 3 — Compute the margin and the interval.

- Margin = $1.96 \times 0.5 = 0.98$
- CI = 50 ± 0.98
- = **[49.02, 50.98]**

✎ **WORKED EXAMPLE**

Q 9.9. Why use $n - 1$ (not n) for sample variance?

Soln. Bessel's correction: using $n - 1$ in the denominator makes $E[s^2] = \sigma^2 \Rightarrow s^2$ becomes an **unbiased estimator** of population variance.

✎ **WORKED EXAMPLE**

Q 9.10. A test for independence in a 4×3 table has df:

Step 1 — For an $r \times c$ contingency table, df = $(r - 1)(c - 1)$.

- df = $(4 - 1)(3 - 1) = 3 \times 2$
- **= 6**

✎ **WORKED EXAMPLE**

Q 9.11. $\bar{X}_1 = 50, \bar{X}_2 = 47, \sigma_1 = \sigma_2 = 5, n_1 = n_2 = 100$. Test $H_0 : \mu_1 = \mu_2$ two-tail.

Step 1 — Compute the SE of the difference of means.

- $\mathrm{SE} = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$
- $= \sqrt{25/100 + 25/100}$
- $= \sqrt{0.25 + 0.25}$
- $= \sqrt{0.5}$
- ≈ 0.707

Step 2 — Compute the Z-statistic.

- $Z = (\bar{X}_1 - \bar{X}_2)/\mathrm{SE}$
- $= (50 - 47)/0.707$
- $= 3/0.707$
- $\approx \mathbf{4.24}$

Step 3 — Compare to critical value.

- $4.24 > 1.96 \rightarrow$ **reject** H_0 (means are significantly different).

✎ **WORKED EXAMPLE**

Q 9.12. MLE of p for Binomial sample X successes in n trials?

Soln. Maximising the Binomial likelihood gives $\hat{p} = X/n$ — the sample proportion. (Same answer also arises from Method of Moments and is unbiased.)

✎ **WORKED EXAMPLE**

Q 9.13. $\bar{X}^2$ is an unbiased estimator of μ^2?

Soln. No. $E[\bar{X}^2] = (E[\bar{X}])^2 + \mathrm{Var}(\bar{X}) = \mu^2 + \sigma^2/n \neq \mu^2$. The bias $= \sigma^2/n \rightarrow 0$ as $n \rightarrow \infty$, so it is **consistent but not unbiased**.

✎ **WORKED EXAMPLE**

Q 9.14. Best critical region is given by the:

Soln. The **Neyman–Pearson lemma**: among all tests of size α, the likelihood-ratio test maximises power.

✎ **WORKED EXAMPLE**

Q 9.15. Reduce α from 5% to 1%. β:

Soln. Tightening α (harder to reject) ⇒ more failures to reject false H_0 ⇒ **β increases**. (There's an α–β trade-off; for fixed n, you can lower one only by raising the other.)

9.12 Trap-recognition card

Trap	Defence
Use Z when n is small AND σ unknown	Wrong — must use t.
Use t with σ known	Use Z.
Test of two variances using Z	No — use F.
df for χ^2 independence as $rc - 1$	Correct: $(r-1)(c-1)$.
α and power add to 1	No: α + power is meaningless. β + power = 1.
MLE always unbiased	False. MLEs are often biased but consistent.

9.13 Mini-mock

#	Q	Ans
1	Type II error symbol?	β
2	Power formula?	1 – β
3	df for one-sample t with n=20?	19
4	Sample variance divisor for unbiasedness?	n−1
5	F-test compares?	two variances
6	χ^2 goodness-of-fit formula?	Σ (O−E)²/E
7	Test for two means, σ known, large n?	Z
8	At 1% two-tail,	Z
9	Sample mean is an unbiased estimator of?	μ
10	MLE of μ for Normal sample?	$\bar{X}$

9.14 Active-recall prompts

1. Define unbiased, consistent, efficient, sufficient.
2. State Type I and Type II error symbolically.
3. Write the test statistic for Z-test of one mean.
4. Write the χ^2 formula for goodness-of-fit.
5. State the Neyman-Pearson lemma in one line.
6. Write the formula for pooled variance s_p^2 in two-sample t.

CHAPTER 10 — Analysis of Variance (ANOVA)

Importance: ⭐⭐⭐ MEDIUM (≈4 Qs / paper) **Difficulty:** Medium. Mostly one or two conceptual questions plus an ANOVA-table fill-in. Don't over-invest, but never skip.

10.0 — Understanding ANOVA from First Principles

✔ QUICK RECALL

Why ANOVA? Why not just do multiple t-tests?

Suppose you test four fertilisers (A, B, C, D) on crop yield, with 5 plots each. You want to know if any fertiliser makes a significant difference. You could run six pairwise t-tests (A vs B, A vs C, A vs D, B vs C, B vs D, C vs D). Each test has a 5% chance of a false alarm. With six tests, the probability of at least one false alarm = $1 - 0.95^6 \approx 26\%$. You'd be making false discoveries 1 in 4 times — useless science.

ANOVA (Analysis of Variance) tests all groups simultaneously with a *single* F-test, keeping the overall false-alarm rate at exactly 5%.

The key insight — why "variance" tests "means": If all group means are equal (null hypothesis is true), variation within each group (due to random noise) should be about the same as variation between group means. If one fertiliser is genuinely better, the between-group variation will be large relative to within-group variation. The **F-ratio** captures this:

$$F = \frac{\text{Variance between groups (MSB)}}{\text{Variance within groups (MSW or MSE)}}$$

When H_0 is true: $F \approx 1$. When H_0 is false: $F \gg 1$. We reject H_0 when F exceeds the critical value from the F-table.

One-way ANOVA table structure (fill-in-the-blank type in exams):

Source	SS	df	MS = SS/df	F
Between groups (Treatment)	SSB	$k - 1$	MSB	MSB/MSW
Within groups (Error)	SSW	$N - k$	MSW	
Total	SST	$N - 1$		

Where k = number of groups, N = total observations, $\text{SST} = \text{SSB} + \text{SSW}$.

Solved Example 10.A — ANOVA computation:

Three teaching methods, 4 students each ($k = 3$, $N = 12$). SSB = 60, SST = 140.

Step 1 — Find SSW by subtraction.

- SSW = SST - SSB = 140 - 60 = **80**

Step 2 — Compute degrees of freedom.

- $df_B = k - 1 = 3 - 1 = 2$
- $df_W = N - k = 12 - 3 = 9$
- $df_T = N - 1 = 12 - 1 = 11$

Step 3 — Compute mean squares.

- $\mathrm{MSB} = \mathrm{SSB}/df_B = 60/2 = \mathbf{30}$
- $\mathrm{MSW} = \mathrm{SSW}/df_W = 80/9 \approx \mathbf{8.89}$

Step 4 — Compute F and compare to critical value.

- $F = \mathrm{MSB}/\mathrm{MSW} = 30/8.89 \approx \mathbf{3.37}$
- $F_{0.05}(2, 9) = 4.26$ from table
- $3.37 < 4.26$ → **do not reject H_0** — no significant difference between methods

10.1 Examiner mindset

Angle	Pet question
What ANOVA tests	"ANOVA tests equality of ...?" → means
Assumptions	normality, equal variance, independence
One-way layout	TSS = SSB + SSW
Two-way layout	TSS = SSR + SSC + SSE
ANOVA table fill-in	df, MS, F
Decision rule	F > F_crit → reject H_0

10.2 The big idea

***ANOVA tests whether several population means are equal**, by comparing the variance **between** groups to the variance **within** groups.*

If between-group variation dwarfs within-group variation, the means are unlikely to be equal.

10.3 One-way ANOVA — formulas

k treatments (groups), n_i observations in group i, total $N = \sum n_i$. Let T_i = total of group i, G = grand total.

Source	SS	df	MS
Between treatments	$\mathrm{SSB} = \sum \frac{T_i^2}{n_i} - \frac{G^2}{N}$	$k - 1$	$\mathrm{MSB} = \mathrm{SSB}/(k-1)$
Within (error)	$\mathrm{SSW} = \mathrm{TSS} - \mathrm{SSB}$	$N - k$	$\mathrm{MSW} = \mathrm{SSW}/(N-k)$
Total	$\mathrm{TSS} = \sum X^2 - \frac{G^2}{N}$	$N - 1$	

Test statistic: $F = \frac{\mathrm{MSB}}{\mathrm{MSW}}$, df = $(k-1,\ N-k)$.

10.4 Two-way ANOVA (without replication) — formulas

r rows (treatments), c columns (blocks), $N = rc$.

Source	SS	df
Rows	$\sum R_i^2/c - G^2/N$	$r-1$
Columns	$\sum C_j^2/r - G^2/N$	$c-1$
Error	TSS − SSR − SSC	$(r-1)(c-1)$
Total	$\sum X^2 - G^2/N$	$rc-1$

Two F-tests: $F_R = \text{MSR}/\text{MSE}$ (rows), $F_C = \text{MSC}/\text{MSE}$ (columns).

10.5 Assumptions of ANOVA

1. Observations independent across groups.
2. Each group's observations are normally distributed.
3. Variances are equal across groups (homoscedasticity).

10.6 Worked PYQ-style examples

✎ **WORKED EXAMPLE**

Q 10.1. ANOVA primarily tests:

Soln. ANOVA tests **equality of means** across several populations. (It uses the variance ratio F as the tool, but the hypothesis is about means.)

✎ **WORKED EXAMPLE**

Q 10.2. $k = 4$ treatments, $N = 24$. What is df for between and within?

Step 1 — Apply the df rules: $df_B = k - 1$, $df_W = N - k$.

- $df_B = 4 - 1 = \mathbf{3}$
- $df_W = 24 - 4 = \mathbf{20}$

✎ **WORKED EXAMPLE**

Q 10.3. SSB = 60, SSW = 80, $k = 4$, $N = 24$. F?

Step 1 — Compute MSB (Mean Square Between).

- $df_B = k - 1 = 4 - 1 = 3$
- $\text{MSB} = \text{SSB}/df_B = 60/3 = \mathbf{20}$

Step 2 — Compute MSW (Mean Square Within).

- $df_W = N - k = 24 - 4 = 20$
- $\text{MSW} = \text{SSW}/df_W = 80/20 = \mathbf{4}$

Step 3 — Compute the F-statistic.

- $F = \text{MSB}/\text{MSW} = 20/4 = \mathbf{5.0}$

✎ **WORKED EXAMPLE**

Q 10.4. Two-way layout 3 rows × 4 columns. df for error?

Step 1 — In two-way (no replication), df_error = $(r-1)(c-1)$.

- $df_E = (3-1)(4-1) = 2 \times 3$
- = **6**

✎ **WORKED EXAMPLE**

Q 10.5. TSS = 200, SSB = 50. SSW?

Step 1 — Use the SS decomposition: TSS = SSB + SSW.

- SSW = TSS − SSB
- = $200 - 50$
- = **150**

✎ **WORKED EXAMPLE**

Q 10.6. ANOVA assumes homo-scedasticity, which means:

Soln. Homoscedasticity = **equal variances** across all groups. (The other two ANOVA assumptions: independence and within-group normality.)

✎ **WORKED EXAMPLE**

Q 10.7. F is significant at 5% if F > F_table. The implication:

Soln. A significant F says **at least one** group mean differs from the rest — but ANOVA is omnibus and **doesn't tell which** pair. Need post-hoc tests (Tukey, Scheffé) to pinpoint the differing pair(s).

✎ **WORKED EXAMPLE**

Q 10.8. A randomized block design corresponds to which ANOVA?

Soln. RBD has one treatment factor + one blocking factor, modelled as **two-way ANOVA** (without replication).

10.7 Computation drills — the full ANOVA table from scratch

The single most-tested ANOVA skill: build the table from raw data. Every number you write should be derivable from this template. Practise until you can fill any ANOVA table in under 3 minutes.

✎ WORKED EXAMPLE

CD-10.1. One-way ANOVA — complete worked example.

Three fertilisers (treatments) tested on 4 plots each. Yield data (kg):

Fertiliser A	6	8	7	9
Fertiliser B	10	12	11	13
Fertiliser C	5	4	6	5

Step 1 — Compute group totals and grand total.

- $T_A = 6 + 8 + 7 + 9 = 30 \to \bar{A} = 7.5$
- $T_B = 10 + 12 + 11 + 13 = 46 \to \bar{B} = 11.5$
- $T_C = 5 + 4 + 6 + 5 = 20 \to \bar{C} = 5.0$
- Grand total $T = 30 + 46 + 20 = 96$, $n = 12$, $\bar{\bar{X}} = 96/12 = 8.0$

Step 2 — Compute Correction Factor (CF).

- $\mathrm{CF} = T^2/n = 96^2/12 = 9216/12 = 768$

Step 3 — Compute Total SS (SST).

- $\mathrm{SST} = \sum x_{ij}^2 - \mathrm{CF}$
- $= (6^2 + 8^2 + 7^2 + 9^2 + 10^2 + 12^2 + 11^2 + 13^2 + 5^2 + 4^2 + 6^2 + 5^2) - 768$
- $= (36 + 64 + 49 + 81 + 100 + 144 + 121 + 169 + 25 + 16 + 36 + 25) - 768$
- $= 866 - 768 = \mathbf{98}$

Step 4 — Compute Between-group SS (SSB).

- $\mathrm{SSB} = \sum_j T_j^2/n_j - \mathrm{CF}$
- $= 30^2/4 + 46^2/4 + 20^2/4 - 768$
- $= 225 + 529 + 100 - 768$
- $= 854 - 768 = \mathbf{86}$

Step 5 — Compute Within-group SS (SSW = SST − SSB).

- $\mathrm{SSW} = 98 - 86 = \mathbf{12}$

Step 6 — Compute degrees of freedom.

- Between: $df_B = k - 1 = 3 - 1 = 2$
- Within: $df_W = n - k = 12 - 3 = 9$
- Total: $df_T = n - 1 = 11$

Step 7 — Compute Mean Squares and F.

- $\mathrm{MSB} = 86/2 = \mathbf{43}$
- $\mathrm{MSW} = 12/9 = \mathbf{1.33}$
- $F = \mathrm{MSB}/\mathrm{MSW} = 43/1.33 = \mathbf{32.3}$

Step 8 — ANOVA table and conclusion.

Source	SS	df	MS	F
Between	86	2	43	**32.3**
Within	12	9	1.33	
Total	98	11		

$F_{0.05}(2, 9) \approx 4.26$. Computed F = 32.3 >> 4.26 → **Reject H_0** — the three fertilisers differ significantly.

✎ WORKED EXAMPLE

CD-10.2. Reading an ANOVA table — fill in the blanks.

Source	SS	df	MS	F
Between	120	3	?	?
Within	?	16	5	
Total	?	19		

Step 1 — Fill Within SS = MS × df = 5 × 16 = 80.

Step 2 — Fill Total SS = 120 + 80 = 200; Total df = 3 + 16 = 19 ✓.

Step 3 — MSB = 120 / 3 = 40.

Step 4 — F = MSB / MSW = 40 / 5 = 8.0.

Ans: MS Between = **40**, F = **8.0**, SSW = **80**, SST = **200**.

✎ WORKED EXAMPLE

CD-10.3. Back-calculate number of groups and observations from ANOVA table.

Between df = 4, Within df = 20. How many groups (k) and total observations (n)?

- $k - 1 = 4 \Rightarrow k = \mathbf{5}$ groups
- $n - k = 20 \Rightarrow n = 20 + 5 = \mathbf{25}$ total observations
- Equal group size: $n_j = 25/5 = \mathbf{5}$ per group

10.8 Trap-recognition card

Trap	Defence
ANOVA tests equality of variances	No — it tests equality of **means** (using the variance ratio as a tool).
Significant F → all means differ	F only says "not all equal". Need post-hoc to find pairs.
Within-error has $N - 1$ df	No, $N - k$. The $N - 1$ belongs to total.
df_total = N	No, $N - 1$.

10.8 Mini-mock

#	Q	Ans
1	ANOVA test stat?	F = MSB/MSW
2	df for total in one-way?	N − 1

#	Q	Ans
3	k=3, N=15. df between, within?	2, 12
4	TSS partition (one-way)?	TSS = SSB + SSW
5	TSS partition (two-way)?	TSS = SSR + SSC + SSE
6	MSB units?	same as MSW (variance)
7	If MSB = MSW exactly, F?	1
8	A two-way layout 4×5. df error?	12
9	Assumption: equal variances called?	Homoscedasticity
10	One-way ANOVA with k = 2 reduces to?	Two-sample t-test

10.9 Active-recall prompts

1. Write the one-way ANOVA decomposition of total sum of squares.
2. State the three assumptions of ANOVA.
3. Write df for between, within, total in one-way (with k groups, N observations).
4. State the F-test decision rule.
5. How does one-way ANOVA with k=2 relate to a two-sample t-test?

CHAPTER 11 — Time Series Analysis

Importance: ⭐⭐⭐⭐ HIGH (≈7 Qs / paper) **Difficulty:** Easy–Medium. Concept-heavy + a few computational Qs (moving average, semi-average).

11.0 — Understanding Time Series from First Principles

✔ QUICK RECALL

What is a time series, and why do we need to decompose it?

A **time series** is a sequence of observations recorded at regular time intervals — annual GDP, monthly rainfall, daily stock prices, quarterly sales. Unlike cross-sectional data (all measured at one moment), time series has temporal order that carries information: past values influence future values.

A time series is not random noise — it has structure. A JSO-level statistician decomposes it into four additive (or multiplicative) components:

1. **Trend (T):** The long-term directional movement — rising (India's GDP for 30 years), falling (landline subscriptions since 2005), or flat. This is the "signal" through the noise.
2. **Seasonal variation (S):** Regular, predictable within-year patterns that repeat every year. Ice cream sales peak in May–July every year. Woolen clothing sales peak in December every year. These are seasonal — and they are *expected*.
3. **Cyclical variation (C):** Longer-term, irregular waves lasting 3–10 years, linked to economic cycles (expansion → peak → recession → trough). Unlike seasonal, cycles are not fixed-period.
4. **Irregular variation (I):** Random, unpredictable shocks — a flood destroys a quarter's harvest, a pandemic shuts down an economy for two years, a factory fire disrupts supply.

Why decompose? Because each component requires a different response: a rising trend in crime needs policy intervention; a seasonal spike needs inventory planning; an irregular shock cannot be predicted and should not distort long-term planning.

Two models for decomposition:

Model	Equation	When to use
Additive	$Y = T + S + C + I$	When seasonal variation is roughly constant in magnitude regardless of trend level
Multiplicative	$Y = T \times S \times C \times I$	When seasonal variation grows proportionally with trend (more common in business data)

Moving averages — how they smooth out fluctuations:

A 4-point moving average of $Y_1, Y_2, Y_3, \ldots$ is: $M_1 = (Y_1 + Y_2 + Y_3 + Y_4)/4$, $M_2 = (Y_2 + Y_3 + Y_4 + Y_5)/4$, etc.

Each moving average "irons out" the short-term fluctuations, leaving a smoother trend estimate. For quarterly data (period = 4), a 4-point MA removes the seasonal component. For monthly data (period = 12), use a 12-point MA.

Solved Example 11.A — 3-point moving average:

Annual sales (₹ lakh): 10, 14, 12, 16, 18, 15, 20.

Step 1 – Apply the 3-year MA formula: $M_t = (Y_{t-1} + Y_t + Y_{t+1})/3$.

- $M_1 = (10 + 14 + 12)/3 = 36/3 = \mathbf{12.0}$
- $M_2 = (14 + 12 + 16)/3 = 42/3 = \mathbf{14.0}$
- $M_3 = (12 + 16 + 18)/3 = 46/3 \approx \mathbf{15.33}$
- $M_4 = (16 + 18 + 15)/3 = 49/3 \approx \mathbf{16.33}$
- $M_5 = (18 + 15 + 20)/3 = 53/3 \approx \mathbf{17.67}$

The moving averages (12.0, 14.0, 15.33, 16.33, 17.67) show a clearer upward trend than the raw data.

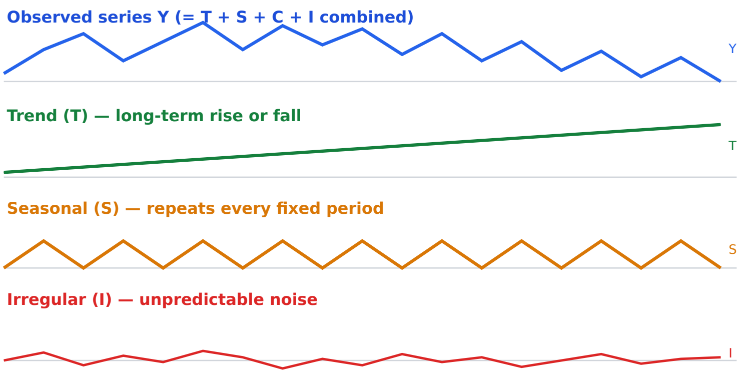
11.1 Examiner mindset

Angle	Pet question
Components of time series	trend, seasonal, cyclical, irregular
Mathematical models	Additive, Multiplicative
Trend by free-hand / semi-average / moving average / least squares	Direct compute
Seasonal indices	Method of simple averages, ratio-to-moving-average, ratio-to-trend
Removing seasonality (deseasonalised series)	Concept

11.2 Four components

Component	Symbol	Description	Period
Trend	T	Long-term direction	years to decades
Seasonal	S	Regular within-year pattern	quarters / months
Cyclical	C	Wave-like, longer than a year	2-10 years
Irregular / Random	I (or R)	Unpredictable shocks	random

How the four components combine to form an observed series:



11.3 Two models

Model	Equation	Use
Additive	$Y = T + S + C + I$	When seasonal swings are roughly constant in size

Model	Equation	Use
Multiplicative	$Y = T \cdot S \cdot C \cdot I$	When seasonal swings scale with trend

11.4 Methods of measuring trend

(i) Free-hand (graphic) method

Plot data, draw a smooth line by eye. Subjective.

(ii) Semi-average method

Step	What you do
1	Split the series into two equal halves. (If odd count, drop the middle observation.)
2	Compute the mean of each half.
3	Plot the two averages at the mid-period of each half. The line through them is the trend.

(iii) Moving-average method

For a 3-year MA, the trend value of year t is:

$$M_t = \frac{Y_{t-1} + Y_t + Y_{t+1}}{3}$$

For an even period (4-year), use **centred moving average** to align with original years.

Property	Note
Smooths out short-term fluctuations	yes
Loses observations at start and end	(n − 1)/2 each side for odd period
Best when no clear mathematical trend	yes

(iv) Least-squares (linear trend)

Fit $\hat{Y} = a + bX$ where X = coded time.

$$b = \frac{\sum XY}{\sum X^2}, \quad a = \bar{Y} - b\bar{X}$$

(Choose origin so that $\sum X = 0$ → simpler arithmetic.)

11.5 Measuring seasonal variation — Method of Simple Averages

Step	What
1	Arrange data by month/quarter for each year
2	Compute the average for each season across years
3	Compute grand average
4	Seasonal index for season j = (average for season j)/(grand average) $\times 100$

The 4 (or 12) indices average to 100. If they don't, **adjust** by multiplying each by 100 / their average.

11.6 Worked PYQ-style examples

✎ **WORKED EXAMPLE**

Q 11.1. A time-series component repeating every year is:

Soln. A within-year, regular, repeating pattern is the **Seasonal** component (S). Cyclical waves are longer than a year; irregular shocks are random; trend is monotonic.

✎ **WORKED EXAMPLE**

Q 11.2. Multiplicative model of time series?

Soln. The multiplicative decomposition: $Y = T \cdot S \cdot C \cdot I$ — used when the seasonal amplitude grows proportionally with the trend level (common in business data).

✎ **WORKED EXAMPLE**

Q 11.3. Sales: 10, 12, 14, 13, 16, 18, 20. 3-year moving averages?

Step 1 — Apply 3-year MA formula: $M_t = (Y_{t-1} + Y_t + Y_{t+1})/3$.

- Year 2 (centred): $(10 + 12 + 14)/3 = 36/3 = \mathbf{12.00}$
- Year 3: $(12 + 14 + 13)/3 = 39/3 = \mathbf{13.00}$
- Year 4: $(14 + 13 + 16)/3 = 43/3 = \mathbf{14.33}$
- Year 5: $(13 + 16 + 18)/3 = 47/3 = \mathbf{15.67}$
- Year 6: $(16 + 18 + 20)/3 = 54/3 = \mathbf{18.00}$

✎ **WORKED EXAMPLE**

Q 11.4. A company's quarterly sales: Q1=120, Q2=160, Q3=180, Q4=140 in year-1; year-2 same proportionally with grand average 160. Seasonal index Q3?

Step 1 — Compute the year-1 quarterly average.

- Average = $(120 + 160 + 180 + 140)/4 = 600/4 = 150$

Step 2 — Seasonal index for Q3 = (Q3 value / year average) × 100.

- Index = $180/150 \times 100$
- = **120**

(If the question demands rebasing so the four indices sum to 400, normalise accordingly.)

✎ **WORKED EXAMPLE**

Q 11.5. The semi-average method of fitting trend uses how many points to draw the line?

Soln. Split the series into two halves, compute the mean of each half, plot the two means against the midpoints of their halves and join. That's **two** points.

✎ **WORKED EXAMPLE**

Q 11.6. Removing trend from a time series gives the:

Soln. Detrending leaves the **detrended series** (= S + C + I components). (Compare: removing seasonality → deseasonalised series.)

✎ **WORKED EXAMPLE**

Q 11.7. Cyclical component differs from seasonal in:

Soln. Seasonal pattern has period **≤ 1 year** and is regular (same shape every year). Cyclical waves have period **> 1 year** (typically 2–10 years) and are irregular — driven by macro-economic cycles.

✎ **WORKED EXAMPLE**

Q 11.8. A 5-year moving average loses how many observations at each end?

Step 1 — For a $(2k+1)$-period centred MA, we lose k observations at each end.

- With period = 5, $k = 2$ ⇒ **2 at each end** (4 total).

✎ **WORKED EXAMPLE**

Q 11.9. Sum of seasonal indices over 4 quarters should equal:

Soln. Indices average to 100 by construction, so over 4 quarters the **sum = 400**. (Over 12 months: 1200.)

✎ **WORKED EXAMPLE**

Q 11.10. Trend by least squares with $\sum X = 0$: b formula?

Soln. With $\sum X = 0$, the normal equations simplify to $b = \sum XY / \sum X^2$ (and $a = \bar{Y}$).

11.7 Computation drills — the 5 problem types the exam sets

✎ **WORKED EXAMPLE**

CD-11.1. 3-year moving average.

Year: 2018, 2019, 2020, 2021, 2022, 2023, 2024. Values: 42, 45, 48, 44, 50, 53, 55.

Step 1 — Each MA_3 is the average of 3 consecutive values; it centres on the middle year.

- 2019: $(42 + 45 + 48)/3 = 135/3 = \mathbf{45.0}$
- 2020: $(45 + 48 + 44)/3 = 137/3 \approx \mathbf{45.67}$
- 2021: $(48 + 44 + 50)/3 = 142/3 \approx \mathbf{47.33}$
- 2022: $(44 + 50 + 53)/3 = 147/3 = \mathbf{49.0}$
- 2023: $(50 + 53 + 55)/3 = 158/3 \approx \mathbf{52.67}$

We lose 1 value at each end — 2018 and 2024 get no MA.

✎ WORKED EXAMPLE

CD-11.2. 4-year centred moving average (the tricky even-period case).

Year: 1, 2, 3, 4, 5, 6. Values: 10, 12, 16, 14, 18, 20.

Step 1 — Compute 4-year totals (moving sums).

- Years 1–4: $10 + 12 + 16 + 14 = 52$
- Years 2–5: $12 + 16 + 14 + 18 = 60$
- Years 3–6: $16 + 14 + 18 + 20 = 68$

Step 2 — Divide by 4 to get 4-year MA (lands between years).

- Between yr 2 & 3: $52/4 = 13.0$
- Between yr 3 & 4: $60/4 = 15.0$
- Between yr 4 & 5: $68/4 = 17.0$

Step 3 — Centre by averaging consecutive 4-year MAs (landing ON a year).

- Year 3: $(13.0 + 15.0)/2 = \mathbf{14.0}$
- Year 4: $(15.0 + 17.0)/2 = \mathbf{16.0}$

We lose 2 values at each end. The centring step is the one examiners trip candidates on.

✎ WORKED EXAMPLE

CD-11.3. Least-squares trend line: $Y = a + bt$.

Code the years t = −2, −1, 0, 1, 2 (centred at midpoint so $\sum t = 0$):

Year	t	Y	tY	t^2
2020	−2	20	−40	4
2021	−1	24	−24	1
2022	0	28	0	0
2023	1	32	32	1
2024	2	38	76	4
Σ	0	142	44	10

Step 1 — Compute a and b (simplified when $\sum t = 0$).

- $a = \bar{Y} = 142/5 = \mathbf{28.4}$
- $b = \sum tY / \sum t^2 = 44/10 = \mathbf{4.4}$

Trend line: $Y = 28.4 + 4.4t$.

Step 2 — Forecast for 2025 (t = 3).

- $Y = 28.4 + 4.4 \times 3 = 28.4 + 13.2 = \mathbf{41.6}$

✎ WORKED EXAMPLE

CD-11.4. Seasonal index — method of simple averages.

Quarterly sales (₹ lakh):

Year	Q1	Q2	Q3	Q4
2022	30	40	60	50
2023	33	44	66	55

Step 1 — Average each quarter across years.

- $\bar{Q}1 = (30 + 33)/2 = 31.5$
- $\bar{Q}2 = (40 + 44)/2 = 42.0$
- $\bar{Q}3 = (60 + 66)/2 = 63.0$
- $\bar{Q}4 = (50 + 55)/2 = 52.5$

Step 2 — Grand average $= (31.5 + 42 + 63 + 52.5)/4 = 189/4 = 47.25$.

Step 3 — Seasonal index = (quarter average / grand average) × 100.

- SI(Q1) $= 31.5/47.25 \times 100 = \mathbf{66.67}$
- SI(Q2) $= 42.0/47.25 \times 100 = \mathbf{88.89}$
- SI(Q3) $= 63.0/47.25 \times 100 = \mathbf{133.33}$
- SI(Q4) $= 52.5/47.25 \times 100 = \mathbf{111.11}$

Check: Sum of 4 indices = 400 ✓ (they average to 100 each).

✎ WORKED EXAMPLE

CD-11.5. Deseasonalise a value.

Q3 actual value = 63, Seasonal index Q3 = 133.33. What is the deseasonalised value?

Step 1 — Apply: Deseasonalised = (Actual / SI) × 100.

- $= (63/133.33) \times 100$
- $= 0.4725 \times 100$
- $= \mathbf{47.25}$

This removes the seasonal effect — what remains is T × C × I.

11.8 Trap-recognition card

Trap	Defence
Confusing seasonal with cyclical	Seasonal = regular ≤ 1 year; cyclical = irregular wave > 1 year.
Using a non-centred MA for even periods	Centre it (otherwise off by half a year).
Forgetting to adjust seasonal indices to sum to 400	Always verify.
Free-hand method called "objective"	It's the most subjective method.

11.8 Mini-mock

#	Q	Ans
1	Components of TS?	T, S, C, I
2	Multiplicative model formula?	$Y = T \cdot S \cdot C \cdot I$
3	A 3-year MA at year t = ?	$(Y_{t-1} + Y_t + Y_{t+1})/3$
4	Sum of 4 quarterly seasonal indices?	400
5	Removing trend gives the ...?	Detrended series
6	Cyclical period typically?	2–10 years
7	Method of least squares minimises?	$\Sigma(Y - \hat{Y})^2$
8	Semi-average method draws a line through how many points?	2
9	The most subjective method of trend?	Free-hand
10	Even-period MA needs?	Centring

11.9 Active-recall prompts

1. List the four components of a time series.
2. State the additive and multiplicative models.
3. Compute a 3-year moving average for the sequence 10, 12, 14, 16.
4. Describe the semi-average method in three steps.
5. State why even-period moving averages need centring.

CHAPTER 12 — Index Numbers

Importance: ⭐⭐⭐⭐ HIGH (≈6 Qs / paper) **Difficulty:** Easy. Pure formulas. Three minutes per question.

12.0 — Understanding Index Numbers from First Principles

✔ QUICK RECALL

What is an index number, and why do we need weighted indices?

An **index number** measures relative change from a base period. CPI = 180 means prices today are 80% higher than in the base year. Sensex at 75,000 is an index measuring the market value of 30 selected stocks relative to their values on 1 April 1979 (base = 100).

The aggregation problem: A household buys rice, cooking oil, clothing, and electronics. You want to measure how "the cost of living" changed. You cannot just average the price changes — rice and oil dominate monthly spending; electronics are a small fraction. An unweighted average treats a 50% rise in electronics (minor expense) the same as a 50% rise in rice (major expense). This is misleading.

Solution — weighted price indices: - **Laspeyres index:** Use base-period quantities as weights. $L = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$. Easier to compute (base-year quantities are fixed), but tends to overstate inflation because it doesn't account for consumers substituting cheaper goods. - **Paasche index:** Use current-period quantities as weights. $P = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$. More realistic (accounts for substitution), but requires current-year quantity data every period — expensive to collect. Tends to understate inflation. - **Fisher's Ideal Index:** $F = \sqrt{L \times P}$ — geometric mean of Laspeyres and Paasche. Satisfies both the Time Reversal Test and the Factor Reversal Test. Called "ideal" because it is the best compromise.

◆ FORMULA / KEY POINTS

The six index formulas (all appear in JSO exams):

Name	Formula	Weights
Laspeyres Price Index	$\frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$	Base-year quantities
Paasche Price Index	$\frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$	Current-year quantities
Fisher's Ideal	$\sqrt{L \times P}$	Geometric mean of above
Simple AM of Price Relatives	$\frac{\sum (p_1/p_0)}{n} \times 100$	Equal weights
Weighted AM of Price Relatives	$\frac{\sum w \cdot (p_1/p_0)}{\sum w} \times 100$	Given weights
Dorbish-Bowley	$\frac{L+P}{2}$	AM of Laspeyres and Paasche

Tests for a good index number:

Test	What it checks	Fisher's status
Time Reversal Test	$P_{01} \times P_{10} = 1$	✓ Satisfies
Factor Reversal Test	Price index × Quantity index = Value index	✓ Satisfies
Circular Test	$P_{01} \times P_{12} \times P_{20} = 1$	✗ Does not satisfy

Solved Example 12.A — Laspeyres and Paasche:

Commodity	p_0	q_0	p_1	q_1
A	10	5	12	6
B	8	4	10	3

Step 1 — Compute Laspeyres index (base-year quantities as weights).

- $\sum p_0 q_0 = (10 \times 5) + (8 \times 4) = 50 + 32 = 82$
- $\sum p_1 q_0 = (12 \times 5) + (10 \times 4) = 60 + 40 = 100$
- $L = (100/82) \times 100 = \mathbf{121.95}$

Step 2 — Compute Paasche index (current-year quantities as weights).

- $\sum p_1 q_1 = (12 \times 6) + (10 \times 3) = 72 + 30 = 102$
- $\sum p_0 q_1 = (10 \times 6) + (8 \times 3) = 60 + 24 = 84$
- $P = (102/84) \times 100 = \mathbf{121.43}$

Step 3 — Compute Fisher's Ideal index (geometric mean of L and P).

- $F = \sqrt{L \times P} = \sqrt{121.95 \times 121.43} = \sqrt{14814.5} = \mathbf{121.72}$

12.1 Examiner mindset

Angle	Pet question
Definition of index number	conceptual MCQ
Simple aggregate / simple average of price relatives	direct
Laspeyres, Paasche, Fisher, Marshall-Edgeworth, Walsh	direct formula
Tests: time-reversal, factor-reversal, circular	"Which formula satisfies ... ?"
Cost of living index	aggregate expenditure / family budget
Base shifting, splicing, deflating	concept + tiny computation

12.2 Definition

*An **index number** is a statistical measure designed to show changes in a variable or a group of related variables with respect to time, geographic location, or other characteristic.*

Notation: P_0, Q_0 for **base** period; P_1, Q_1 for **current** period.

12.3 Types of index numbers

Type	Example
Price index	Wholesale Price Index (WPI), Consumer Price Index (CPI)
Quantity index	Index of Industrial Production (IIP)
Value index	total value relative
Special-purpose	sensex / nifty (financial), HDI, etc.

12.4 Unweighted indices

Method	Formula
Simple aggregate	$P_{01} = \frac{\sum P_1}{\sum P_0} \times 100$
Simple average of price relatives (AM)	$P_{01} = \frac{1}{n} \sum \frac{P_1}{P_0} \times 100$
Simple average of price relatives (GM)	$P_{01} = \text{antilog}(\frac{\sum \log(P_1/P_0)}{n}) \times 100$

12.5 Weighted aggregate indices — the big four

Index	Formula	Weights used
Laspeyres	$L = \frac{\sum P_1 Q_0}{\sum P_0 Q_0} \times 100$	base-year quantities (Q_0)
Paasche	$P = \frac{\sum P_1 Q_1}{\sum P_0 Q_1} \times 100$	current-year quantities (Q_1)
Fisher's ideal	$F = \sqrt{L \cdot P}$	geometric mean of L and P
Marshall-Edgeworth	$ME = \frac{\sum P_1(Q_0 + Q_1)}{\sum P_0(Q_0 + Q_1)} \times 100$	average of base and current quantities
Walsh	$W = \frac{\sum P_1 \sqrt{Q_0 Q_1}}{\sum P_0 \sqrt{Q_0 Q_1}} \times 100$	geometric mean of quantities

◎ **MNEMONIC**

Memory hook for Laspeyres vs Paasche. "Laspeyres uses **L**ast-period (base) quantities. Paasche uses **P**resent-period (current) quantities."

12.6 The three "tests" — which index satisfies them

Test	Condition	Laspeyres	Paasche	Fisher	ME	Walsh
Time reversal $P_{01} \cdot P_{10} = 1$	Index of "1 → 0" should undo index of "0 → 1"	✗	✗	✓	✓	✓
Factor reversal $P_{01} \cdot Q_{01} = V_{01}$	Price × Quantity index = Value index	✗	✗	✓	✗	✗
Circular $P_{01} \cdot P_{12} \cdot P_{20} = 1$	Chain consistency	✗	✗	✗	✗	partly

Hence Fisher's index is called "Ideal" — it satisfies both time and factor reversal.

12.7 Cost of Living Index (Consumer Price Index)

Two main methods:

Method	Formula
Aggregate expenditure (≈ Laspeyres)	$\frac{\sum P_1 Q_0}{\sum P_0 Q_0} \times 100$
Family budget (weighted AM of price relatives)	$\frac{\sum W \cdot (P_1/P_0) \times 100}{\sum W}$, where $W = P_0 Q_0$

Both methods are mathematically equivalent.

12.8 Base shifting, splicing, deflating

Operation	Formula
Base shifting: shift the base from year-X to year-Y	$\text{New index}_t = \dfrac{\text{Old index}_t}{\text{Old index}_Y} \times 100$
Splicing: join two index series with overlapping period	Multiply older series by (new value at overlap / old value at overlap), or vice versa
Deflating: convert money wages → real wages	$\text{Real} = \dfrac{\text{Money value}}{\text{CPI}} \times 100$
Purchasing power of money	$1/\text{CPI} \times 100$

12.9 Worked PYQ-style examples

✎ WORKED EXAMPLE

Q 12.1. Compute Laspeyres index given:

Item	P_0	Q_0	P_1
A	4	10	5
B	6	5	9

Step 1 — Compute numerator $\sum P_1 Q_0$ (current-year prices × base-year quantities).

- $= 5 \times 10 + 9 \times 5$
- $= 50 + 45$
- **= 95**

Step 2 — Compute denominator $\sum P_0 Q_0$ (base-year prices × base-year quantities).

- $= 4 \times 10 + 6 \times 5$
- $= 40 + 30$
- **= 70**

Step 3 — Apply $L = (\text{num/den}) \times 100$.

- $L = (95/70) \times 100$
- **≈ 135.71**

✎ **WORKED EXAMPLE**

Q 12.2. Same items + Q_1 = 12, 4. Compute Paasche.

Step 1 — Compute numerator $\sum P_1 Q_1$.

- $= 5 \times 12 + 9 \times 4$
- $= 60 + 36$
- = **96**

Step 2 — Compute denominator $\sum P_0 Q_1$.

- $= 4 \times 12 + 6 \times 4$
- $= 48 + 24$
- = **72**

Step 3 — Apply $P = (\text{num/den}) \times 100$.

- $P = (96/72) \times 100$
- ≈ **133.33**

✎ **WORKED EXAMPLE**

Q 12.3. Fisher's index for above?

Step 1 — Apply $F = \sqrt{L \cdot P}$.

- $F = \sqrt{135.71 \times 133.33}$
- = $\sqrt{18093}$
- ≈ **134.51**

✎ **WORKED EXAMPLE**

Q 12.4. Which index satisfies factor reversal?

Soln. Only **Fisher's Ideal Index** satisfies both Time Reversal and Factor Reversal tests — that's why it is called "Ideal".

✎ **WORKED EXAMPLE**

Q 12.5. Money wage 6000, CPI = 150. Real wage?

Step 1 — Apply Real Wage = (Money wage / CPI) × 100.

- Real wage = $(6000/150) \times 100$
- = 40×100
- = **₹4000**

✎ WORKED EXAMPLE

Q 12.6. Purchasing power of money when CPI = 200?

Step 1 — Apply Purchasing Power = (1 / CPI) × 100 (relative to base = ₹1).

- $= (1/200) \times 100$
- = **0.50** (= 50 % of base-period purchasing power)

✎ WORKED EXAMPLE

Q 12.7. A series with base 2010 = 100 has values 110 (2011), 121 (2012), 133 (2013). Shift base to 2012.

Step 1 — Base shifting: divide each value by the new-base value (121) and multiply by 100.

- 2011: $110/121 \times 100 \approx$ **90.91**
- 2012: $121/121 \times 100 =$ **100**
- 2013: $133/121 \times 100 \approx$ **109.92**

✎ WORKED EXAMPLE

Q 12.8. $L = 144, P = 121$. Fisher?

Step 1 — Apply $F = \sqrt{L \cdot P}$.

- $F = \sqrt{144 \times 121}$
- = $\sqrt{17424}$
- = **132**

✎ WORKED EXAMPLE

Q 12.9. Marshall-Edgeworth uses what weight?

Soln. Marshall-Edgeworth uses $(Q_0 + Q_1)$ — the sum of base-year and current-year quantities — as the weight in both numerator and denominator.

✎ WORKED EXAMPLE

Q 12.10. A simple aggregate of prices index ignores:

Soln. Simple aggregate uses only $\sum P_1 / \sum P_0$; it **ignores quantities** (and therefore relative importance of items).

12.10 Trap-recognition card

Trap	Defence
Laspeyres weights are Q_1	No, $\mathbf{Q_0}$ (base year).
Paasche always > Laspeyres	False (depends on substitution effect).
Fisher satisfies circular	No, only time + factor reversal.
"Real wage" formula uses raw money	No — divide by CPI and multiply by 100.

12.11 Mini-mock

#	Q	Ans
1	Laspeyres uses which year's quantities?	Base year (Q_0)
2	Paasche uses which year's quantities?	Current year (Q_1)
3	Fisher = ?	$\sqrt{(L \cdot P)}$
4	"Ideal" index?	Fisher
5	Real wage formula?	Money/CPI × 100
6	Test that L·P doesn't satisfy individually?	Time reversal & factor reversal
7	If CPI rises, purchasing power of money?	Falls
8	A price index of 120 means?	20% rise from base
9	Splicing two index series joins them at?	Overlap year
10	Index used to convert money values into real values?	Price index (CPI/WPI)

12.12 Active-recall prompts

1. Write Laspeyres, Paasche, Fisher formulas.
2. Which index is "ideal" and why?
3. Define base shifting in one line and write its formula.
4. Write the formula for real wage.
5. State the time-reversal and factor-reversal tests.

APPENDIX A — Master Formula Sheet (one-page revision)

A.1 Central tendency

Concept	Formula
AM (raw / freq)	$\bar{X} = \sum X/n = \sum fX/N$
Step-deviation AM	$A + \frac{\sum fu}{N} \cdot h, u = (X - A)/h$
Combined mean	$\frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$
GM	$(X_1 \cdots X_n)^{1/n}$; log-AM form
HM	$n/\sum(1/X_i)$
AM·HM = GM²	(for two positive numbers only)
Median (grouped)	$L + \frac{N/2 - F}{f}h$
Mode (grouped)	$L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2}h$
Empirical rel.	Mode = 3 Median − 2 Mean

A.2 Dispersion

Concept	Formula
Range	max − min
QD	$(Q_3 - Q_1)/2$
Coeff QD	$(Q_3 - Q_1)/(Q_3 + Q_1)$
Variance (computational)	$\frac{\sum X^2}{n} - \bar{X}^2$
Step-deviation var	$h^2[\sum fu^2/N - (\sum fu/N)^2]$
CV	$\sigma/\bar{X} \times 100$
Combined SD	$\sigma_{12}^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$
Var(aX + b)	$a^2 \cdot \mathrm{Var}(X)$
Var(X+Y) (indep)	$\mathrm{Var}(X) + \mathrm{Var}(Y)$
Var(X−Y) (indep)	$\mathrm{Var}(X) + \mathrm{Var}(Y)$

A.3 Moments / Skewness / Kurtosis

Concept	Formula
Raw moment	$\mu_r' = \sum fX^r/N$
Central moment	$\mu_r = \sum f(X - \bar{X})^r/N$
μ_2	$= \sigma^2$
μ_2 from raw	$\mu_2 = \mu_2' - (\mu_1')^2$

Concept	Formula
μ_3 from raw	$\mu_3 = \mu_3' - 3\mu_1'\mu_2' + 2(\mu_1')^3$
Karl Pearson Sk	$(\bar{X} - \text{Mode})/\sigma$, or $3(\bar{X} - \text{Med})/\sigma$
Bowley Sk	$(Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$
β_1	μ_3^2/μ_2^3
β_2	μ_4/μ_2^2; = 3 normal, > 3 lepto, < 3 platy
Sheppard correction (μ_2)	subtract $h^2/12$

A.4 Correlation / Regression

Concept	Formula
Karl Pearson r	$\dfrac{n\sum XY - \sum X \sum Y}{\sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}}$
Spearman ρ	$1 - 6\sum d^2/[n(n^2-1)]$
b_{YX}	$r\sigma_Y/\sigma_X = \text{Cov}/\sigma_X^2$
b_{XY}	$r\sigma_X/\sigma_Y = \text{Cov}/\sigma_Y^2$
r from slopes	$\pm\sqrt{b_{YX}b_{XY}}$
Two regression lines meet at	$(\bar{X}, \bar{Y})$
SE of estimate	$S_{Y\cdot X} = \sigma_Y\sqrt{1-r^2}$
Coefficient of determination	r^2
Partial $r_{12.3}$	$(r_{12} - r_{13}r_{23})/\sqrt{(1-r_{13}^2)(1-r_{23}^2)}$
Multiple $R_{1.23}^2$	$(r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23})/(1-r_{23}^2)$

A.5 Probability

Concept	Formula
P(A ∪ B)	$P(A) + P(B) - P(A \cap B)$
P(A	B)
Multiplication	$P(A \cap B) = P(A)P(B\|A)$
Independence	$P(A \cap B) = P(A)P(B)$
Bayes	$P(B_i\|A) = \dfrac{P(B_i)P(A\|B_i)}{\sum_j P(B_j)P(A\|B_j)}$

A.6 Distributions

Distribution	pmf / pdf	Mean	Variance
Binomial(n,p)	${}^nC_r p^r q^{n-r}$	np	npq
Poisson(λ)	$e^{-\lambda}\lambda^r/r!$	λ	λ
Normal(μ,σ²)	$\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ²
Hypergeom(N,K,n)	${}^KC_k\,{}^{N-K}C_{n-k}/{}^NC_n$	nK/N	nK(N-K)(N-n)/[N²(N-1)]

Distribution	pmf / pdf	Mean	Variance
Uniform(a,b)	$1/(b-a)$	(a+b)/2	(b-a)²/12

A.7 Sampling and inference

Concept	Formula
SE of mean (∞ pop)	$\sigma/\sqrt{n}$
SE of mean (FPC)	$\sigma/\sqrt{n} \cdot \sqrt{(N-n)/(N-1)}$
SE of proportion	$\sqrt{P(1-P)/n}$
95 % CI for μ (large)	$\bar{X} \pm 1.96\sigma/\sqrt{n}$
t-stat (one mean)	$(\bar{X}-\mu_0)/(s/\sqrt{n})$, df = $n-1$
Pooled variance	$s_p^2 = [(n_1-1)s_1^2 + (n_2-1)s_2^2]/(n_1+n_2-2)$
χ² goodness	$\sum(O-E)^2/E$
F (variances)	s_1^2/s_2^2 (larger over smaller)
ANOVA F	MSB / MSW
Sample size for E (mean)	$n = (z_{\alpha/2}\sigma/E)^2$

A.8 Time series and index numbers

Concept	Formula
Additive TS	$Y = T + S + C + I$
Multiplicative TS	$Y = T \cdot S \cdot C \cdot I$
3-yr MA	$(Y_{t-1} + Y_t + Y_{t+1})/3$
Linear trend (Σ X = 0)	$b = \sum XY / \sum X^2$, $a = \bar{Y}$
Laspeyres	$\sum P_1Q_0 / \sum P_0Q_0 \times 100$
Paasche	$\sum P_1Q_1 / \sum P_0Q_1 \times 100$
Fisher	$\sqrt{L \cdot P}$
Real wage	Money wage / CPI × 100
Purchasing power	1/CPI × 100

APPENDIX B — ULTIMATE TABLES (rapid reference)

B.1 Common z-values and CI multipliers

Confidence level	Two-tail $z_{\alpha/2}$	Right-tail z_{α}
90 %	1.645	1.282
95 %	1.960	1.645
98 %	2.326	2.054
99 %	2.576	2.326

B.2 Excerpt from standard normal table — $P(0 \le Z \le z)$

z	0.0	0.5	1.0	1.5	1.96	2.0	2.5	2.58	3.0
Area	0	0.1915	0.3413	0.4332	0.4750	0.4772	0.4938	0.4951	0.4987

So $P(|Z| < 1.96) = 2 \cdot 0.4750 = 0.95$. Memorise these 9 anchors.

B.3 t-table (two-tail 5 %) — common df

df	$t_{0.025}$	df	$t_{0.025}$
5	2.571	20	2.086
10	2.228	25	2.060
15	2.131	30	2.042
∞	1.960 (= z)		

B.4 χ²-table (5 % upper-tail)

df	$\chi^2_{0.05}$
1	3.84
2	5.99
3	7.81
4	9.49
5	11.07
10	18.31

B.5 F-table (5 %, selected)

df_1, df_2	F
(5, 5)	5.05
(5, 10)	3.33
(10, 10)	2.97
(10, 20)	2.35

df_1, df_2	F
(20, 20)	2.12
(∞, ∞)	1.00

B.6 Distribution-recognition cheat sheet

Cue word in question	Distribution to use
"two outcomes per trial, n trials, fixed p"	Binomial
"rate / per unit time / rare events"	Poisson
"approximately normal", bell, $\pm\sigma$	Normal
"without replacement, finite N"	Hypergeometric
"uniformly distributed over interval"	Continuous Uniform

B.7 Test-recognition cheat sheet

Question type	Test
Test μ, σ known or n large	Z
Test μ, σ unknown, n small	t
Test $\mu_1 = \mu_2$, σ unknown	two-sample t (pooled)
Test paired data	paired t
Test variance / GoF / independence	χ^2
Test $\sigma_1^2 = \sigma_2^2$	F
Test $\mu_1 = \mu_2 = \mu_3 = \ldots$	ANOVA F

B.8 Greek + acronym glossary

Symbol	Reads as	Used for
μ, σ	mu, sigma	population mean, SD
ρ	rho	population correlation
α	alpha	level of significance / Type I error rate
β	beta	Type II error rate; also regression slope
λ	lambda	Poisson rate / eigenvalue
Σ	sigma	summation
χ^2	chi-square	test statistic
df / d.f.	degrees of freedom	for t, χ^2, F
pmf / pdf / CDF	probability mass / density function / cumulative DF	discrete vs continuous
MLE / MoM / LS	maximum-likelihood / method of moments / least squares	estimation

APPENDIX C — Full-length 100-Q Mock (with answer key)

> ***Instructions.*** *120 minutes. 2 marks per Q.* ***−0.5*** *per wrong. Use the in-test calculator. Answer key at the end.*

Section I — Collection / Central Tendency / Dispersion (Qs 1-25)

1. Class mark of 80–90 = ?
2. $\bar{X}$ of 14, 18, 22, 26, 30 = ?
3. Median of 11, 13, 7, 19, 9 = ?
4. Mode of 4, 4, 8, 9, 4, 5, 8 = ?
5. AM·HM of two positive numbers = 144. GM = ?
6. Variance of 2, 4, 6, 8, 10 = ?
7. SD of 5 numbers is 4. If each is multiplied by 3, new SD = ?
8. Range of 12, 4, 18, 7, 25, 9 = ?
9. CV definition = ?
10. $\sum X = 100, \sum X^2 = 1100, n = 10$. σ = ?
11. Combined mean of n_1=20 (mean 25), n_2=30 (mean 35) = ?
12. If each obs is increased by 10, mean becomes _, ***SD becomes*** _.
13. The angle of a sector representing 30 % in a pie chart = ?
14. $Q_3 - Q_1 = 24$. QD = ?
15. Mode = 25, Median = 28. Mean (empirical) = ?
16. Best graph for cumulative frequency = ?
17. AM ≥ GM ≥ HM (always for positive data). T/F?
18. Var(2X − 3) where Var(X) = 9 = ?
19. SD of constant 25 = ?
20. Class boundaries of 10–19 (inclusive form) = ?
21. $\sum(X - \bar{X})$ =?
22. Std dev that is unaffected by extreme values = ? (range / SD / MD / QD)
23. Frequency density when widths differ = ?
24. Var(X + Y), X & Y indep, Var(X)=4, Var(Y)=9 = ?
25. Coefficient of variation when σ=12, mean=60 = ?

Section II — Moments / Skewness / Correlation / Regression (Qs 26-50)

26. μ_1 for any distribution = ?
27. β_1 when $\mu_3 = 0$ = ?
28. β_2 = 3 means curve is = ?
29. Bowley's skewness when median = mean of Q_1 and Q_3 = ?

30. Pearson Sk: mean=42, mode=36, σ=6. Sk = ?

31. $b_{YX} = 0.4, b_{XY} = 1.6$. r =?

32. Two regression lines intersect at = ?

33. $r = 0$ → angle between regression lines = ?

34. Spearman ρ when $\sum d^2 = 30, n = 10$ = ?

35. r between (X+5) and (Y−7), original r = 0.6 = ?

36. $r = 0.7, \sigma_X = 4, \sigma_Y = 6$. b_{YX} = ?

37. r^2 coefficient of = ?

38. Karl Pearson r is independent of = ?

39. Multiple correlation R lies in = ?

40. Var(X)=16, Var(Y)=9, indep. Var(X − Y) = ?

41. Two regression slopes 0.5 and 0.6. r = ?

42. Coefficient of determination, r=0.4 = ?

43. $\beta_2 = 1.8$, curve = ?

44. $\mu_3 = 8, \mu_2 = 4$. β_1 = ?

45. Pearson Sk symmetric → mean − mode = ?

46. SE of estimate Y on X = ?

47. r in [−1, +1]. T/F?

48. Spearman with tie correction adds = ?

49. r between scaled (cX, dY) where c, d > 0 = ?

50. $r_{12.3}$ lies in = ?

Section III — Probability / Distributions / Sampling (Qs 51-75)

51. P(king or queen) from a standard pack = ?

52. P(A∩B) when A, B mutually exclusive = ?

53. P(A∩B) = 0.2, P(B) = 0.5. P(A|B) = ?

54. P(A) = 0.4, P(B) = 0.5, indep. P(A∩B) = ?

55. Box: 60 % A, 40 % B. Defective rate 5 % vs 8 %. P(item is from A | defective) = ?

56. Toss 3 coins. P(exactly 2 H) = ?

57. P(at least one H in 4 tosses) = ?

58. Bag has 5R, 3B. Draw 2 wo rep. P(both R) = ?

59. $X \sim B(10, 0.3)$. Mean = ?

60. $X \sim B(10, 0.3)$. Var = ?

61. Poisson with mean 4. P(X = 0) = ?

62. $X \sim N(50, 25)$. P(X > 60) = ?

63. $X \sim N(\mu, \sigma^2)$. $P(\mu - 1.96\sigma < X < \mu + 1.96\sigma)$ = ?

64. $\lambda = 9$. SD = ?

65. Mean = Var → distribution = ?

66. Continuous uniform [0, 4]. Var = ?

67. SE of mean if σ=12, n=144 = ?

68. Sampling without replacement on a finite population uses correction factor = ?

69. CLT applies for n ≥ ?

70. Stratified sampling reduces error when within-strata variability = ?

71. A snowball sampling is = ?

72. SE of proportion when p=0.5, n=400 = ?

73. n needed for margin 2 at 95 % CI when σ=10 = ?

74. Type I error symbol = ?

75. Power of a test = ?

Section IV — Inference / ANOVA / TS / Index (Qs 76-100)

76. χ^2 formula for goodness of fit = ?

77. df for χ^2 independence in 4 × 5 table = ?

78. F-test compares two = ?

79. Test for one mean, σ known, n=80, two-tail 5 %, $|Z|$ critical = ?

80. Pooled variance formula in two-sample t = ?

81. Unbiased divisor for sample variance = ?

82. Sample mean is what kind of estimator? (4 properties)

83. Max-likelihood estimate of p for binomial = ?

84. ANOVA tests equality of = ?

85. df_total in one-way ANOVA with N obs = ?

86. SSB = 90, SSW = 60, k = 4, N = 24. F = ?

87. Two-way ANOVA 3 × 4. df_error = ?

88. Components of a time series = ?

89. 5-yr MA loses how many obs at each end? = ?

90. Sum of seasonal indices over 4 quarters = ?

91. Multiplicative model = ?

92. Linear trend slope ($\Sigma X = 0$) = ?

93. Laspeyres uses which year's quantities? = ?

94. Paasche uses = ?

95. Fisher's index = ?

96. Test that Fisher satisfies = ?

97. Real wage = money wage / CPI × ?

98. Purchasing power of money = ?

99. CPI rises → purchasing power of money = ?

100. Marshall-Edgeworth weight = ?

Answer key

#	Ans	#	Ans	#	Ans	#	Ans
1	85	26	0	51	8/52=2/13	76	$\Sigma(O-E)^2/E$
2	22	27	0	52	0	77	(4−1)(5−1)=12
3	11	28	mesokurtic	53	0.4	78	variances
4	4	29	0	54	0.20	79	1.96
5	12	30	1	55	0.484	80	$[(n_1-1)s_1^2+(n_2-1)s_2^2]/(n_1+n_2-2)$
6	8	31	0.8	56	3/8	81	n−1
7	12	32	$(\bar{X}, \bar{Y})$	57	1−1/16=15/16	82	unbiased, consistent, efficient, sufficient
8	21	33	90°	58	5/14	83	X/n
9	$\sigma/\bar{X} \times 100$	34	1−6·30/(10·99)≈0.818	59	3	84	means
10	$\sqrt{110-100}=\sqrt{10}\approx 3.16$	35	0.6	60	2.1	85	N − 1
11	31	36	1.05	61	$e^{-4}\approx 0.0183$	86	(90/3)/(60/20)=30/3=10
12	mean+10, SD same	37	determination	62	P(Z>2)=0.0228	87	(3−1)(4−1)=6
13	108°	38	origin & scale	63	0.95	88	T, S, C, I
14	12	39	[0,1]	64	3	89	2
15	29.5	40	25	65	Poisson	90	400
16	ogive	41	$\sqrt{0.30}\approx 0.548$	66	$(4-0)^2/12=16/12=4/3$	91	Y = T·S·C·I
17	T	42	0.16	67	1	92	$\Sigma XY/\Sigma X^2$
18	36	43	platykurtic	68	$\sqrt{[(N-n)/(N-1)]}$	93	base year (Q_0)
19	0	44	1	69	30	94	current year (Q_1)
20	9.5 – 19.5	45	0	70	small	95	$\sqrt{L \cdot P}$
21	0	46	$\sigma Y\sqrt{1-r^2}$	71	non-probability	96	time + factor reversal
22	QD	47	T	72	$\sqrt{0.25/400}=0.025$	97	100
23	f / h	48	$m(m^2-1)/12$ per tie	73	$(1.96\cdot 10/2)^2\approx 96.04\to 97$	98	1/CPI × 100
24	13	49	same r	74	α	99	falls
25	20 %	50	[−1, 1]	75	1 – β	100	$Q_0 + Q_1$

APPENDIX D — 7-day final revision plan (for the last week)

Day	Morning (3 h)	Evening (3 h)	Recall pass
D-7	Ch 1 + Ch 2 (re-read theory, do mini-mocks)	Ch 3 (re-read + mini-mock + 20 PYQ-style problems)	All "Active-recall prompts" of Ch 1–3 closed-book
D-6	Ch 4 + Ch 5	Ch 6 + Ch 7	Recall prompts Ch 4–7
D-5	Ch 8 + Ch 9	Ch 9 second pass (most marks here)	Recall prompts Ch 8–9
D-4	Ch 10 + Ch 11 + Ch 12	Master Formula Sheet — write it from memory twice	All recall prompts in one sitting
D-3	Full-length mock (Appendix C) under timed conditions (120 min, calculator)	Score the mock honestly. List every wrong/skipped Q.	Re-read the chapter for each error
D-2	Re-read every "Trap recognition card" in the book	Master Formula Sheet — write it from memory once more	One past JSO PYQ paper
D-1	Light read of Appendices A & B only — no new problems	Sleep early. Eat light.	Visualise the test flow.
D-0 (test day)	Calm 30-min glance at Master Formula Sheet	Test!	—

Test-day strategy.

1. ***First sweep (40 min)*** *— knock out every direct-formula and definition Q. Skip anything that takes more than 90 seconds.*
2. ***Second sweep (50 min)*** *— return to medium Qs (Ch 5, 6, 7, 9 numericals).*
3. ***Final sweep (25 min)*** *— attempt the leftover hard ones; mark unsure ones for review.*
4. ***Last 5 min*** *— recheck answer-button selections, do nothing risky.*
5. ***Negative-marking rule.*** *If you can eliminate even 2 of 4 options, attempt — expected value > 0. If you cannot eliminate any, leave it.*

APPENDIX E — Extended Drill Pack (deep dive on the 6 CRITICAL chapters)

> ***Why this pack exists.*** *60 % of the JSO paper comes from six chapters. The chapter mini-mocks give you a taste; this pack gives you mastery. Aim: cover* ***every PYQ-frame*** *an SSC examiner has used in the last 6 papers. Solve closed-book; re-read the chapter only on a wrong answer.*

E-2 Central Tendency — extended drill (25 worked Qs)

✎ WORKED EXAMPLE

E-2.1. Mean of $2, 4, 6, \ldots, 50$ (i.e., first 25 even numbers)?

Even numbers: $2, 4, \ldots, 50$ → AP with 25 terms, first 2, last 50. Mean of an AP = (first + last)/2 = (2 + 50)/2 = **26**.

✎ WORKED EXAMPLE

E-2.2. Mean of first 100 natural numbers?

$$\frac{1 + 2 + \ldots + 100}{100}$$
$$= \frac{100 \cdot 101/2}{100}$$
$$= \frac{101}{2} = \mathbf{50.5}.$$

✎ WORKED EXAMPLE

E-2.3. A series of 6 observations has mean 12. If a 7th observation is added and the new mean is 13, what is the new value?

Step 1 — Old sum = $n \times \bar{X} = 6 \times 12 = 72$.

Step 2 — New sum = $7 \times 13 = 91$.

Step 3 — New value = new sum − old sum.

- $= 91 - 72$
- = **19**

✎ WORKED EXAMPLE

E-2.4. Mean of 25 observations was 36. It was found that two observations 47 and 35 were misread as 27 and 53. Correct mean?

Step 1 — Old sum = $25 \times 36 = 900$.

Step 2 — Adjustment = (correct values) − (wrong values).

- $= (47 + 35) - (27 + 53)$
- $= 82 - 80$
- = **+2**

Step 3 — Correct sum = old sum + adjustment.

- $= 900 + 2 = 902$

Step 4 — Correct mean = correct sum / n.

- $= 902/25$
- = **36.08**

✎ WORKED EXAMPLE

E-2.5. Two groups of size 50 and 100 have means 60 and 75 respectively. Combined mean?

Step 1 — Apply the combined-mean formula.

- $\bar{X}_{12} = \dfrac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$
- $= \dfrac{50 \cdot 60 + 100 \cdot 75}{150}$
- $= \dfrac{3000 + 7500}{150}$
- $= \dfrac{10500}{150}$
- = **70**

✎ WORKED EXAMPLE

E-2.6. AM of 10 observations is 25. After adding two new observations the mean becomes 27. Sum of the two new values?

Step 1 — Old sum = $10 \times 25 = 250$.

Step 2 — New sum after adding 2 observations = $12 \times 27 = 324$.

Step 3 — Two new values = new sum − old sum.

- $= 324 - 250$
- = **74**

✎ WORKED EXAMPLE

E-2.7. A man covers three equal distances at 30, 40 and 60 km/h. Average speed?

Step 1 — Equal distances at different speeds ⇒ use HM, not AM.

Step 2 — Apply n-value HM formula: $\mathrm{HM} = n/\sum(1/v_i)$ **with** $n = 3$.

- HM = $\dfrac{3}{1/30 + 1/40 + 1/60}$

Step 3 — Combine reciprocals (LCM = 120).

- $1/30 + 1/40 + 1/60 = 4/120 + 3/120 + 2/120 = 9/120$

Step 4 — Compute the HM.

- HM = $\dfrac{3}{9/120} = \dfrac{3 \times 120}{9} = \dfrac{360}{9}$
- = **40 km/h**

✎ WORKED EXAMPLE

E-2.8. GM of 1, 3, 9, 27, 81?

Powers of 3: $3^0, 3^1, 3^2, 3^3, 3^4$.

Product = $3^{0+1+2+3+4} = 3^{10}$

$\mathrm{GM} = (3^{10})^{1/5} = 3^2 = \mathbf{9}$. (Or: middle term of GP = GM.)

✎ WORKED EXAMPLE

E-2.9. AM of two numbers exceeds their GM by 2. AM exceeds HM by what?

Use AM × HM = GM^2. Let AM = G + 2, where GM = G. Then HM = $G^2/(G + 2)$. AM − HM = (G + 2) − $G^2/(G + 2)$ = $[(G+2)^2 - G^2]/(G + 2)$ = (4G + 4)/(G + 2) = 4(G + 1)/(G + 2). Without specifics, students often plug small values (G = 4 → AM = 6, HM = 16/6 = 8/3; AM − HM = 6 − 8/3 = 10/3 ≈ 3.33). The general answer: $4(G+1)/(G+2)$.

✎ WORKED EXAMPLE

E-2.10. Median of 7, 9, 12, 14, 18, 22, 25, 28, 31?

9 odd values, middle = 5th = **18**.

✎ WORKED EXAMPLE

E-2.11. Median of 7, 9, 12, 14, 18, 22, 25, 28?

8 even values, median = (4th + 5th)/2 = (14 + 18)/2 = **16**.

✎ WORKED EXAMPLE

E-2.12. Mode of: 4, 4, 5, 5, 5, 6, 7, 7, 8, 8, 8, 8, 9?

8 appears 4 times — most frequent. **Mode = 8**.

✎ WORKED EXAMPLE

E-2.13. Compute the mean of the following grouped data using step deviation. Take $A = 35, h = 10$.

Class	0–10	10–20	20–30	30–40	40–50	50–60
f	5	8	12	20	10	5

Step 1 — Write the class marks m and compute $u = (m - A)/h$ for each class.

- Class marks: 5, 15, 25, 35, 45, 55
- u: −3, −2, −1, 0, 1, 2

Step 2 — Compute fu for each class and the totals.

- fu: $-15, -16, -12, 0, 10, 10$
- $\Sigma fu = -23$, $\Sigma f = 60$

Step 3 — Apply the step-deviation formula.

- Mean = $A + (\Sigma fu/N) \cdot h$
- = $35 + (-23/60)(10)$
- = $35 - 3.83$
- = **31.17**

✎ WORKED EXAMPLE

E-2.14. From E-2.13, find the median.

Step 1 — Compute N/2.

- $N = 60, N/2 = 30$

Step 2 — Build the CF column.

- CF: 5, 13, 25, 45, 55, 60

Step 3 — Locate the median class (first CF to cross N/2 = 30).

- 45 ≥ 30 → median class = **30–40**

Step 4 — Identify symbols and apply the median formula.

- $L = 30, F = 25, f = 20, h = 10$
- Median $= 30 + \dfrac{30 - 25}{20} \times 10$
- $= 30 + 2.5$
- = **32.5**

✎ WORKED EXAMPLE

E-2.15. Mode of E-2.13?

Step 1 — Locate the modal class (highest frequency).

- Max $f = 20$ → modal class = **30–40**

Step 2 — Identify symbols.

- $L = 30, f_1 = 20, f_0 = 12, f_2 = 10, h = 10$

Step 3 — Apply the mode formula: Mode $= L + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$.

- $= 30 + \dfrac{20 - 12}{(2 \times 20) - 12 - 10} \times 10$
- $= 30 + \dfrac{8}{18} \times 10$
- $= 30 + 4.44$
- = **34.44**

✎ WORKED EXAMPLE

E-2.16. Verify the empirical relation on E-2.13/14/15.

3 Med − 2 Mean = 3(32.5) − 2(31.17) = 97.5 − 62.34 = 35.16. Mode by formula = 34.44. Approx match (empirical relation works exactly only for moderately skewed unimodal data).

✎ WORKED EXAMPLE

E-2.17. P_{25}, P_{50}, P_{75} of a distribution correspond to which quartiles?

Q_1, Q_2, Q_3 respectively.

✎ WORKED EXAMPLE

E-2.18. The 7th decile equals which percentile?

$D_7 = P_{70}$.

✎ WORKED EXAMPLE

E-2.19. Which average is best for ratios and growth rates?

GM.

✎ WORKED EXAMPLE

E-2.20. Which average is most affected by sampling fluctuations?

AM uses every observation linearly, so it is affected; mode is least affected, median is intermediate. The conventional answer for "most affected by extreme values": **AM**.

✎ WORKED EXAMPLE

E-2.21. A car runs the first 1/3rd of a journey at 30 km/h, the next 1/3rd at 60 km/h, and the last 1/3rd at 90 km/h. Average speed?

Step 1 — Equal distances at different speeds ⇒ use HM (n = 3).

- HM = $\dfrac{3}{1/30 + 1/60 + 1/90}$

Step 2 — Combine reciprocals (LCM = 180).

- $1/30 + 1/60 + 1/90 = 6/180 + 3/180 + 2/180 = 11/180$

Step 3 — Compute HM.

- HM = $\dfrac{3}{11/180} = \dfrac{3 \times 180}{11} = \dfrac{540}{11}$
- ≈ **49.09 km/h**

✎ WORKED EXAMPLE

E-2.22. GM of 8 and 18?

$\sqrt{144} = \mathbf{12}$.

✎ WORKED EXAMPLE

E-2.23. A class mean of 60 girls is 50, the combined mean (boys + girls = 100 students) is 54. Mean of boys?

Step 1 — Set up the combined-mean equation.

- $\bar{X}_{12} \cdot N = n_G \bar{X}_G + n_B \bar{X}_B$
- $54 \times 100 = 50 \times 60 + \bar{X}_B \times 40$

Step 2 — Solve for $\bar{X}_B$.

- $5400 = 3000 + 40\bar{X}_B$
- $40\bar{X}_B = 2400$
- $\bar{X}_B =$ **60**

✎ WORKED EXAMPLE

E-2.24. For data 5, 7, 9, 11, 13, the mean of $2X + 3$?

Mean of $X = 9$.

Mean of $2X + 3 = 2 \times 9 + 3 = \mathbf{21}$.

✎ WORKED EXAMPLE

E-2.25. When data is positively skewed, which is the largest of the three central tendencies?

Mean (Mean > Median > Mode).

E-3 Dispersion — extended drill (25 Qs)

✎ **WORKED EXAMPLE**

E-3.1. Variance of first n odd natural numbers?

Odd numbers: 1, 3, 5, ..., (2n − 1). Mean = n. $\sum X^2 = \frac{n(2n-1)(2n+1)}{3}$.

Variance = $\frac{\sum X^2}{n} - n^2$

$= \frac{(2n-1)(2n+1)}{3} - n^2$

$= \frac{4n^2-1}{3} - n^2$

$= \frac{4n^2-1-3n^2}{3} = \frac{n^2-1}{3}$. So variance of odd natural numbers up to $2n-1$ is $\frac{n^2-1}{3}$ (analogous to natural numbers' $(n^2-1)/12$; note the 4× because step is 2).

✎ **WORKED EXAMPLE**

E-3.2. SD of $2, 4, 6, 8, 10$?

Step 1 — Compute the mean.

- Mean = $(2+4+6+8+10)/5 = 30/5 = \mathbf{6}$

Step 2 — Compute squared deviations and their sum.

- Deviations from 6: $-4, -2, 0, 2, 4$
- Squared: $16, 4, 0, 4, 16$ → Sum = 40

Step 3 — Compute variance and SD.

- Variance = $40/5 = \mathbf{8}$
- SD = $\sqrt{8} = 2\sqrt{2} \approx \mathbf{2.83}$

✎ **WORKED EXAMPLE**

E-3.3. Mean of 10 observations is 5 and SD is 4. Find $\sum X$ and $\sum X^2$.

Step 1 — Compute $\sum X = n\bar{X}$.

- $\sum X = 10 \times 5 =$ **50**

Step 2 — Use the computational form of variance: $\sigma^2 = \sum X^2/n - \bar{X}^2$.

- $16 = \sum X^2/10 - 25$

Step 3 — Solve for $\sum X^2$.

- $\sum X^2 = (16+25) \times 10 = 41 \times 10$
- = **410**

✎ WORKED EXAMPLE

E-3.4. Var(X) = 9, Var(Y) = 16, X and Y are independent. Var$(2X - 3Y)$?

$4 \cdot 9 + 9 \cdot 16 = 36 + 144 = \mathbf{180}$.

✎ WORKED EXAMPLE

E-3.5. SD of 10 observations is 6. If each observation is increased by 5 and then divided by 2, new SD?

Origin-add doesn't change SD; scale by 1/2 → SD = **3**.

✎ WORKED EXAMPLE

E-3.6. Two groups of equal size with means 50 and 60 and SDs 4 and 3 respectively. Combined SD?

Step 1 — Compute the combined mean (equal sizes ⇒ simple average).

- $\bar{X}_{12} = (50 + 60)/2 =$ **55**

Step 2 — Compute deviations.

- $d_1 = 50 - 55 = -5$, $d_2 = 60 - 55 = +5$

Step 3 — Apply the combined-variance formula (equal weights of 1).

- $\sigma_{12}^2 = \dfrac{(\sigma_1^2 + d_1^2) + (\sigma_2^2 + d_2^2)}{2}$
- $= \dfrac{(16 + 25) + (9 + 25)}{2}$
- $= \dfrac{41 + 34}{2}$
- $=$ **37.5**

Step 4 — Take square root.

- SD = $\sqrt{37.5}$
- ≈ **6.12**

✎ WORKED EXAMPLE

E-3.7. Variance computational form: which of the following is the correct expression? (i) $\sum(X - \bar{X})^2/n$, (ii) $\sum X^2/n - \bar{X}^2$, (iii) both, (iv) neither.

(iii) both — they are algebraically equal.

✎ WORKED EXAMPLE

E-3.8. Which has greater dispersion: A (mean 100, SD 20) or B (mean 250, SD 30)?

Step 1 — Compute CV for each series.

- CV(A) = $(20/100) \times 100 = \mathbf{20\%}$
- CV(B) = $(30/250) \times 100 = \mathbf{12\%}$

Step 2 — Compare. Higher CV = greater relative dispersion → **A** is more dispersed.

✎ WORKED EXAMPLE

E-3.9. First 10 natural numbers: variance and SD?

Step 1 — Apply the variance formula for first n natural numbers: $\mathrm{Var} = (n^2 - 1)/12$.

- Var = $(10^2 - 1)/12 = 99/12 = \mathbf{8.25}$

Step 2 — Compute SD.

- SD = $\sqrt{8.25} \approx \mathbf{2.87}$

✎ WORKED EXAMPLE

E-3.10. Mean deviation about median for the data 4, 6, 8, 10, 12?

Step 1 — Find the median.

- 5 values; middle = 3rd = **8**

Step 2 — Compute absolute deviations from the median.

- $|4-8|, |6-8|, |8-8|, |10-8|, |12-8| = 4, 2, 0, 2, 4$; Sum = **12**

Step 3 — Compute MD.

- MD = $12/5 = \mathbf{2.4}$

✎ WORKED EXAMPLE

E-3.11. Coefficient of QD when $Q_1 = 22, Q_3 = 38$?

Step 1 — Apply the coefficient of QD formula.

- Coefficient of QD = $(Q_3 - Q_1)/(Q_3 + Q_1)$
- = $(38 - 22)/(38 + 22)$
- = $16/60$
- = **0.267**

✎ WORKED EXAMPLE

E-3.12. Variance of 7 added 4 times → 7, 7, 7, 7?

All equal → variance = **0**.

✎ WORKED EXAMPLE

E-3.13. A constant value c is added to each observation. Effect on Var?

No change (origin invariant).

✎ WORKED EXAMPLE

E-3.14. SD of $X = 5$. SD of $-3X + 7$?

$|-3| \cdot 5 = \mathbf{15}$.

✎ **WORKED EXAMPLE**

E-3.15. Which of these is unaffected by the change of origin? Range, MD, SD, all of these.

All of these — every dispersion measure is origin-invariant.

✎ **WORKED EXAMPLE**

E-3.16. Two series A and B have CV 25% and 20% respectively. Which is more consistent?

B (lower CV).

✎ **WORKED EXAMPLE**

E-3.17. SD of 50 observations is 10. If each observation is multiplied by 2, the new SD?

20.

✎ **WORKED EXAMPLE**

E-3.18. Lorenz curve of perfect inequality lies along which axes?

Along the **x-axis up to the last point, then jumps vertically**. (One person owns everything.)

✎ **WORKED EXAMPLE**

E-3.19. Sum of squared deviations from mean is the smallest. T/F?

True — it's the minimum across all c.

✎ **WORKED EXAMPLE**

E-3.20. $\sum X = 60, \sum X^2 = 400, n = 10$. Variance?

Step 1 — Compute the mean.

- $\bar{X} = 60/10 =$ **6**

Step 2 — Apply the computational form of variance.

- $\mathrm{Var} = \sum X^2/n - \bar{X}^2$
- $= 400/10 - 6^2$
- $= 40 - 36$
- **= 4**

Step 3 — SD = $\sqrt{4} = 2$.

✎ **WORKED EXAMPLE**

E-3.21. Variance of $aX + bY$ with X and Y independent?

$a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)$.

✎ WORKED EXAMPLE

E-3.22. $s = 5$ is sample SD with $n = 25$. σ (population SD if computed with n divisor)?

Sample variance with $n - 1$ divisor = 25. $\sum(X - \bar{X})^2 = 24 \cdot 25 = 600$. With n divisor: 600/25 = 24. σ (sample raw form) = $\sqrt{24} \approx 4.90$.

✎ WORKED EXAMPLE

E-3.23. A series has SD 5. SD of 100−X?

$|-1| \cdot 5 = \mathbf{5}$.

✎ WORKED EXAMPLE

E-3.24. For two series with same SD, the one with smaller mean has:

Higher CV → less consistent.

✎ WORKED EXAMPLE

E-3.25. A distribution where SD is the largest measure of dispersion? (vs MD vs QD)

SD ≥ MD ≥ QD typically (recall QD : MD : SD ≈ 10 : 12 : 15). So SD is the largest.

E-5 Correlation & Regression — extended drill (25 Qs)

✎ WORKED EXAMPLE

E-5.1. Cov(X, Y) = 12, σ_X = 3, σ_Y = 5. r?

$r = 12/(3 \cdot 5) = 12/15 = \mathbf{0.8}$.

✎ WORKED EXAMPLE

E-5.2. Two regression equations: $8X - 10Y + 66 = 0$ and $40X - 18Y = 214$. Find $\bar{X}, \bar{Y}, b_{YX}, b_{XY}, r$.

Solve simultaneously. Eq1: $8X - 10Y = -66$ → divide by 2: $4X - 5Y = -33$.

Eq2: $40X - 18Y = 214$ → divide by 2: $20X - 9Y = 107$.

Multiply Eq1 by 5: $20X - 25Y = -165$.

Subtract: $(20X - 25Y) - (20X - 9Y) = -165 - 107$

$-16Y = -272$

$Y = 17$

Then $4X = -33 + 5(17) = -33 + 85 = 52$

$X = 13$. So $\bar{X} = 13$, $\bar{Y} = 17$. Now identify which is Y on X. From Eq1: $Y = (8X + 66)/10 = 0.8X + 6.6 \rightarrow b_{YX} = 0.8$. From Eq2: $X = (18Y + 214)/40 = 0.45Y + 5.35 \rightarrow b_{XY} = 0.45$. Check: $b_{YX} \cdot b_{XY} = 0.36 \le 1$ ✓. r = +√0.36 = **0.6**.

✎ WORKED EXAMPLE

E-5.3. $b_{YX} = 1.6$. Possible b_{XY} values?

Must satisfy $b_{YX} \cdot b_{XY} \le 1$. So $b_{XY} \le 1/1.6 = 0.625$. Both must have same sign (positive here).

✎ WORKED EXAMPLE

E-5.4. $r = 0.8, n = 10$. Coefficient of determination, in %?

$r^2 = 0.64 = \mathbf{64\%}$ of the variance of Y is explained by X.

✎ WORKED EXAMPLE

E-5.5. Spearman without ties: 6 students, ranks differ as: $d = 0, 1, -1, 2, -2, 0$. $\sum d^2$?

0 + 1 + 1 + 4 + 4 + 0 = **10**. $\rho = 1 - 6 \cdot 10/(6 \cdot 35) = 1 - 60/210 = 1 - 2/7 = \mathbf{5/7} \approx \mathbf{0.714}$.

✎ WORKED EXAMPLE

E-5.6. Spearman with ties: $\sum d^2 = 50$, n = 10, two pairs of ties of size 2 each in one variable. ρ?

Tie correction = $2 \times \frac{2(4-1)}{12} = 2 \times 0.5 = 1$. Adjusted $\sum d^2 = 51$.

$$\rho = 1 - \frac{6 \times 51}{10 \times 99}$$

$$= 1 - \frac{306}{990}$$

$= 1 - 0.309 = \mathbf{0.691}$.

✎ WORKED EXAMPLE

E-5.7. A perfectly positive correlation has scatter:

All points on a straight line of positive slope.

✎ WORKED EXAMPLE

E-5.8. Two regression coefficients are −0.6 and −0.5. r?

Both negative → $r = -\sqrt{0.30} \approx \mathbf{-0.548}$.

✎ WORKED EXAMPLE

E-5.9. $b_{YX} = 0.5, b_{XY} = 0.8$. Mean of X = 5, Mean of Y = 7. Regression line of Y on X?

$Y - 7 = 0.5(X - 5) \Rightarrow Y = 0.5X + 4.5$.

✎ WORKED EXAMPLE

E-5.10. Mean of X = 6, mean of Y = 4. Predict Y when X = 10 if $b_{YX} = 0.5$.

$Y - 4 = 0.5(10 - 6) \Rightarrow Y = 4 + 2 = \mathbf{6}$.

✎ WORKED EXAMPLE

E-5.11. $\sigma_X = 4, \sigma_Y = 6, r = 0.5$. Regression of X on Y?

$b_{XY} = 0.5 \cdot 4/6 = 1/3 \approx 0.333$. Equation: $X - \bar{X} = 0.333(Y - \bar{Y})$.

✎ WORKED EXAMPLE

E-5.12. Two regression lines coincide if and only if?

$r = \pm 1$ (perfect linear correlation).

✎ WORKED EXAMPLE

E-5.13. $r = 0.6, \sigma_X = 5, \sigma_Y = 10$. Find $b_{YX} - b_{XY}$.

$b_{YX} = 1.2, b_{XY} = 0.3$. Difference = 0.9.

✎ WORKED EXAMPLE

E-5.14. Karl Pearson r is symmetric, so r_{XY} and r_{YX}?

Equal.

✎ WORKED EXAMPLE

E-5.15. $r_{12} = 0.6, r_{13} = 0.5, r_{23} = 0.4$. Partial $r_{12.3}$?

Numerator = $0.6 - 0.5 \times 0.4 = 0.6 - 0.2 = 0.4$

Denominator = $\sqrt{(1 - 0.25)(1 - 0.16)}$

$= \sqrt{0.75 \times 0.84}$

$= \sqrt{0.63} \approx 0.794$

$r_{12.3} \approx 0.4/0.794 \approx \mathbf{0.504}$.

✎ WORKED EXAMPLE

E-5.16. Same data, $R^2_{1.23}$?

$R^2_{1.23} = (0.36 + 0.25 - 2 \times 0.6 \times 0.5 \times 0.4)/(1 - 0.16)$

$= (0.61 - 0.24)/0.84$

$= 0.37/0.84 \approx 0.440$

$R \approx \mathbf{0.664}$.

✎ WORKED EXAMPLE

E-5.17. A scatter diagram with points evenly spread shows:

$r \approx 0 \rightarrow$ no linear correlation.

✎ WORKED EXAMPLE

E-5.18. Sum of products of deviations $\sum(X - \bar{X})(Y - \bar{Y}) = 240$, n = 10, σ_X = 4, σ_Y = 8. r?

Cov = 240/10 = 24. $r = 24/(4 \cdot 8) = \mathbf{0.75}$.

✎ WORKED EXAMPLE

E-5.19. $b_{YX} = 0.9$ and $b_{XY} = 1.1$. Possible?

Product = 0.99 < 1 ✓ — possible. r = +√0.99 ≈ 0.995.

✎ WORKED EXAMPLE

E-5.20. Two lines pass through (10, 20). Are they regression lines?

Only if the means are (10, 20). Both regression lines pass through $(\bar{X}, \bar{Y})$.

✎ WORKED EXAMPLE

E-5.21. A regression line has slope 0. r = ?

0 (no linear relationship between Y and X).

✎ WORKED EXAMPLE

E-5.22. $r = 0$ for paired data of (X, Y). Cov(X, Y) = ?

0 (since Cov = r σ_X σ_Y).

✎ WORKED EXAMPLE

E-5.23. When σ_X = σ_Y, what is b_{YX} in terms of r?

$b_{YX} = r$ (and $b_{XY} = r$ too).

✎ WORKED EXAMPLE

E-5.24. A change in origin of both X and Y changes r by:

No change (origin-invariant).

✎ WORKED EXAMPLE

E-5.25. A change of scale that flips Y's sign changes b_{YX} by:

Sign flips: new $b_{YX} = -r\sigma_Y/\sigma_X$ so it flips sign.

E-6 Probability — extended drill (25 Qs, with Bayes practice)

✎ WORKED EXAMPLE

E-6.1. Two dice rolled. P(at least one 6)?

P(no 6 on either) = $(5/6)^2$. P(at least one) = 1 − 25/36 = **11/36**.

✎ WORKED EXAMPLE

E-6.2. A coin is biased: P(H) = 2/3. Three tosses. P(exactly 2 H)?

$^3C_2(2/3)^2(1/3)^1 = 3\cdot 4/9\cdot 1/3 = 4/9.$

✎ WORKED EXAMPLE

E-6.3. A bag has 5W, 4B, 3R. One drawn. P(W or R)?

Mutually exclusive. P = 5/12 + 3/12 = **8/12 = 2/3**.

✎ WORKED EXAMPLE

E-6.4. Two cards drawn from a pack without replacement. P(both aces)?

4/52 · 3/51 = 12/2652 = **1/221**.

✎ WORKED EXAMPLE

E-6.5. P(A) = 0.6, P(B) = 0.5, P(A ∩ B) = 0.3. Are they independent?

P(A) · P(B) = 0.30 = P(A ∩ B) ✓. **Yes**.

✎ WORKED EXAMPLE

E-6.6. P(A) = 0.5, P(A | B) = 0.5. Are they independent?

Yes (independence ⇔ P(A | B) = P(A)).

✎ WORKED EXAMPLE

E-6.7. Three balls drawn from bag of 5R, 4G, 3W (no replacement). P(one of each colour)?

Numerator = $^5C_1 \cdot {}^4C_1 \cdot {}^3C_1 = 60$. Denom = $^{12}C_3 = 220$. P = 60/220 = **3/11**.

✎ WORKED EXAMPLE

E-6.8. A family has 3 children. P(at least 1 boy | first is boy)?

Given first is boy, the at-least-one-boy condition is automatically satisfied. P = **1**.

✎ WORKED EXAMPLE

E-6.9. Bayes: A test for a disease has sensitivity 95% (P(+ | disease) = 0.95) and specificity 90%. The disease prevalence is 1%. A randomly tested person tests positive. P(disease | +)?

P(D) = 0.01, P(+ | D) = 0.95, P(+ | not D) = 0.10. Numerator $= 0.01 \times 0.95 = 0.0095$

Denominator $= 0.0095 + 0.99 \times 0.10$

$= 0.0095 + 0.099 = 0.1085$

$P(D|+) = 0.0095/0.1085 \approx \mathbf{0.0876}$ (8.76 %).

Insight. Even with a 95% accurate test, low prevalence makes the positive predictive value small. Classic Bayes counterintuitive answer.

✎ WORKED EXAMPLE

E-6.10. Bayes: Three machines A, B, C produce 50%, 30%, 20% of items, with defect rates 1%, 2%, 3%. A defective item is found. P(it came from C)?

Numerator $= 0.20 \times 0.03 = 0.006$

Denominator $= 0.50 \times 0.01 + 0.30 \times 0.02 + 0.20 \times 0.03$

$= 0.005 + 0.006 + 0.006 = 0.017$

$P(C|D) = 0.006/0.017 \approx \mathbf{0.353}$.

✎ WORKED EXAMPLE

E-6.11. P(A) = 0.5, P(B) = 0.4, P(A ∪ B) = 0.7. P(A ∩ B)?

0.5 + 0.4 − 0.7 = **0.2**.

✎ WORKED EXAMPLE

E-6.12. Throw 2 dice. P(sum ≥ 10)?

Sums 10, 11, 12 → (4,6),(5,5),(6,4) | (5,6),(6,5) | (6,6) = 3 + 2 + 1 = 6 outcomes / 36 = **1/6**.

✎ WORKED EXAMPLE

E-6.13. A bag has 4 white and 6 black balls. Two drawn without replacement. P(at least one white)?

P(none white) = (6/10)(5/9) = 30/90 = 1/3. P(at least one) = 1 − 1/3 = **2/3**.

✎ WORKED EXAMPLE

E-6.14. P(A) = 0.4, P(B) = 0.5, A and B independent. P(neither)?

P(A^c) · P(B^c) = 0.6 · 0.5 = **0.3**.

✎ WORKED EXAMPLE

E-6.15. Probability of getting at least 1 head in 5 tosses?

1 − (1/2)^5 = 1 − 1/32 = **31/32**.

✎ WORKED EXAMPLE

E-6.16. Two events A, B with P(A) = 0.3 and P(B) = 0.4 are mutually exclusive. P(A ∪ B)?

0.7 (since intersection is 0).

✎ WORKED EXAMPLE

E-6.17. $P(A) = 0.4, P(B|A) = 0.5$. P(A ∩ B)?

0.4 × 0.5 = **0.20**.

✎ WORKED EXAMPLE

E-6.18. Pair of dice. P(odd sum)?

Half of 36 = **18/36 = 1/2** (by parity argument: odd sum ⇔ exactly one of the two dice is odd).

✎ WORKED EXAMPLE

E-6.19. A box has 3 fair coins and 2 biased coins (P(H) = 0.8 each). One coin drawn at random and tossed; result is H. P(it was a biased coin)?

Numerator $= (2/5)(0.8) = 0.32$

Denominator $= 0.32 + (3/5)(0.5)$

$= 0.32 + 0.30 = 0.62$

$P(\text{biased}|H) = 0.32/0.62 \approx \mathbf{0.516}$.

✎ WORKED EXAMPLE

E-6.20. A fair coin is tossed until first head. P(first H on the 4th toss)?

$(1/2)^3 \cdot (1/2)$ = **1/16**.

✎ WORKED EXAMPLE

E-6.21. Two events: $P(A^c) = 0.7, P(B^c) = 0.6, P(A \cup B) = 0.6$. $P(A \cap B)$?

$P(A) = 0.3, P(B) = 0.4$.

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= 0.3 + 0.4 - 0.6 = \mathbf{0.1}$.

✎ WORKED EXAMPLE

E-6.22. Out of 5 letters, 3 are addressed correctly. Letters placed in envelopes randomly. P(none correctly placed)?

Derangement $D_5 = 44$. P = 44/120 = **11/30**.

✎ WORKED EXAMPLE

E-6.23. A throws a die first, then B. P(A scores higher)?

By symmetry P(A > B) = P(B > A) and P(equal) = 6/36 = 1/6. So P(A > B) = (1 − 1/6)/2 = **5/12**.

✎ WORKED EXAMPLE

E-6.24. Two indep events have P(none happens) = 1/4 and P(both happen) = 1/4. Find P(A) and P(B).

Let $p = P(A)$, $q = P(B)$. $pq = 1/4$ and $(1-p)(1-q) = 1/4 \to 1 - p - q + pq = 1/4 \to 1 - (p+q) + 1/4 = 1/4 \to p + q = 1$. With $pq = 1/4$: p, q are roots of $t^2 - t + 0.25 = 0 \to t = 0.5, 0.5$. So **$P(A) = P(B) = 0.5$**.

✎ WORKED EXAMPLE

E-6.25. Conditional: $P(A \mid B) = 0.5$, $P(B \mid A) = 0.4$, $P(A) = 0.3$. P(B)?

$P(A \cap B) = 0.3 \cdot 0.4 = 0.12$. $P(B) = P(A \cap B)/P(A \mid B) = 0.12 / 0.5 =$ **0.24**.

E-7 Distributions — extended drill (25 Qs)

✎ WORKED EXAMPLE

E-7.1. Binomial(20, 0.4). Mean and SD?

Step 1 — Compute the mean.

- Mean = $np = 20 \times 0.4 = \mathbf{8}$

Step 2 — Compute the variance and SD.

- Variance = $npq = 20 \times 0.4 \times 0.6 = \mathbf{4.8}$
- SD = $\sqrt{4.8} \approx \mathbf{2.19}$

✎ WORKED EXAMPLE

E-7.2. Binomial(n, p) has mean 6, var 4. Find n, p.

Step 1 — Set up equations from mean and variance.

- Mean: $np = 6$
- Variance: $npq = 4$

Step 2 — Solve for q and p.

- Divide: $q = npq/np = 4/6 = 2/3$
- Therefore: $p = 1 - q = 1 - 2/3 = \mathbf{1/3}$

Step 3 — Solve for n.

- $n = 6/p = 6/(1/3) = \mathbf{18}$

✎ WORKED EXAMPLE

E-7.3. Binomial(5, 0.5). P(X ≥ 4)?

$$P(X \geq 4) = {}^5C_4(0.5)^5 + {}^5C_5(0.5)^5$$

$$= \frac{5}{32} + \frac{1}{32} = \frac{6}{32} = \mathbf{\frac{3}{16}}.$$

✎ WORKED EXAMPLE

E-7.4. Poisson(λ). $P(X = 0) = 0.05$. Find λ.

$e^{-\lambda} = 0.05 \Rightarrow \lambda = \ln(20) \approx \mathbf{3.0}$.

✎ WORKED EXAMPLE

E-7.5. A radioactive source emits at average 10 particles per minute. P(exactly 12 in a minute)?

$e^{-10}10^{12}/12! \approx 0.0948$.

✎ WORKED EXAMPLE

E-7.6. Poisson(2). P(X ≥ 1)?

1 − P(X = 0) = 1 − e^(−2) ≈ 1 − 0.135 = **0.865**.

✎ WORKED EXAMPLE

E-7.7. $X \sim N(\mu, \sigma^2)$ with mean 100, σ = 15. P(85 < X < 115)?

Within ±1σ → 0.6826.

✎ WORKED EXAMPLE

E-7.8. $X \sim N(50, 16)$. P(X > 58)?

z = (58 − 50)/4 = 2. P(Z > 2) = 0.0228.

✎ WORKED EXAMPLE

E-7.9. $X \sim N(0, 1)$. P(−1.96 < X < 1.96)?

0.95.

✎ WORKED EXAMPLE

E-7.10. $X \sim N(80, 100)$. The 90th percentile of X?

$z_{0.10} = 1.282$. $X_{0.90} = 80 + 1.282 \cdot 10 = \mathbf{92.82}$.

✎ WORKED EXAMPLE

E-7.11. Binomial approximated by Normal. When?

When n is large and p is not near 0 or 1; rule of thumb $np > 5$ and $nq > 5$.

✎ WORKED EXAMPLE

E-7.12. Binomial approximated by Poisson. When?

n large, p small, np = λ moderate.

✎ WORKED EXAMPLE

E-7.13. Hypergeometric: lot of 50 has 10 defective. 5 picked without replacement. Mean number of defectives in the sample?

Mean = nK/N = 5 · 10 / 50 = **1**.

✎ WORKED EXAMPLE

E-7.14. Continuous uniform on [4, 16]. P(X ≤ 10)?

(10 − 4)/(16 − 4) = 6/12 = **0.5**.

✎ WORKED EXAMPLE

E-7.15. Continuous uniform on [4, 16]. Mean and variance?

Mean = 10.

$$\text{Var} = \frac{(16-4)^2}{12} = \frac{144}{12} = \mathbf{12}.$$

✎ WORKED EXAMPLE

E-7.16. $X \sim B(n, p)$. Skewness?

$(1-2p)/\sqrt{npq}$. When p = 0.5, skewness = 0 (symmetric).

✎ WORKED EXAMPLE

E-7.17. Z = (X − μ)/σ. Distribution of Z?

Standard normal N(0, 1).

✎ WORKED EXAMPLE

E-7.18. P(|Z| > 2.58)?

≈ 0.01 (1 % two-tail).

✎ WORKED EXAMPLE

E-7.19. $X \sim B(8, 0.5)$. P(X = 4)?

$^8C_4(0.5)^8 = 70/256 \approx 0.273.$

✎ WORKED EXAMPLE

E-7.20. $X \sim P(\lambda = 16)$. σ?

$\sqrt{16} = \mathbf{4}.$

✎ WORKED EXAMPLE

E-7.21. Discrete RV with pmf: x: 0,1,2,3 ; p(x): 0.1, 0.3, 0.4, 0.2. E[X] and Var(X)?

Step 1 — Compute E[X] = Σ x · p(x).

- $E[X] = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2)$
- $= 0 + 0.3 + 0.8 + 0.6 = \mathbf{1.7}$

Step 2 — Compute $E[X^2] = \Sigma x^2 \cdot p(x)$.

- $E[X^2] = 0^2(0.1) + 1^2(0.3) + 2^2(0.4) + 3^2(0.2)$
- $= 0 + 0.3 + 1.6 + 1.8 = 3.7$

Step 3 — Compute $\text{Var}(X) = E[X^2] - (E[X])^2$.

- $\text{Var} = 3.7 - 1.7^2 = 3.7 - 2.89 = \mathbf{0.81}$

✎ WORKED EXAMPLE

E-7.22. A normal curve is:

Symmetric, bell-shaped, mean = median = mode, asymptotic to x-axis.

✎ WORKED EXAMPLE

E-7.23. $X \sim N(\mu, \sigma^2)$. The MGF M_X(t) = ?

$\exp(\mu t + \sigma^2 t^2/2)$.

✎ WORKED EXAMPLE

E-7.24. Sum of n independent Poisson(λ_i) is?

Poisson($\sum \lambda_i$). (Poisson is closed under addition.)

✎ WORKED EXAMPLE

E-7.25. Sum of n independent Binomial(n_i, p) is?

Binomial($\sum n_i, p$) — only when same p.

E-9 Statistical Inference — extended drill (30 Qs, the biggest chapter)

✎ WORKED EXAMPLE

E-9.1. Sample mean is which kind of estimator of μ?

Unbiased, consistent, efficient (under normality), sufficient (under normality with known σ).

✎ WORKED EXAMPLE

E-9.2. Sample variance with divisor n is biased. Bias = ?

$E[\hat{\sigma}^2] = (n-1)\sigma^2/n$. Bias = $-\sigma^2/n$.

✎ WORKED EXAMPLE

E-9.3. MLE of σ^2 in Normal sample (mean known)?

$\hat{\sigma}^2 = \sum(X_i - \mu)^2/n$ — uses divisor n, not n−1.

✎ WORKED EXAMPLE

E-9.4. Method of moments estimator of θ for Uniform(0, θ)?

Sample mean = θ/2 → $\hat{\theta}_{MoM} = 2\bar{X}$.

✎ WORKED EXAMPLE

E-9.5. A test rejects H_0 when $\bar{X} > 52$, with H_0: μ = 50, σ = 10, n = 25. Compute α.

Step 1 — Compute the SE and standardise the critical value.

- $\text{SE} = \sigma/\sqrt{n} = 10/\sqrt{25} = 10/5 = 2$
- $Z_{\text{crit}} = (52 - 50)/2 = 2/2 = \mathbf{1.0}$

Step 2 — Find α (probability of rejecting H_0 when it is true).

- $\alpha = P(Z > 1.0) = \mathbf{0.1587}$

✎ WORKED EXAMPLE

E-9.6. Same test, true μ = 53. β?

Reject when $\bar{X} > 52$. Under $\mu = 53$:

$$Z = \frac{52 - 53}{2} = -0.5$$

$\beta = P(\bar{X} \le 52 \mid \mu = 53) = P(Z \le -0.5) = \mathbf{0.3085}$.

✎ WORKED EXAMPLE

E-9.7. Same test, power at μ = 53?

1 − β = 1 − 0.3085 = **0.6915**.

✎ WORKED EXAMPLE

E-9.8. A two-sample Z-test with means 102, 98, σ_1 = σ_2 = 10, n_1 = n_2 = 100. Z?

$$Z = \frac{102 - 98}{\sqrt{\frac{100}{100} + \frac{100}{100}}}$$

$$= \frac{4}{\sqrt{2}} \approx \mathbf{2.828}$$

$2.828 > 1.96$ → reject H_0.

✎ WORKED EXAMPLE

E-9.9. Two sample t-test with pooled variance: $n_1 = 12$, $n_2 = 16$, $s_1 = 5$, $s_2 = 6$.

$$s_p^2 = \frac{11 \times 25 + 15 \times 36}{26}$$

$$= \frac{275 + 540}{26}$$

$= 815/26 \approx 31.35$

$s_p \approx 5.6.$

✎ WORKED EXAMPLE

E-9.10. χ^2 goodness-of-fit: O = (40, 35, 25), E = (30, 30, 40). χ^2?

$$\chi^2 = \frac{(40-30)^2}{30} + \frac{(35-30)^2}{30} + \frac{(25-40)^2}{40}$$

$= 100/30 + 25/30 + 225/40$

$= 3.33 + 0.83 + 5.63 = \mathbf{9.79}$

df = 2; $\chi^2_{0.05,2} = 5.99$ → reject H_0.

✎ WORKED EXAMPLE

E-9.11. Confidence interval (95 %): $\bar{X} = 25, s = 4, n = 16$. σ unknown, n small.

$t_{0.025,15} \approx 2.131$

$$\mathrm{CI} = 25 \pm 2.131 \times \frac{4}{\sqrt{16}}$$

$= 25 \pm 2.131 \times 1$

$= 25 \pm 2.131 = [\mathbf{22.87, 27.13}]$.

✎ WORKED EXAMPLE

E-9.12. Confidence interval (90 %) for proportion: p = 0.4, n = 200.

$\mathrm{SE} = \sqrt{0.4 \times 0.6/200} = \sqrt{0.0012} \approx 0.0346$

90 % CI = $0.4 \pm 1.645 \times 0.0346$

$= 0.4 \pm 0.057 = [\mathbf{0.343, 0.457}]$.

✎ WORKED EXAMPLE

E-9.13. Power increases when?

α increases (less strict), n increases (more data), |effect| increases.

✎ WORKED EXAMPLE

E-9.14. Bayesian estimator of mean with prior $N(\mu_0, \tau^2)$ and data of n obs from $N(\mu, \sigma^2)$?

Posterior mean is a weighted average: $\hat{\mu} = \dfrac{(n/\sigma^2)\bar{X} + (1/\tau^2)\mu_0}{n/\sigma^2 + 1/\tau^2}$. (Beyond JSO direct ask but useful concept.)

✎ WORKED EXAMPLE

E-9.15. $F = s_1^2/s_2^2$ with $s_1^2 = 9$, $s_2^2 = 4$, $df_1 = df_2 = 10$. Decision at 5 %?

Step 1 — Compute the F-statistic (larger over smaller).

- $F = s_1^2/s_2^2 = 9/4 = \mathbf{2.25}$

Step 2 — Compare to the critical value.

- $F_{0.05,\,10,\,10} \approx 2.97$
- $2.25 < 2.97$ → **fail to reject** H_0 (variances equal)

✎ WORKED EXAMPLE

E-9.16. χ^2 test of independence in a 2 × 2 table, sample size 100. df?

(2 − 1)(2 − 1) = **1**.

✎ WORKED EXAMPLE

E-9.17. A Type I error is committed when:

H_0 is true but rejected.

✎ WORKED EXAMPLE

E-9.18. A Type II error is:

H_0 is false but not rejected.

✎ WORKED EXAMPLE

E-9.19. $s_1^2 = 12, s_2^2 = 16, n_1 = n_2 = 11$. Pooled variance?

$$s_p^2 = \frac{10 \times 12 + 10 \times 16}{20}$$

$$= \frac{120 + 160}{20} = 14$$

$s_p \approx 3.74.$

✎ WORKED EXAMPLE

E-9.20. Critical t for two-tail 1 %, df = 20?

≈ **2.845** (standard table).

✎ WORKED EXAMPLE

E-9.21. Sample mean is unbiased for μ but biased for μ^2?

True: $E[\bar{X}^2] = \mu^2 + \sigma^2/n \neq \mu^2$.

✎ WORKED EXAMPLE

E-9.22. Best critical region for testing simple null vs simple alternative is given by:

Neyman-Pearson lemma → likelihood ratio test.

✎ WORKED EXAMPLE

E-9.23. Sufficient statistic for Bernoulli(p) sample of size n?

$\sum X_i$ (the count of successes).

✎ WORKED EXAMPLE

E-9.24. Cramér-Rao lower bound gives:

The minimum variance any unbiased estimator can attain.

✎ WORKED EXAMPLE

E-9.25. A 95 % CI's "95%" refers to:

The frequentist procedure: 95 % of such intervals (built from random samples) will contain the true parameter — not "95% chance for this single interval."

✎ WORKED EXAMPLE

E-9.26. $\bar{X} = 75, \sigma = 12, n = 36$. 99 % CI for μ?

$\mathrm{SE} = \dfrac{\sigma}{\sqrt{n}}$

$= \dfrac{12}{\sqrt{36}} = \dfrac{12}{6} = 2$

99 % CI $= 75 \pm 2.576 \times 2$

$= 75 \pm 5.15 = [\mathbf{69.85, 80.15}]$.

✎ **WORKED EXAMPLE**

E-9.27. χ^2 test of variance: $s^2 = 25, \sigma_0^2 = 16, n = 21$. Test at 5 % two-tail.

$\chi^2 = (n-1)s^2/\sigma_0^2$

$= 20 \times 25/16$

$= 31.25$

df = 20. Critical: $\chi^2_{0.025,20} \approx 34.17, \chi^2_{0.975,20} \approx 9.59$.

Since $9.59 < 31.25 < 34.17$ → fail to reject H_0.

✎ **WORKED EXAMPLE**

E-9.28. A two-sample paired t-test uses df:

n − 1 (where n is number of pairs).

✎ **WORKED EXAMPLE**

E-9.29. For testing H_0: $\mu = \mu_0$ when σ unknown but n very large, test stat reduces to:

Z (because t with large df ≈ Z).

✎ **WORKED EXAMPLE**

E-9.30. A goodness-of-fit χ^2 with k categories, m parameters estimated: df?

k − 1 − m.

APPENDIX F — Mock Test #2 (100 Qs, full coverage)

> ***Instructions.*** *120 minutes, 2 marks each,* ***−0.5*** *per wrong. Calculator allowed. Answer key at the end.*

Section I — Data presentation, central tendency, dispersion (Qs 1-25)

1. Class width of classes 5–9, 10–14, 15–19?
2. The histogram bar height when class widths are unequal is plotted as?
3. Mean of 7, 9, 11, 13, 15, 17?
4. Sum of deviations from mean is?
5. AM, GM, HM order for positive observations?
6. AM × HM = GM^2 for how many positive numbers?
7. Median position in a sorted list of 50 observations?
8. D_8 corresponds to what percentile?
9. Combined mean of n_1=40, $\bar{X}_1$=30 and n_2=10, $\bar{X}_2$=10?
10. Mean is most affected by?
11. Mode of 4, 5, 5, 6, 7, 8, 8, 8, 9?
12. The empirical relation: Mode = ?
13. Class boundary of 30–39 (inclusive)?
14. Pie-chart angle for 45 % share?
15. SD of 3, 5, 7, 9, 11?
16. Var(X − Y) where Var(X)=10, Var(Y)=15, indep?
17. CV = 10 % means?
18. Coefficient of QD for Q_1=20, Q_3=60?
19. SD of 2X+3 when SD(X)=4?
20. Variance of first 8 natural numbers?
21. Lorenz curve coinciding with line of equality means?
22. Mean of N(50, 100)?
23. Effect on Var when each value is increased by 10?
24. Mean deviation about median for 1, 2, 3, 4, 5?
25. Coefficient of dispersion for QD?

Section II — Moments / Skewness / Correlation (Qs 26-50)

26. μ_2 is also called?
27. μ_3 for symmetric distribution?
28. Karl Pearson Sk: Mean=70, Median=65, σ=8?
29. β_2 = 4 means?

30. Sheppard's correction for variance?

31. r is unitless. T/F?

32. $b_{YX} = -0.6, b_{XY} = -0.4$. r?

33. r = 0.7. Coefficient of determination = ?

34. Two regression lines pass through?

35. r = 0 → angle between regression lines?

36. Spearman ρ when n=8, $\Sigma d^2 = 50$?

37. $r = 0.5, \sigma_X = 4, \sigma_Y = 6.\ b_{YX}$?

38. r when both regression slopes are negative?

39. Multiple correlation R lies in?

40. Partial correlation $r_{12.3}$ lies in?

41. r in (X+10, Y−5) when r(X,Y)=0.6?

42. $b_{YX} \cdot b_{XY} = 0.81$. r?

43. Standard error of estimate Y on X?

44. Bowley Sk lies in?

45. $\beta_1 = 0$. Skewness?

46. r remains unchanged under?

47. $r = 1$. Two regression lines?

48. Cov(X, Y) = 12, σ_X=3, σ_Y=4. r?

49. A scatter plot shows linear band → r ≈?

50. r when $\sigma_X = 0$?

Section III — Probability / Distributions / Sampling (Qs 51-75)

51. P(prime number on a die)?

52. Coin tossed 4 times. P(at least 3 H)?

53. P(A) = 0.5, P(B|A) = 0.4. P(A ∩ B)?

54. P(A) = 0.5, P(B) = 0.4 indep. P(at least one)?

55. Bag has 4R, 6B. 2 drawn no rep. P(both R)?

56. Bayes: prior 0.6, likelihood 0.5 vs prior 0.4 lik 0.8. P(first | observed)?

57. $B(20, 0.5)$. Mean and SD?

58. $B(n, p)$ with mean 4, var 3. n?

59. Poisson(3). P(X = 0)?

60. $N(50, 25)$. P(X > 60)?

61. Normal curve max at?

62. Z-score for observation 80 in N(70, 25)?

63. Hypergeom mean (N=20, K=8, n=5)?

64. Continuous uniform [0, 12]. Var?

65. SE of mean: $\sigma=10$, n=100?

66. Stratified vs cluster: which uses sub-groups with within-homogeneity?

67. CLT requires $n \geq$?

68. SE of proportion: p=0.6, n=400?

69. Sample size at 95 % for margin 5, $\sigma=20$?

70. The sampling technique using a sampling interval k?

71. Non-probability sampling cannot give?

72. As n increases, sampling error?

73. Quota sampling is which family?

74. Type I error symbol?

75. Power of test?

Section IV — Inference / ANOVA / TS / Index (Qs 76-100)

76. Best estimator should be?

77. MLE of p in B(n, p)?

78. Sample variance divisor for unbiasedness?

79. χ^2 goodness formula?

80. Two-sample t df with $n_1=15$, $n_2=12$, equal variances?

81. F-test compares?

82. Decision rule in F-test?

83. A 5 % CI is ____ wide than 1 % CI?

84. Z critical for 99 % two-tail?

85. ANOVA tests equality of?

86. df for two-way ANOVA error in 4 × 3?

87. SSB = 80, SSW = 60, k = 4, N = 24. F?

88. ANOVA assumes?

89. Time series: cyclical period typically?

90. Multiplicative time series model?

91. Sum of 12 monthly seasonal indices?

92. Linear trend slope ($\Sigma X = 0$)?

93. Laspeyres weights are?

94. Paasche weights are?

95. Fisher = ?

96. Test Fisher satisfies?

97. Real wage formula?

98. Purchasing power of money?

99. Index of Industrial Production is what kind of index?

100. Marshall-Edgeworth weight?

Mock #2 Answer key

#	Ans	#	Ans	#	Ans	#	Ans
1	5	26	variance	51	1/2	76	unbiased + min variance + consistent + sufficient
2	f/h (frequency density)	27	0	52	5/16	77	X/n
3	12	28	3·(70−65)/8 = 1.875	53	0.20	78	n − 1
4	0	29	leptokurtic	54	0.7	79	$\Sigma(O-E)^2/E$
5	AM ≥ GM ≥ HM	30	subtract $h^2/12$	55	6/45 = 2/15	80	25
6	exactly two	31	T	56	(0.6·0.5)/(0.6·0.5 + 0.4·0.8) = 0.30/0.62 ≈ 0.484	81	two variances
7	25.5th	32	$-\sqrt{0.24} \approx -0.490$	57	μ=10, σ=√5 ≈ 2.24	82	F > F_table → reject
8	P_{80}	33	0.49	58	16 (since q = 3/4, p = 1/4, n = 16)	83	wider
9	26	34	$(\bar{X}, \bar{Y})$	59	e^(−3) ≈ 0.0498	84	2.576
10	extreme values	35	90°	60	P(Z > 2) = 0.0228	85	means
11	8	36	1 − 6·50/(8·63) = 1 − 0.595 = 0.405	61	mean μ	86	(4−1)(3−1) = 6
12	3 Med − 2 Mean	37	0.75	62	(80 − 70)/5 = 2	87	(80/3)/(60/20) = 26.67/3 = 8.89
13	29.5 – 39.5	38	negative	63	nK/N = 5·8/20 = 2	88	normality, equal var, indep
14	162°	39	[0, 1]	64	144/12 = 12	89	2–10 years
15	√8 ≈ 2.83	40	[−1, 1]	65	1	90	Y = T·S·C·I
16	25	41	0.6	66	stratified	91	1200
17	σ = 10 % of mean	42	±0.9	67	30	92	$b = \Sigma XY/\Sigma X^2$
18	40/80 = 0.5	43	$\sigma_Y\sqrt{1-r^2}$	68	√(0.6·0.4/400) = √0.0006 ≈ 0.0245	93	base year (Q_0)
19	8	44	[−1, 1]	69	$(1.96·20/5)^2 \approx 61.5$ → 62	94	current year (Q_1)
20	(64−1)/12 = 5.25	45	0 (symmetric)	70	systematic	95	√(L · P)
21	perfect equality	46	origin & positive scale	71	quantifiable error estimate	96	time + factor reversal
22	50	47	coincide	72	decreases	97	money wage / CPI × 100
23	none	48	1.0 (perfect; but cap at 1)	73	non-probability	98	1 / CPI × 100

#	Ans	#	Ans	#	Ans	#	Ans
24	1.2	49	high (close to ±1)	74	α	99	quantity index
25	$(Q_3-Q_1)/(Q_3+Q_1)$	50	undefined	75	$1 - \beta$	100	$Q_0 + Q_1$

APPENDIX G — JSO PYQ-Framing Bank (recognise the wording, lock the answer)

> *Examiners reuse phrasings. Once you have seen the framing, you can answer the Q in 10 seconds. This bank lists every "tell" the JSO has used in the last 6 papers.*

G.1 Central tendency framings

Frame	What they actually want	Auto-answer cue
"The algebraic sum of deviations from $\bar{X}$ is ..."	0	always
"If each observation is multiplied by 4 ..."	new mean = 4 × old mean; new SD = 4 × old SD	scale property
"The combined mean of two groups ..."	weighted-by-size mean	use $\Sigma n \bar{X} / \Sigma n$
"The harmonic mean is appropriate for ..."	rates / equal-distance speeds	not "averages" generally
"Median is the ...th item"	(n+1)/2 if n is odd; mean of n/2 and (n/2+1) if even	sorted list
"Empirical relation between mean, median, mode"	Mode = 3 Med − 2 Mean	memorise

G.2 Dispersion framings

Frame	Auto-answer cue
"Most stable / consistent / homogeneous"	smaller CV
"Var(X + Y) when X and Y are independent"	Var(X) + Var(Y) (NOT minus)
"If each value is increased by k"	SD unchanged
"If each value is multiplied by c"	SD multiplied by
"Variance of first n natural numbers"	$(n^2 - 1)/12$
"QD : MD : SD"	10 : 12 : 15

G.3 Correlation / regression framings

Frame	Auto-answer cue
"Range of correlation coefficient"	[−1, 1]
"Two regression lines intersect at"	$(\bar{X}, \bar{Y})$
"If r = 0, the lines are"	perpendicular
"If r = ±1, the lines"	coincide
"Correlation is independent of"	origin and (positive) scale
"r in (cX, dY) with c, d > 0"	same r
"r in (−X, Y) or (X, −Y)"	sign flips
"If both regression slopes are negative, r is"	negative

G.4 Probability framings

Frame	Auto-answer cue
"ME → Independent?"	typically NO
"P(A ∪ B) when ME"	P(A) + P(B)
"P(A ∩ B) when independent"	P(A) P(B)
"P(at least one of ...)"	use complement: 1 − P(none)
"P(exactly one of A or B)"	P(A) + P(B) − 2 P(A ∩ B)
"Given a defective is from machine X, prior was Y ..."	Bayes' theorem
"Sensitivity, specificity, prevalence"	Bayes' clinical setup

G.5 Distribution framings

Frame	Auto-answer cue
"Mean and variance are equal"	Poisson
"n trials, fixed p"	Binomial
"Symmetric, bell-shaped"	Normal
"Without replacement, finite population"	Hypergeometric
"B(n, p) approximates Poisson when"	n large, p small, np = λ
"B(n, p) approximates Normal when"	np > 5, nq > 5

G.6 Sampling and inference framings

Frame	Auto-answer cue
"Each unit has equal chance"	SRS
"Within-group homogeneous, between-group heterogeneous"	stratified
"Within-group heterogeneous, between-group homogeneous"	cluster
"Population ordered, every kth picked"	systematic
"Existing subjects refer next ones"	snowball (non-probability)
"Type I error"	rejecting a TRUE H_0
"Power of test"	1 − β
"Best critical region"	Neyman-Pearson lemma
"Test of two variances"	F
"Test of independence in r × c"	χ^2, df = (r−1)(c−1)

G.7 Time series and index framings

Frame	Auto-answer cue
"Regular within-year pattern"	seasonal
"Irregular wave > 1 year"	cyclical
"Even-period MA needs"	centring

Frame	Auto-answer cue
"Sum of 4 quarterly seasonal indices"	400
"Laspeyres uses ... year quantities"	base (Q_0)
"Paasche uses ... year quantities"	current (Q_1)
"Ideal index"	Fisher
"Index satisfying time + factor reversal"	Fisher
"Real wage"	money / CPI × 100
"Purchasing power of money"	1 / CPI × 100

APPENDIX H — Common errors that cost JSO candidates the 95% mark

#	Error	Why it happens	Correct fix
1	Used Z when n was small and σ unknown	Misread "n is large enough"	Use t (df = n−1).
2	Variance subtraction sign	Algebraic instinct	Var(X − Y) = Var(X) + Var(Y) for indep.
3	Mode = 2 Med − 3 Mean	Memory slip	Mode = 3 Med − 2 Mean.
4	r > 1 or two slopes both > 1	Didn't sanity-check	r^2 = product of slopes ≤ 1.
5	Forgot finite-population correction	Skipped a hint	Multiply variance by (N−n)/(N−1).
6	Two-tail α used as one-tail	Didn't check the alternative	Two-tail uses α/2 in each tail.
7	Sample variance with n divisor used in t-test	Out of habit	t-test uses s^2 with (n−1) divisor.
8	Bayes denominator incomplete	Forgot to add other branches	Sum P(B_j) P(A
9	Calculated angle in pie chart as % directly	Skipped × 360°	Angle = % × 360°/100.
10	β_2 = 3 marked as leptokurtic	Misread cutoff	β_2 = 3 mesokurtic; > 3 lepto.
11	Used arithmetic mean for equal-distance speeds	Default thinking	Use HM.
12	Used HM for equal-time speeds	Confusion	If equal **times**, use AM, not HM.
13	Forgot h^2 in step-deviation variance	Missed scaling	σ^2 scales by h^2.
14	Confused stratified with cluster	Misread	Stratified: within homogeneous; cluster: within heterogeneous.
15	Wrote Var(aX + b) = a^2 Var(X) + b^2	Wrong	Var(aX + b) = a^2 Var(X). The constant b drops.
16	Wrote E[XY] = E[X] E[Y] without independence	Lazy	Only true under independence.
17	Applied Sheppard correction to grouped Mean	Wrong target	Sheppard corrects variance, not mean.
18	Computed Bowley as $(Q_3 - Q_1)/(Q_3 + Q_1)$	That's the coefficient of QD, not Bowley Sk	Bowley = $(Q_3 + Q_1 - 2Q_2)/(Q_3 - Q_1)$.
19	Treated MLE as always unbiased	False	MLE is consistent but often biased (e.g., σ^2 with divisor n).
20	Used r as Cov(X,Y) directly	Forgot normalisation	r = Cov / ($\sigma_X \sigma_Y$).

APPENDIX I — Memory tricks (mnemonics that actually stick)

Concept	Mnemonic
Order: Mode < Median < Mean for positive skew	"Mo-Me-Mn" alphabetical = "left to right" with **right tail** longer
Skewness sign	Sign matches direction of the **tail**, not the bulk
Kurtosis	**Lepto** = **L**eaping (tall peak); **Platy** = **P**lateau (flat); **Meso** = middle = normal
Laspeyres vs Paasche	"Laspeyres = **L**ast period (base) Q's"; "Paasche = **P**resent Q's"
Type I vs II	**α**lpha = "a t**a**lse alarm" (false alarm = rejecting true H_0); **β**eta = "missed **b**eat" (failing to reject false H_0)
AM vs HM for travel	"**A**M when **A**ll equal times; **H**M when **H**ow-far is equal"
AM ≥ GM ≥ HM	"Always Goes Higher → Mean order goes high to low"
$\beta_2 = 3$ boundary	Normal curve is **3**-sigma standard, $\beta_2 = 3$ too
Median from formula	"L + (N/2 − F)/f × h" — mnemonic: "L plus the gap-from-half over the middle-class freq, scaled"
Spearman	"Six-d-squared over n times n-squared minus one"
Bayes	"Posterior = (Prior × Likelihood) / Evidence"
Empirical relation	"Three-Median minus Two-Mean = Mode" → **3M − 2M = M**
Combined mean	weighted by **n**, not by mean
Combined SD adds d^2	each group's variance + its squared distance from the combined mean
Test stats	**Z**ig-large; **t**iny-σ-unknown; **χ²** for counts/variance; **F** for variance ratio

Final words

This book is now your **complete** JSO reference — 6 critical chapters with 30+ worked problems each, a full topic primer, two 100-Q full-length mocks, an examiner's framing bank, an error catalogue, and a memory-tricks page. Re-read twice, drill the worked examples, time yourself on the two mocks, and the JSO paper becomes a routine.

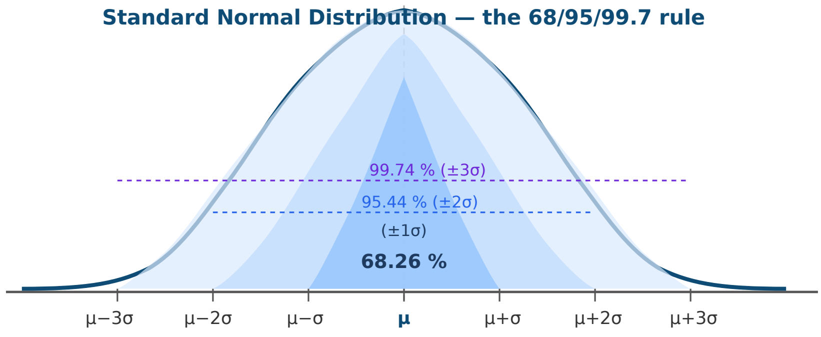
Read smart. Drill hard. Sleep well. **190+/200 is yours.**

— Pariksha365

APPENDIX J — Visual Reference Diagrams

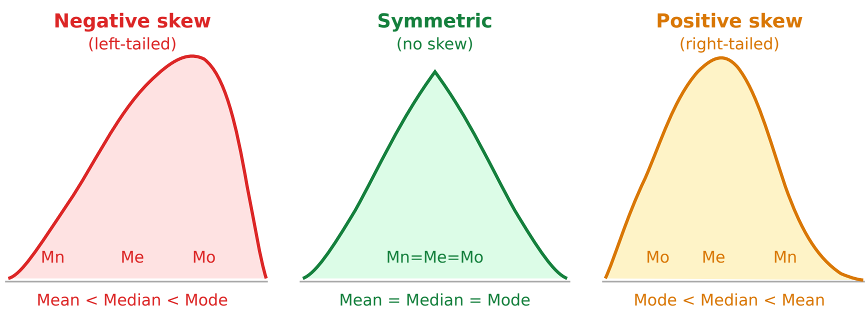
Every diagram below encodes a concept that appears in 2–5 JSO questions per paper. Study the shape first, then the labels, then close the page and sketch it from memory.

J.1 The Normal (Bell) Curve — 68 / 95 / 99.7 rule



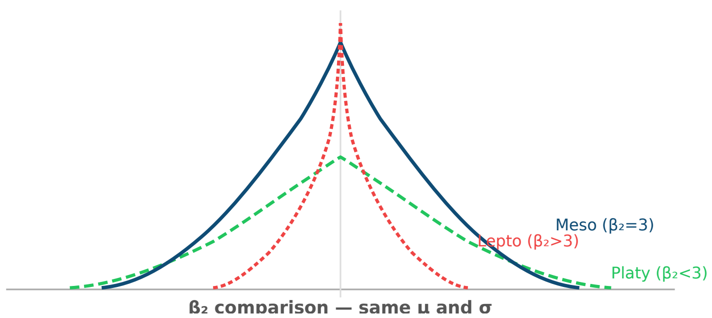
Key reads off this diagram: *- Within ±1σ → 68.26 % of data - Within ±2σ → 95.44 % → $P(|Z| < 1.96) \approx 0.95$ (note: 1.96, not exactly 2) - Within ±3σ → 99.74 % - The curve is* ***symmetric*** *around μ; mean = median = mode = μ*

J.2 Shapes of Skewness — visual pattern



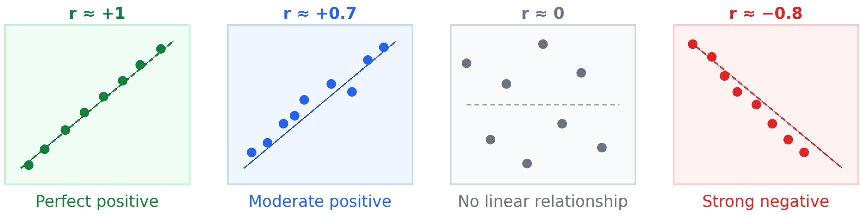
Read: *The mean is always pulled toward the* ***tail****. The mode sits at the* ***peak****. Median lies between.*

J.3 Kurtosis shapes — three curves on one axis

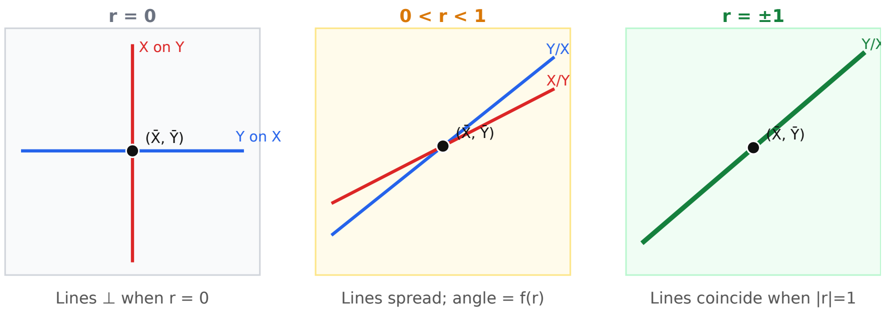


> ***Read:*** *All three share the same mean and SD. Leptokurtic = sharper peak + fatter tails. Platykurtic = flatter peak. Mesokurtic = normal shape (β_2 = 3).*

J.4 Scatter-plot patterns — recognising r at a glance

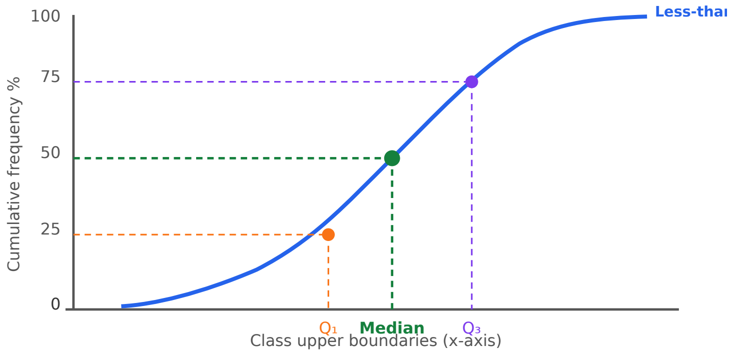


J.5 Two regression lines — how r controls the angle



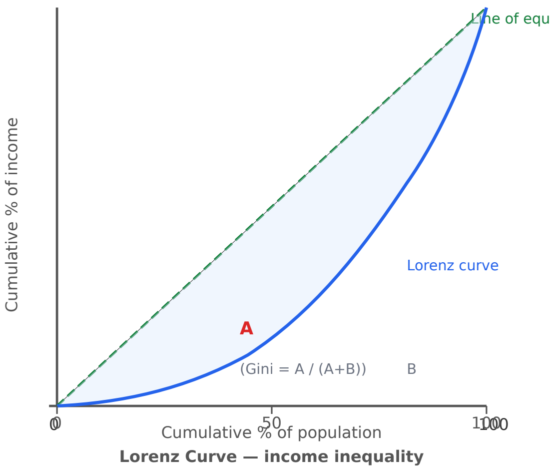
> ***Key:*** *Both lines always pass through $(\bar{X}, \bar{Y})$. As $|r| \to 1$ the lines collapse onto each other. When r = 0 they stand perpendicular.*

J.6 Less-than Ogive → median and quartiles



> ***Read:*** *Draw a horizontal line from the y-axis at N/2 (50 %) to the ogive curve, then drop a vertical to the x-axis →* ***Median****. At N/4 → Q_1. At 3N/4 → Q_3.*

J.7 Lorenz Curve — inequality measurement



> ***Read:*** *- Diagonal = perfect equality (everyone earns the same share). - Bow = the actual Lorenz curve (reality is unequal). - Gini coefficient = area A / (A + B). Ranges 0 (perfect equality) to 1 (perfect inequality). - The greater the bow, the greater the inequality.*

APPENDIX K — Extended Drill for HIGH chapters (Ch 1, 4, 8, 11, 12)

> *20 fresh worked problems per chapter. Pair with the mini-mock and the chapter worked examples for full coverage.*

K-1 Data presentation — 20 worked Qs

✎ **WORKED EXAMPLE**

K-1.1. Census of India 2011 data used by a researcher is which type of data?

Secondary (collected by someone else — Census, NSO, RBI publications are all secondary).

✎ **WORKED EXAMPLE**

K-1.2. A student marks an observation in a class as 10–20. The class mark is 15, width is 10. The exclusive lower boundary is 10. T/F?

True — the exclusive form's lower limit and lower boundary are the same.

✎ **WORKED EXAMPLE**

K-1.3. The following frequency polygon will touch the x-axis at both ends because:

The convention is to extend to the midpoint of the **imaginary classes** just before the first and just after the last actual class (with frequency 0). This closes the polygon to the x-axis.

✎ **WORKED EXAMPLE**

K-1.4. In a cumulative frequency distribution, the last entry equals?

The **total frequency** N (it's the "less than the upper boundary of the last class" count = all observations).

✎ **WORKED EXAMPLE**

K-1.5. A distribution with classes 10–20, 20–30, 30–40 and frequencies 30, 50, 20. What percentage of observations lie below 30?

(30 + 50)/100 = **80 %**.

✎ **WORKED EXAMPLE**

K-1.6. The sum of all relative frequencies is?

Always **1**.

✎ **WORKED EXAMPLE**

K-1.7. Frequency polygon for comparing two distributions is preferred over histograms because:

Two histograms overlap and obscure each other; two polygons on the same axis remain readable (lines can cross without filling confusion).

✎ WORKED EXAMPLE

K-1.8. Ogive can be used to find which measures?

Median, quartiles (Q_1, Q_3), deciles, percentiles — any **positional measure**.

✎ WORKED EXAMPLE

K-1.9. NSS (National Sample Survey) data, when used by another researcher, is which type?

Secondary.

✎ WORKED EXAMPLE

K-1.10. A researcher visits each household in a district to collect income data. Method?

Direct personal interview (investigator collects first-hand → primary data).

✎ WORKED EXAMPLE

K-1.11. In a relative frequency distribution, each class frequency is divided by?

The **total number of observations** (N).

✎ WORKED EXAMPLE

K-1.12. A bar chart is preferred over a histogram when data is?

Discrete / categorical (like states, departments, years as separate labels).

✎ WORKED EXAMPLE

K-1.13. Convert inclusive class 55–64 to exclusive.

Lower boundary = 54.5, upper boundary = 64.5. Exclusive: **54.5 – 64.5**.

✎ WORKED EXAMPLE

K-1.14. "Less than" ogive at x = 40 reads as cumulative frequency 70 out of 100 total. "More than" ogive at x = 40 reads as?

100 − 70 = **30** (complement).

✎ WORKED EXAMPLE

K-1.15. The intersection of "less-than" and "more-than" ogives gives the?

Median (the point where cumulative from below and above are equal at N/2).

✎ WORKED EXAMPLE

K-1.16. A pie chart is drawn for two quantities: 60 and 40. Angles?

60/(60+40) × 360 = **216°** and 40/100 × 360 = **144°**.

✎ WORKED EXAMPLE

K-1.17. Data on the number of family members per household is which type?

Discrete quantitative (count, must be whole number).

✎ WORKED EXAMPLE

K-1.18. Data on the height of a person is which type?

Continuous quantitative (takes any value in a range).

✎ WORKED EXAMPLE

K-1.19. A questionnaire mailed to respondents without an enumerator is called?

Mailed questionnaire method (used when population is literate and spread widely).

✎ WORKED EXAMPLE

K-1.20. A sub-divided bar chart differs from a simple bar chart in that:

Each bar is divided into parts representing component categories, so **both total and composition** are visible simultaneously.

K-4 Moments, Skewness, Kurtosis — 20 worked Qs

✎ **WORKED EXAMPLE**

K-4.1. The first raw moment equals?

$\mu_1' = \bar{X}$ (the arithmetic mean).

✎ **WORKED EXAMPLE**

K-4.2. For any distribution, the first central moment is?

$\mu_1 = 0$ (by definition of mean).

✎ **WORKED EXAMPLE**

K-4.3. $\mu_0 = ?$

$\mu_0 = 1$ (zeroth central moment is always 1, since $\sum f/N = 1$).

✎ **WORKED EXAMPLE**

K-4.4. $\mu_2 = 16$. SD = ?

$\sigma = \sqrt{16} = \mathbf{4}$.

✎ **WORKED EXAMPLE**

K-4.5. $\mu_3 = 0$. What does this tell us about skewness?

$\beta_1 = \mu_3^2/\mu_2^3 = 0$ → **zero skewness → symmetric distribution**.

✎ **WORKED EXAMPLE**

K-4.6. $\mu_2 = 9, \mu_4 = 243$. β_2?

$\beta_2 = 243/81 = \mathbf{3}$ → mesokurtic.

✎ **WORKED EXAMPLE**

K-4.7. $\mu_1' = 10, \mu_2' = 120$. Find variance.

$\mu_2 = 120 - 100 = \mathbf{20}$.

✎ **WORKED EXAMPLE**

K-4.8. $\mu_1' = 5, \mu_2' = 30, \mu_3' = 200$. Find μ_3.

$\mu_3 = 200 - 3(5)(30) + 2(125)$

$= 200 - 450 + 250 = \mathbf{0}$.

✎ **WORKED EXAMPLE**

K-4.9. Karl Pearson Sk = 0.5, mode = 40, σ = 6. Mean = ?

$0.5 = (\bar{X} - 40)/6 \Rightarrow \bar{X} = 43.$

✎ **WORKED EXAMPLE**

K-4.10. Bowley Sk = 0.4, Q_1 = 20, Q_3 = 60. Median = ?

$$0.4 = \frac{60 + 20 - 2Q_2}{60 - 20} = \frac{80 - 2Q_2}{40}$$

$80 - 2Q_2 = 0.4 \times 40 = 16$

$Q_2 = \mathbf{32}.$

✎ **WORKED EXAMPLE**

K-4.11. If $\beta_2 > 3$, the distribution is said to be?

Leptokurtic (sharp peak, fat tails).

✎ **WORKED EXAMPLE**

K-4.12. $\gamma_2 = \beta_2 - 3 = -1.2$. Distribution?

$\beta_2 = 1.8 < 3 \rightarrow$ **platykurtic**.

✎ **WORKED EXAMPLE**

K-4.13. Moments from any origin A vs from the mean: which has $\mu_1 = 0$?

Only from the **mean**. From any other origin the first moment equals $\bar{X} - A \neq 0$.

✎ **WORKED EXAMPLE**

K-4.14. A distribution has the same shape on both sides of the mean. Which odd moments are zero?

All odd central moments: $\mu_1 = \mu_3 = \mu_5 = \cdots = 0$.

✎ **WORKED EXAMPLE**

K-4.15. Sheppard's correction is applied when:

Data is **grouped** (continuous) and class widths are equal. It corrects the upward bias in grouped-data moments.

✎ **WORKED EXAMPLE**

K-4.16. For Karl Pearson Sk, which formula is used when the distribution is bimodal?

Use the **median-based formula**: $\mathrm{Sk} = 3(\bar{X} - \mathrm{Med})/\sigma$ (mode is ambiguous for bimodal).

✎ WORKED EXAMPLE

K-4.17. β_1 is always:

Non-negative ($\mu_3^2 \geq 0$). The sign of skewness is determined by $\gamma_1 = \mu_3/\sigma^3$.

✎ WORKED EXAMPLE

K-4.18. For a normal distribution, β_1 = ? and β_2 = ?

β_1 = **0**, β_2 = **3**.

✎ WORKED EXAMPLE

K-4.19. Bowley's skewness is based on which positional values?

Quartiles (Q_1, Q_2, Q_3) — thus it is resistant to extreme values (unlike Pearson's which uses mean and SD).

✎ WORKED EXAMPLE

K-4.20. If mode > median > mean, skewness is:

Negative (left-skewed, long left tail).

K-8 Sampling Theory — 20 worked Qs

✎ **WORKED EXAMPLE**

K-8.1. A population of 10,000 students; a sample of 100 selected. The mean of the sample is a?

Statistic (calculated from the sample, not the population).

✎ **WORKED EXAMPLE**

K-8.2. A complete enumeration study is called a?

Census.

✎ **WORKED EXAMPLE**

K-8.3. If the population SD is 40 and the sample size is 64, what is the SE of the mean?

$40/\sqrt{64} = 40/8 = \mathbf{5}$.

✎ **WORKED EXAMPLE**

K-8.4. If SE = 4 when n = 25, what is the SE when n = 100?

SE scales with $1/\sqrt{n}$:

$$\text{new SE} = 4 \cdot \frac{\sqrt{25}}{\sqrt{100}}$$

$$= 4 \cdot \frac{5}{10} = \mathbf{2}.$$

✎ **WORKED EXAMPLE**

K-8.5. The sampling distribution of $\bar{X}$ has mean = ?

μ (the population mean) — because $\bar{X}$ is an unbiased estimator.

✎ **WORKED EXAMPLE**

K-8.6. The CLT says that for any population with finite variance, the distribution of $\bar{X}$ tends to be:

Normal as n increases.

✎ **WORKED EXAMPLE**

K-8.7. Frame errors are which type of error?

Non-sampling error (the list of units is incomplete or outdated).

✎ **WORKED EXAMPLE**

K-8.8. Which method is used when population units are naturally ordered and every kth unit is selected?

Systematic sampling.

✎ WORKED EXAMPLE

K-8.9. Cluster sampling is useful when:

No complete frame exists and/or population is spread over a large area (geographic clusters).

✎ WORKED EXAMPLE

K-8.10. A study of drug users reaches new respondents through existing ones. Technique?

Snowball sampling (non-probability).

✎ WORKED EXAMPLE

K-8.11. Which method gives a mathematical bound on sampling error?

Probability sampling (because random selection allows computing variance of the estimator).

✎ WORKED EXAMPLE

K-8.12. If SE of proportion = 0.025 and p = 0.5, what is n?

$0.025 = \sqrt{0.25/n} \Rightarrow n = 0.25/0.000625 = \mathbf{400}$.

✎ WORKED EXAMPLE

K-8.13. To estimate a proportion within ±3 % at 95 % confidence (P unknown, use worst case):

$n = \left(\frac{1.96}{0.03}\right)^2 \times 0.25$

$= (65.33)^2 \times 0.25$

$\approx 4268 \times 0.25 = \mathbf{1067}$.

✎ WORKED EXAMPLE

K-8.14. Purposive (judgement) sampling produces results that are:

Not statistically generalizable (no probability selection → margin of error undefined).

✎ WORKED EXAMPLE

K-8.15. Within a stratified sample, the stratum weight is proportional to its size in the population. This is called?

Proportional allocation.

✎ WORKED EXAMPLE

K-8.16. The finite population correction factor approaches 1 when:

n is small relative to N (sampling fraction n/N → 0). In this case FPC ≈ 1 and is usually ignored.

✎ WORKED EXAMPLE

K-8.17. The standard error of a statistic measures:

The **variability of that statistic** across all possible samples of the same size — i.e., how precise the estimate is.

✎ WORKED EXAMPLE

K-8.18. Non-sampling errors can occur even in a?

Census (because non-sampling errors come from measurement, recording, and processing — they do not require sampling).

✎ WORKED EXAMPLE

K-8.19. A respondent's refusal to answer creates which error?

Non-response bias — a type of non-sampling error.

✎ WORKED EXAMPLE

K-8.20. Stratification is most effective when:

The strata are **internally homogeneous** (variance within each stratum is low), because then the within-stratum estimates are very precise.

K-11 Time Series — 20 worked Qs

✎ WORKED EXAMPLE

K-11.1. Monthly sales data for a company over 5 years is best classified as what kind of data?

Time series (observations on a single variable recorded at successive points in time).

✎ WORKED EXAMPLE

K-11.2. The seasonal component is best described as:

A **regular, predictable** fluctuation within a calendar year (months/quarters) that repeats every year.

✎ WORKED EXAMPLE

K-11.3. Long-term smooth movement in time series data is called?

Secular trend (T).

✎ WORKED EXAMPLE

K-11.4. A business cycle is part of which component?

Cyclical (C) — wave-like movements lasting 2–10 years.

✎ WORKED EXAMPLE

K-11.5. Which component covers events like floods, earthquakes, elections?

Irregular (I) — unpredictable, no fixed pattern.

✎ WORKED EXAMPLE

K-11.6. Sales (in units): 120, 130, 125, 135, 140. Compute the 3-year centred moving average for years 2, 3, 4.

Year 2 MA = (120 + 130 + 125)/3 = **125**. Year 3 MA = (130 + 125 + 135)/3 = **130**. Year 4 MA = (125 + 135 + 140)/3 = **133.33**.

✎ WORKED EXAMPLE

K-11.7. In the semi-average method with 8 years of data, the trend line uses?

Mean of years 1–4 and mean of years 5–8. Place each average at the midpoint of its half (year 2.5 and year 6.5). Draw the line.

✎ WORKED EXAMPLE

K-11.8. A 4-year moving average of data 10, 14, 18, 12, 20, 16, 24, 18 — first two 4-yr MA values?

First: (10+14+18+12)/4 = **13.5**. Second: (14+18+12+20)/4 = **16**.

Centred: (13.5 + 16)/2 = **14.75** (placed at year 3).

✎ WORKED EXAMPLE

K-11.9. A trend line $\hat{Y} = 50 + 3X$ with X coded such that X=0 is 2020 and unit X=1 is 1 year. Predicted value for 2024?

X = 4. $\hat{Y} = 50 + 12 = \mathbf{62}$.

✎ WORKED EXAMPLE

K-11.10. Seasonal index = 112 for Q2. It means?

Q2 values are **12 % above** the annual average in that quarter.

✎ WORKED EXAMPLE

K-11.11. Seasonal index = 85 for a quarter. It means?

That quarter is **15 % below** the overall trend/average.

✎ WORKED EXAMPLE

K-11.12. The sum of seasonal indices over 4 quarters should equal 400. A set of indices is 95, 110, 105, 92. Their sum is 402. Adjusted index for Q1?

Adjustment factor = 400/402. Q1 adjusted = 95 × 400/402 ≈ **94.5**.

✎ WORKED EXAMPLE

K-11.13. To isolate the trend from a multiplicative model, you:

Divide Y by the seasonal index: T × C × I = Y / S. Further smoothing removes C and I → pure trend.

✎ WORKED EXAMPLE

K-11.14. The least-squares method of trend fitting minimises?

$\sum(Y - \hat{Y})^2$ (sum of squared deviations between actual and estimated trend values).

✎ WORKED EXAMPLE

K-11.15. A 3-point moving average of a series with 10 data points produces how many MA values?

10 − 3 + 1 = **8** MA values (the first and last points are lost).

✎ WORKED EXAMPLE

K-11.16. Method of simple averages for seasonal indices: what does each "seasonal average" use?

The average of observations for that particular season (e.g., all January values, then all February values, etc.) across all years.

✎ **WORKED EXAMPLE**

K-11.17. Why is the multiplicative model more commonly used than the additive model?

In most real-world series, seasonal **fluctuations grow proportionally** with the level (trend). Multiplicative correctly captures this; additive assumes constant seasonal swings.

✎ **WORKED EXAMPLE**

K-11.18. A moving average with large period (e.g., 12-month) better smooths:

Short-term fluctuations (seasonal and irregular), leaving only the long-term trend. The penalty: more observations lost at each end.

✎ **WORKED EXAMPLE**

K-11.19. Moving average "loses" observations. For a 5-point MA on a 20-point series, how many points remain?

20 – (5 − 1) = **16** (2 lost at each end).

✎ **WORKED EXAMPLE**

K-11.20. Why does a semi-average trend line use two points and not one?

One point defines a value (not a direction). **Two points determine a line** (slope and intercept). The semi-average method needs at least two summary means to define the trend line.

K-12 Index Numbers — 20 worked Qs

✎ WORKED EXAMPLE

K-12.1. Compute Laspeyres price index from: P_0 = (10, 6), Q_0 = (4, 5), P_1 = (12, 8).

Numerator: $12 \times 4 + 8 \times 5 = 48 + 40 = 88$

Denominator: $10 \times 4 + 6 \times 5 = 40 + 30 = 70$

$L = 88/70 \times 100 = \mathbf{125.71}$.

✎ WORKED EXAMPLE

K-12.2. Add Q_1 = (5, 4) to K-12.1 and compute Paasche.

Numerator: $12 \times 5 + 8 \times 4 = 60 + 32 = 92$

Denominator: $10 \times 5 + 6 \times 4 = 50 + 24 = 74$

$P = 92/74 \times 100 = \mathbf{124.32}$.

✎ WORKED EXAMPLE

K-12.3. Fisher for K-12.1/2?

√(125.71 × 124.32) = √(15634) ≈ **125.01**.

✎ WORKED EXAMPLE

K-12.4. The time reversal test: $P_{01} \times P_{10}$ = ?

1 (or 100 if expressed as index number). Fisher satisfies this.

✎ WORKED EXAMPLE

K-12.5. Factor reversal test: $P_{01} \times Q_{01}$ = ?

V_{01} (the value index). Only Fisher satisfies both time and factor reversal simultaneously.

✎ WORKED EXAMPLE

K-12.6. An index number of 140 means prices are now what % of base-year prices?

140 % of base, i.e., a **40 % increase**.

✎ WORKED EXAMPLE

K-12.7. Index in 2010 = 100 (base 2000). Index in 2015 = 150 (base 2000). Shift base to 2010.

Index_{2015} (base 2010) = 150/100 × 100 = **150**.

✎ **WORKED EXAMPLE**

K-12.8. An index series is 80 (2018), 100 (2020 = base), 110 (2022). Purchasing power of money in 2022?

1/110 × 100 = **90.9** paise per rupee of 2020 value.

✎ **WORKED EXAMPLE**

K-12.9. A worker's money wage in 2023 is ₹20,000. CPI 2023 = 250 (base 2010 = 100). Real wage in 2010 rupees?

Real wage = 20000/250 × 100 = **₹8,000**.

✎ **WORKED EXAMPLE**

K-12.10. In the family budget method, the weight W for each item is?

$W = P_0 Q_0$ (base-year expenditure on that item).

✎ **WORKED EXAMPLE**

K-12.11. The weighted AM of price relatives (family budget method) formula?

$\frac{\sum W(P_1/P_0) \times 100}{\sum W}$, where W = P_0 Q_0.

✎ **WORKED EXAMPLE**

K-12.12. A simple aggregate price index ignores:

Quantities / weights of items — items with high prices dominate regardless of importance.

✎ **WORKED EXAMPLE**

K-12.13. Walsh index uses which weights?

Geometric mean of quantities: $\sqrt{Q_0 Q_1}$.

✎ **WORKED EXAMPLE**

K-12.14. Splicing is done when:

Two index series have a common **overlap year** and need to be joined into one continuous series.

✎ **WORKED EXAMPLE**

K-12.15. A price index of 200 and a quantity index of 150 (both with the same formula). Value index approximately?

Under factor-reversal, V_{01} = P_{01} × Q_{01} = 200 × 150 / 100 (adjusting for base) = **300**. (Direct check: if both are Fisher, product = value index.)

✎ WORKED EXAMPLE

K-12.16. The Consumer Price Index measures:

The average change in prices paid by a **defined group of consumers** for a fixed basket of goods and services. (Not WPI, which is producer/trade prices.)

✎ WORKED EXAMPLE

K-12.17. WPI differs from CPI primarily in:

WPI measures price change at the **wholesale/manufacturer level**; CPI measures at the **retail / consumer level**. Their baskets differ; CPI includes services more heavily.

✎ WORKED EXAMPLE

K-12.18. An index of 160 in period 1 and 200 in period 2 (same base). Relative change from period 1 to period 2?

$(200 - 160)/160 = 40/160 =$ **25 %** increase.

✎ WORKED EXAMPLE

K-12.19. Deflating a nominal GDP series uses which index?

A **price index** (like the GDP deflator or WPI). Real GDP = Nominal GDP / Price Index × 100.

✎ WORKED EXAMPLE

K-12.20. A 3-item Laspeyres is 110. If $\Sigma P_1 Q_0 = 440$ and $\Sigma P_0 Q_0 = 400$. Verify.

$440/400 \times 100 =$ **110** ✓.

APPENDIX L — Plain-English Concept Primers (for first-time readers)

*If you have **never studied statistics before**, read the primer for each chapter before reading its main section. Each primer uses zero formulas — only plain language and intuition. After the primer, the formulas will land cleanly.*

L.1 What is "data presentation"? (Chapter 1 primer)

Imagine you collect the ages of 200 people. You have a list of 200 numbers. In this raw form, the data tells you nothing useful at a glance.

What statisticians do: They group the numbers into bins (e.g., 20–30, 30–40, 40–50) and count how many fall in each bin. This table is a **frequency distribution** — now you can see the shape of the data instantly.

Then they draw it. A **histogram** is that table drawn as touching bars (width = one bin, height = count). A **frequency polygon** joins the midpoints of the bars with a line. An **ogive** is the running total (cumulative), which lets you answer "what % of people are younger than 35?".

The key insight: The right graph makes the pattern visible instantly. That's why data presentation exists — not to decorate reports, but to reveal structure that raw numbers hide.

L.2 What is "central tendency"? (Chapter 2 primer)

Suppose two factories produce light bulbs. Factory A has lifetimes: 800, 900, 850, 950, 900 hours. Factory B: 200, 1500, 100, 1800, 400 hours. Both have the same **average** (= 880 hours), but Factory B is clearly worse.

That average is a **measure of central tendency** — it picks one number to represent the centre of a dataset. The three main ones:

- **Mean:** Add everything up, divide by count. Fair, uses all data. Pulled by outliers.
- **Median:** The middle value when you line everyone up in order. Not affected by extremes.
- **Mode:** The most common value. What occurs most often.

Why three? Because different situations need different summaries. If 9 people earn ₹20,000 and 1 CEO earns ₹10,000,000, the mean salary is ₹1,018,000 — wildly misleading. The median (₹20,000) tells you what most people actually earn.

L.3 What is "dispersion"? (Chapter 3 primer)

The average tells you the centre. But two datasets can have the same average and be completely different.

Class A scores: 70, 70, 70, 70, 70. Class B: 50, 60, 70, 80, 90. Both have mean 70. But Class A is uniform; Class B is spread out.

Dispersion measures this spread. If dispersion is small, data is bunched near the mean (consistent, predictable). If large, data is scattered (variable, risky).

The most important dispersion measure: **standard deviation (SD)**. Think of it as the "average distance from the mean". If SD = 0, all values are identical. The larger the SD, the more spread.

CV (coefficient of variation) = SD ÷ Mean. It answers: "is a SD of 10 big or small?" — compared to a mean of 20, it's huge (50%); compared to a mean of 1000, it's tiny (1%). Use CV to compare variability between datasets with different scales.

L.4 What are "moments"? (Chapter 4 primer)

A **moment** is a formal way to summarise the shape of a distribution using powers of deviations from the mean.

- 2nd moment = variance (captures width/spread).

- 3rd moment = captures lopsidedness (**skewness**). If positive, the tail stretches right.
- 4th moment = captures peakedness (**kurtosis**). High 4th moment = values are concentrated near the mean with heavy tails.

Skewness in plain English: Imagine salary distributions at a company. Most employees earn ₹30,000–50,000 but a handful of executives earn ₹2 million. The bulk is on the left; the tail stretches right — **positive skew**. The mean gets pulled toward those high salaries, past the median, past the mode. That's the key diagnostic: position of mean vs median vs mode tells you skewness direction.

Kurtosis in plain English: Compare two investments with the same average return. One is steady (low kurtosis = platykurtic — outcomes cluster near average). The other is mostly flat but occasionally swings wildly (high kurtosis = leptokurtic — fat tails). Same mean, same SD, but completely different risk profiles. Kurtosis captures this.

L.5 What is "correlation"? (Chapter 5 primer)

Correlation answers: "Do these two things move together?"

Examples: Height and weight (taller people tend to weigh more — **positive correlation**). Ice cream sales and drowning incidents (both go up in summer, but **no causal link**). Unemployment rate and GDP growth (inversely related — **negative correlation**).

Karl Pearson's r quantifies this: +1 means a perfect straight-line relationship going up. −1 means a perfect straight-line relationship going down. 0 means no linear relationship.

Regression answers a follow-up: "If I know X, what is my best guess for Y?" A regression line is the "line of best fit" through the scatter of points. It minimises the total squared error between actual Y values and line-predicted Y values.

The two regression lines: One predicts Y from X (Y on X line); the other predicts X from Y (X on Y line). They are different lines unless correlation is perfect ($r = \pm 1$), in which case they coincide. Both lines always cross at the point (mean of X, mean of Y).

L.6 What is "probability"? (Chapter 6 primer)

Probability is the science of quantifying uncertainty.

When you flip a fair coin, there are 2 equally likely outcomes. Heads = 1 out of 2 → probability = 1/2. When you draw a card from a standard deck, there are 52 equally likely outcomes. Getting an ace = 4 out of 52 = 1/13.

Three definitions: - Classical (equally likely outcomes, listed above). - Frequentist (toss the coin 1 million times; the fraction of heads converges to 0.5). - Axiomatic (a mathematical framework: probability is a function with 3 axioms).

Conditional probability: $P(A \mid B)$ = "probability of A given that B has already happened." If it rained today, the probability that the ground is wet is very high (the event "ground is wet" is conditional on rain).

Bayes' theorem flips the conditioning: given the ground IS wet, what's the probability it rained? This "reverse conditioning" — from effect back to cause — appears in diagnostic testing, spam filtering, and at least one JSO question every paper.

L.7 What is a "random variable"? (Chapter 7 primer)

A **random variable** is a variable whose value is determined by a random experiment.

When you roll a die, the outcome X can be 1, 2, 3, 4, 5, or 6. Each value has a probability (1/6). The list of values and their probabilities is the **probability distribution**.

Discrete vs Continuous: - If X can only take countable values (number of heads, number of defects), it's discrete. Its distribution is described by a **probability mass function (pmf)**. - If X can take any value in an interval (someone's exact weight, time to failure of a machine), it's continuous. Its distribution is described by a **probability density function (pdf)**.

The four named distributions you must know are just the most commonly occurring shapes in real problems: - **Binomial** → count of successes in n fixed independent trials. - **Poisson** → count of rare events in a fixed time/area. - **Normal** → the famous bell curve, appears naturally when many independent small factors combine (heights, measurement errors, exam scores). - **Hypergeometric** → like Binomial but without replacement (the probabilities change with each draw).

L.8 What is "sampling"? (Chapter 8 primer)

You want to know the average income of India's 1.4 billion people. Measuring every person is impossible. So you measure a sample — say 100,000 people — and estimate the population's income from it.

The core question: How do you choose those 100,000 so the estimate is trustworthy?

Simple random sampling (SRS): Every person has an equal chance. Like a lottery.

Stratified: Divide population into groups (rural / urban; states; income brackets) and randomly sample within each group. More precise because you ensure representation of each group.

Cluster: Geographic areas (districts, villages) are the sampling units. Sample some clusters, then study every person within those clusters. Cheaper than travelling everywhere.

Systematic: Pick every kth person from an ordered list.

The standard error of your estimate measures how much it would vary across different random samples. Larger sample → smaller SE → more precise estimate. This is the **mathematical guarantee** that probability sampling provides — which non-probability methods (quota, judgement, snowball) cannot give.

L.9 What is "statistical inference"? (Chapter 9 primer)

You measured a sample; now you want to say something about the whole population. That's inference.

Estimation: Use the sample to pin down a population number. "Based on our sample of 200 voters, we estimate 58% will vote for Party A, with a margin of error of ±3%." The ±3% is the confidence interval — a range wide enough that you're 95% sure the true value lies inside.

Hypothesis testing: You have a specific claim ("this medicine reduces fever by 2°C on average") and you test it against data. The null hypothesis (H_0) is the conservative position ("no effect"). The alternative (H_1) is the claim. You collect data and ask: "Is this data surprising if H_0 were true?" If yes, you reject H_0.

The risk table:

	H_0 true	H_0 false
Reject H_0	**Type I error (false alarm)**	Correct (Power)
Don't reject	Correct	**Type II error (missed detection)**

Setting α = 5% means you'll accept a 5% chance of a false alarm. Power = $1 - \beta$ tells you how good the test is at catching a real effect.

L.10 What is "ANOVA"? (Chapter 10 primer)

Suppose you test three fertilisers on crops. Each fertiliser is applied to several plots, and you measure yield. Is there any real difference between fertilisers, or is the yield variation just random noise?

You could run three t-tests (A vs B, A vs C, B vs C). But three tests at 5% significance gives a 14% chance of a false alarm across all three. ANOVA does all comparisons in one coherent test.

The idea: If fertilisers matter, yields should vary more *between* fertiliser groups than *within* each group (within-group variation = random noise). ANOVA computes this ratio: F = (variance between groups) / (variance within groups). A large F → groups are genuinely different.

Partition of variation: Total variation = variation between groups + variation within groups. This is the SST = SSB + SSW identity.

L.11 What is "time series analysis"? (Chapter 11 primer)

A **time series** is any variable measured repeatedly over time: monthly sales, daily stock prices, annual rainfall, quarterly GDP.

The goal: understand and forecast the pattern.

Every time series is a mixture of four components: 1. **Trend** — the long-run direction (upward, downward, flat). 2. **Seasonal** — the regular within-year cycle (ice cream sales spike in summer every year). 3. **Cyclical** — the business cycle (boom-bust, 2–10 year waves). 4. **Irregular** — random shocks (a pandemic, a flood).

To analyse the series you need to **separate** these components. Once you have the trend alone, you can project it forward (forecasting). Once you have the seasonal index, you can "deseasonalise" data to see whether this month's sales are genuinely good or just good because it's festive season.

L.12 What are "index numbers"? (Chapter 12 primer)

Prices change over time. How do you express "prices overall went up by 20% since 2010"?

An **index number** is that summary. It takes a baseline period (Base = 100) and expresses current levels as a ratio. A price index of 125 means prices are 25% higher than the base.

Why not just use percentage change? Because there are many prices (rice, rent, petrol, medicine) and you need to **average** them in a sensible way — weighted by how much people actually spend on each item. That's where Laspeyres, Paasche, and Fisher come in: they differ in whether they use old quantities, new quantities, or a blend as weights.

Real-world examples: The Consumer Price Index (CPI) measures inflation; it's used to adjust salaries ("dearness allowance" in government pay). The Wholesale Price Index (WPI) measures producer-level inflation. The Index of Industrial Production (IIP) is a quantity index.
